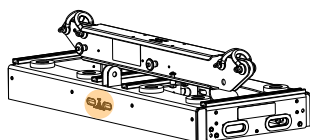


Rigging procedures

Preparing K1-BUMP

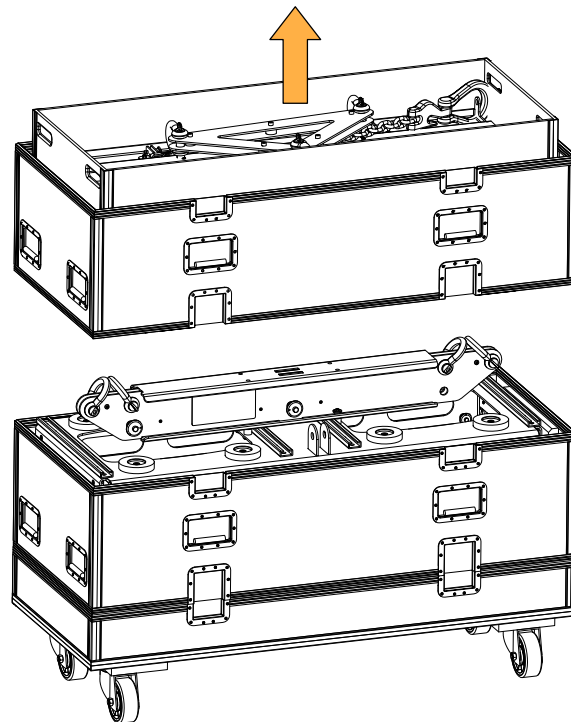
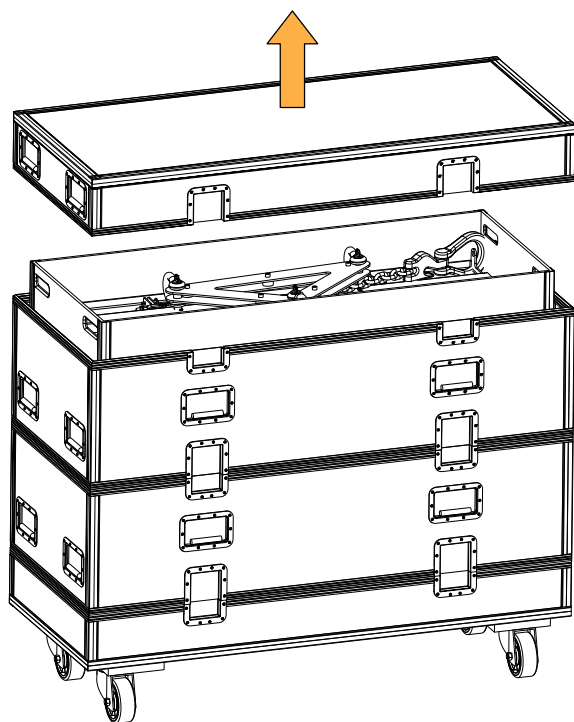
Min number of operators	2
Rigging accessories	K1-BUMP 4 Ø 22 mm shackles WLL 3.25 t K1-DELTA (optional) K1-BPCHAIN (optional)

- 
All along the procedure:
 Systematically ensure that each ball-locking pin is fully inserted by pulling on it.
- 
 To avoid any physical handling, K1-BUMP must be transported and prepared in K-BUMPFLIGHT.
- 
 An identification sticker is present at the front of K1-BUMP to identify the front and the rear sides.

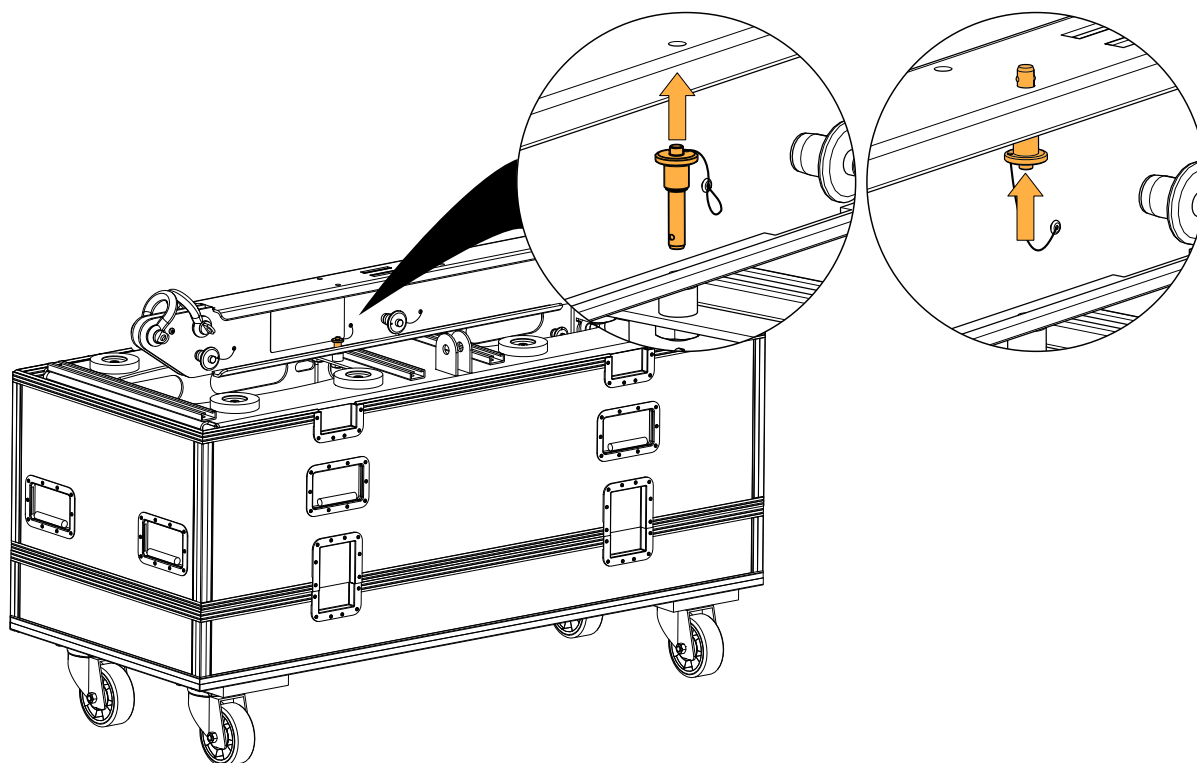


Procedure

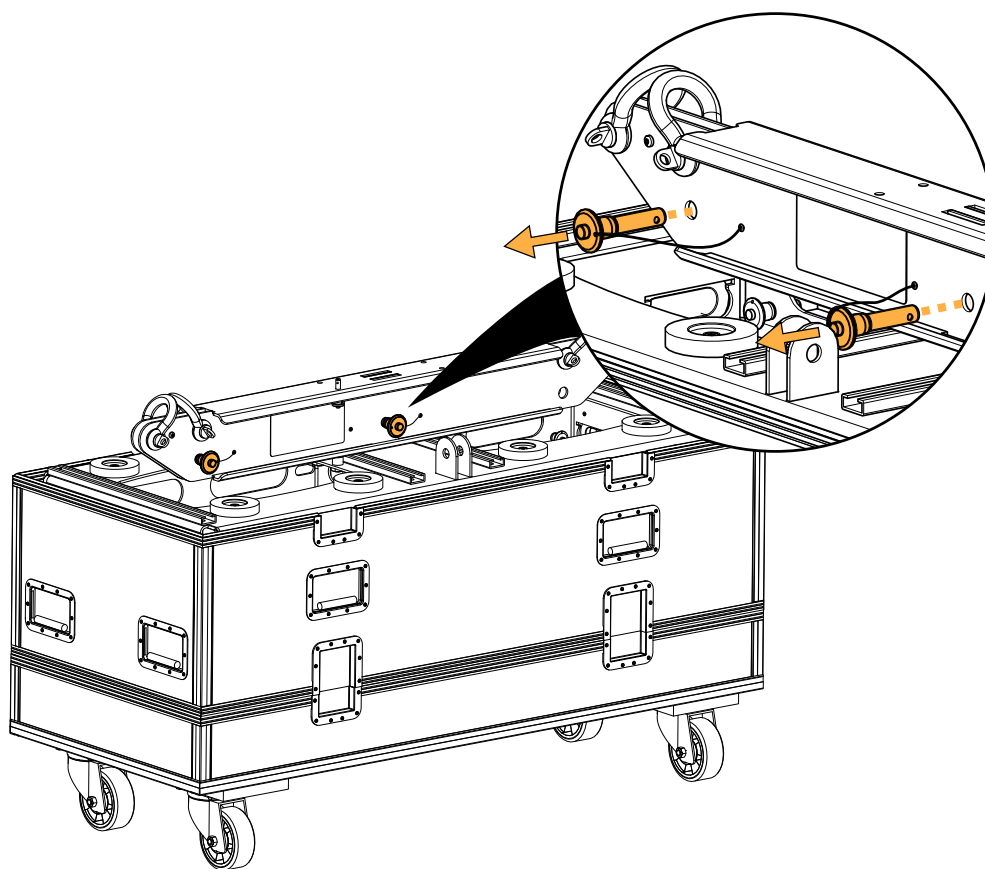
1. Place K-BUMPFLIGHT at the rigging location. Remove the lid and the first compartment.



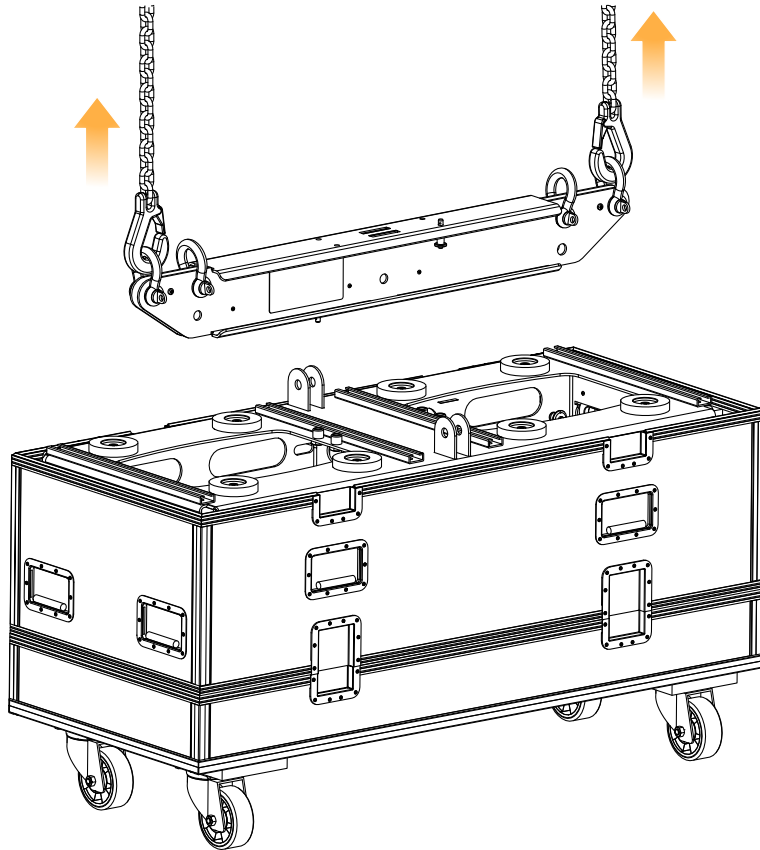
- 2.** On each side of the K1-BUMP rigging bar, remove the storage ball-locking pin. Secure the storage ball-locking pin in the rigging bar.



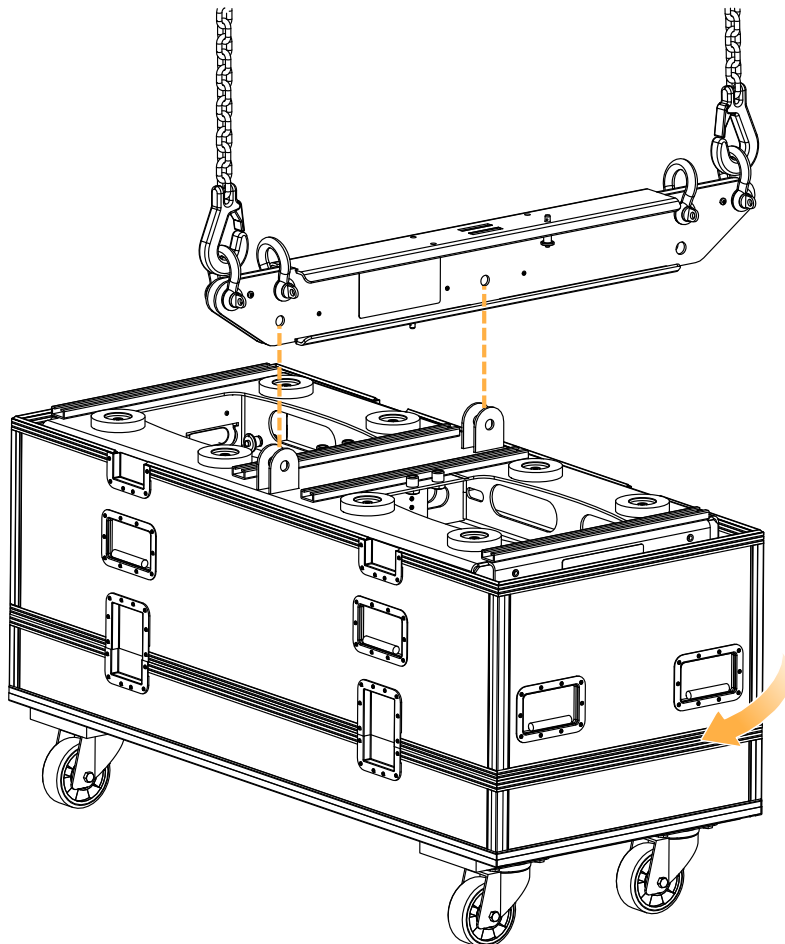
- 3.** Remove the two rigging ball-locking pins.



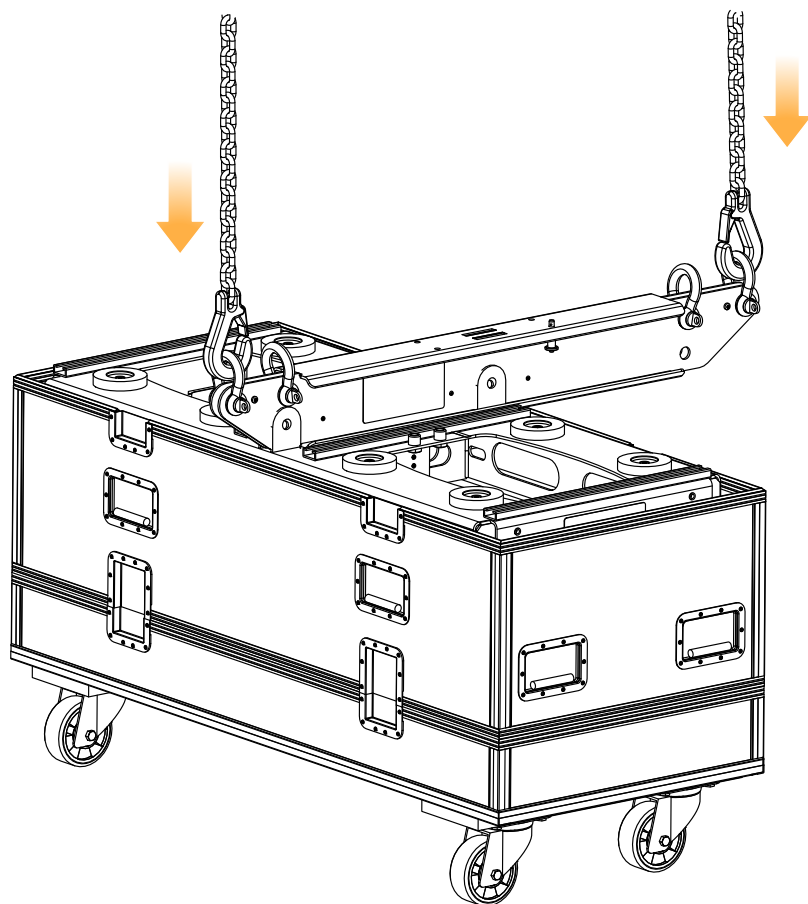
- 4.** Put the K1-BUMP rigging bar in rigging position:
a) Lift the K1-BUMP rigging bar using the motor.



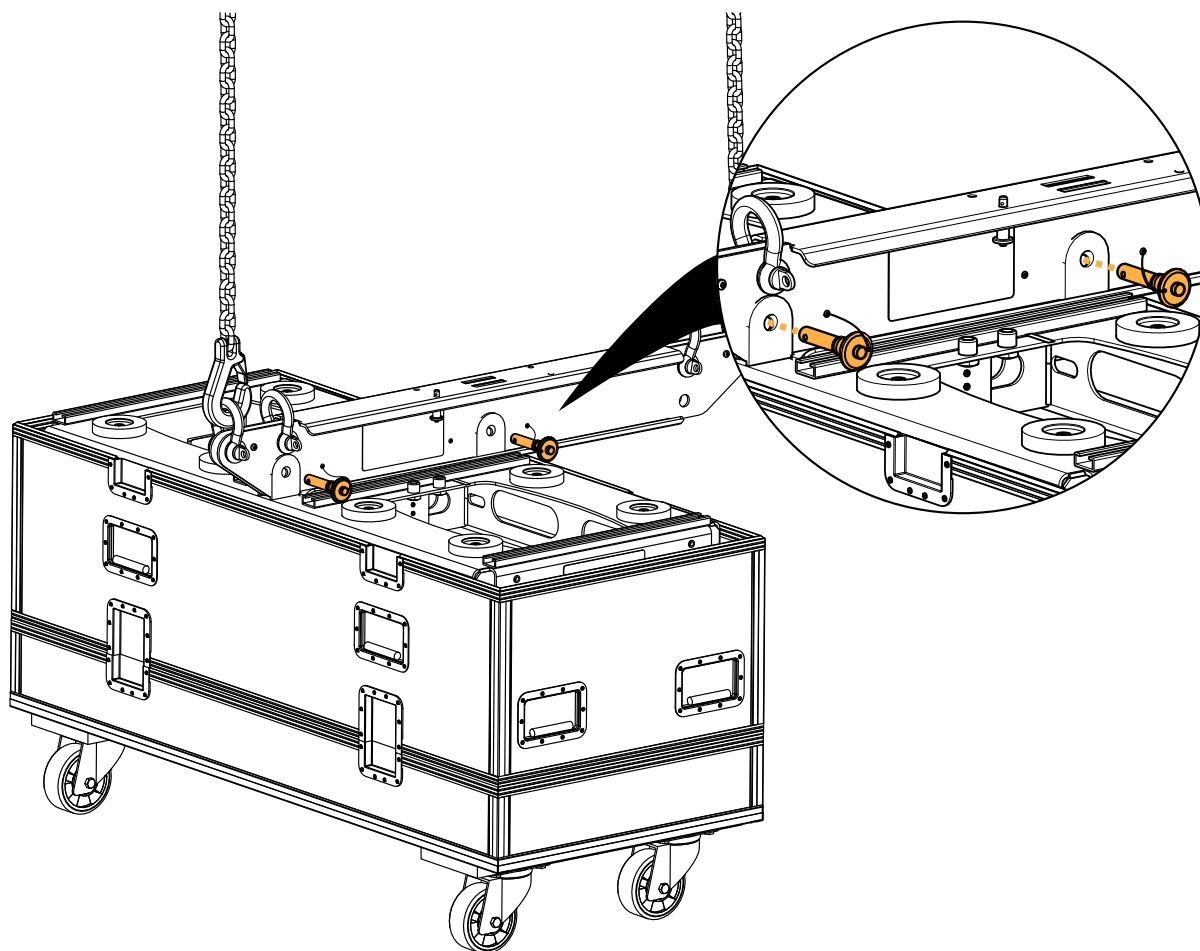
- b) Rotate K-BUMPFLIGHT and align K1-BUMP rigging frame holes with the rigging bar holes.



c) Lower and position the K1-BUMP rigging bar on K1-BUMP rigging frame.

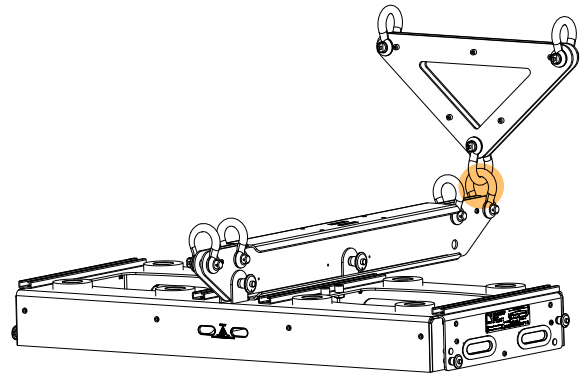
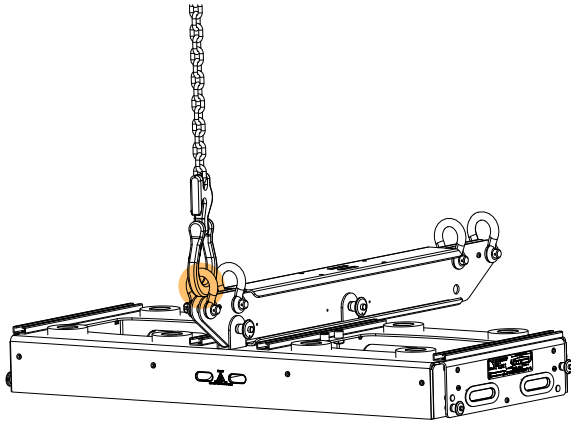


5. Secure the K1-BUMP rigging bar using the two rigging ball-locking pins.

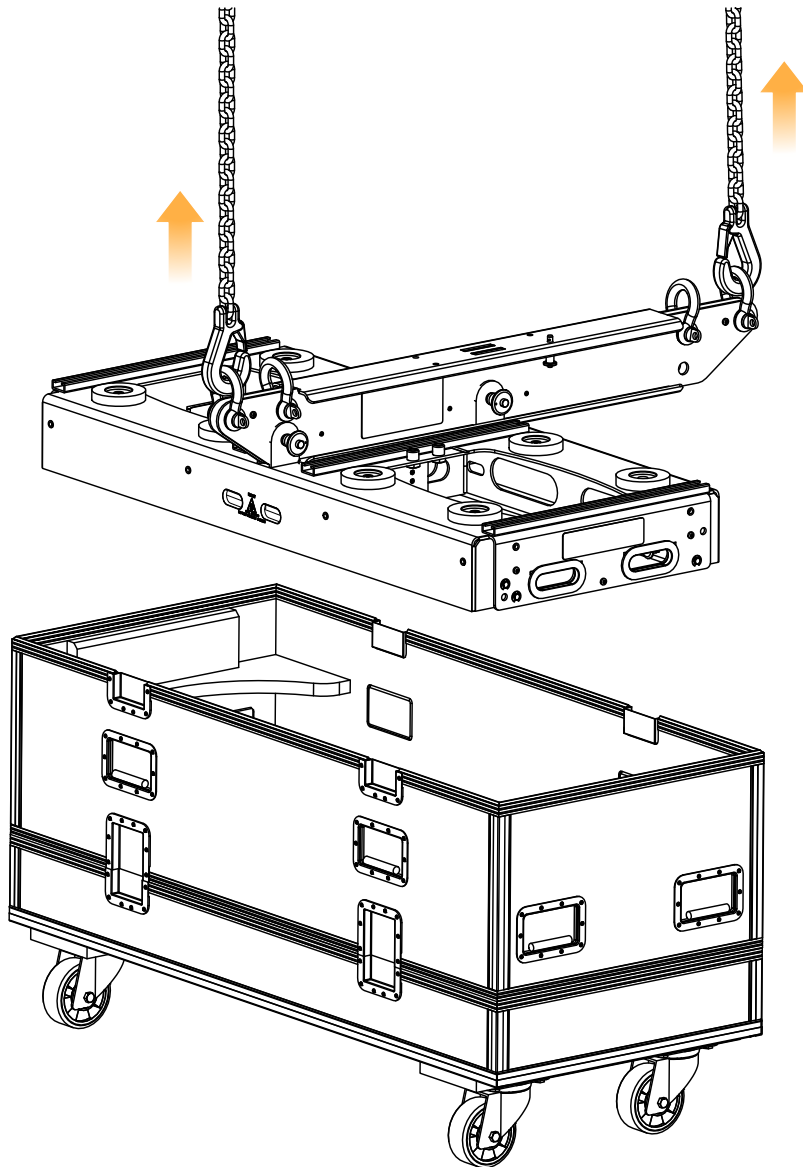


6. Optional:

- Secure K1-BPCHAIN to the front external shackle.
- Secure K1-DELTA to the rear external shackle.



7. Lift K1-BUMP using the motor.



Securing LA-RAK II AVB on K1-BUMP

Min number of operators	2
Rigging accessories	K1-BUMP
	4 Ø 22 mm shackles WLL 3.25 t
	Rack stabilizer



A maximum of four LA-RAK II AVB can be rigged on top of the K1-BUMP.

Prerequisite

Prepare K1-BUMP for rigging and place it on the ground. Refer to [Preparing K1-BUMP](#) (p.62).

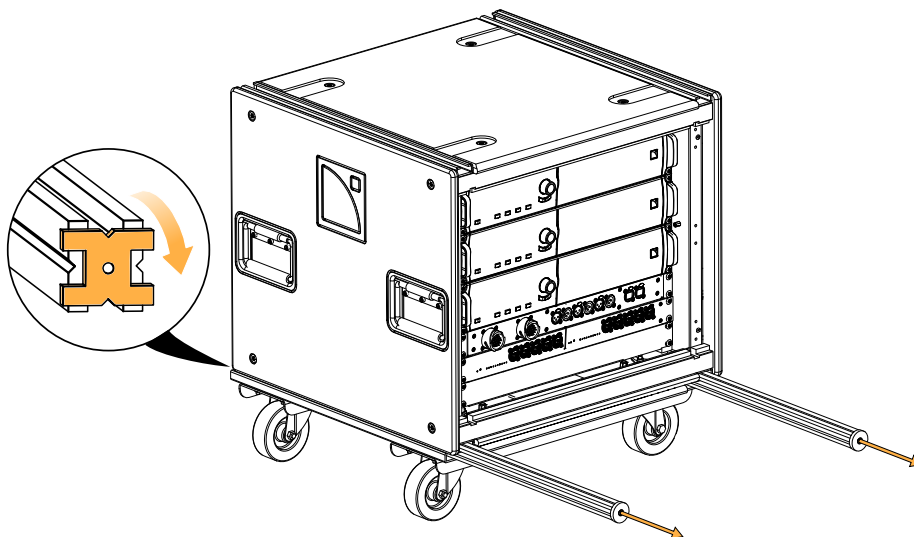
Procedure

1. Remove the coupling bars from LA-RAK II AVB.

Turn the spring-loaded safety mechanisms to release the bars and slide them out.



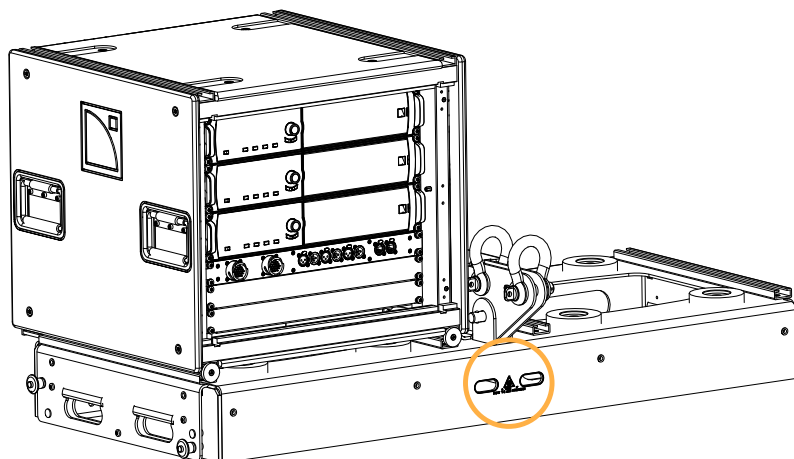
Make sure the spring-loaded safety is intact.



2. Position LA-RAK II AVB on K1-BUMP.



LA-RAK II AVB must be installed with the front panel oriented towards the K1-BUMP laser slits.



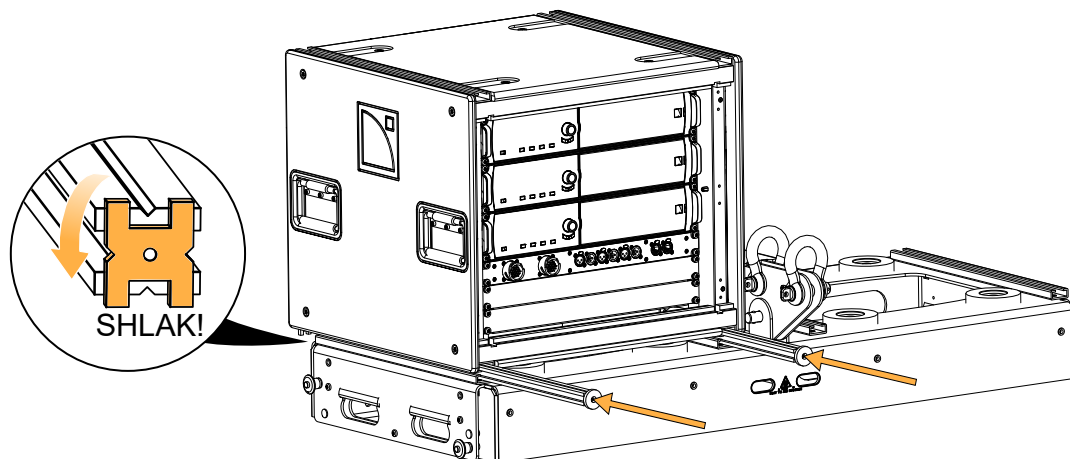
3. Secure LA-RAK II AVB with the coupling bars:

! **Always insert the coupling bars so that the plastic spring-loaded safety points toward the ground (depending on the site angle):**

From the front for a positive site angle.

From the back for a negative site angle.

- a) Insert the spring-loaded safety in the LA-RAK II AVB rails.
- b) Give a quarter turn and slide the bar until the safety locks into place.



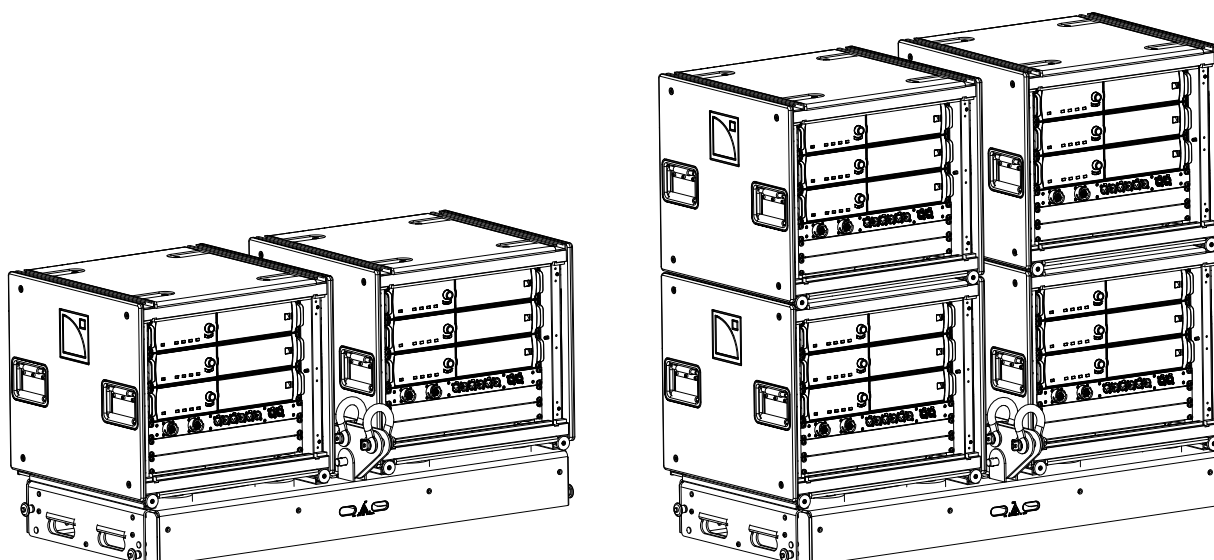
4. Repeat steps from 1 (p.67) to 3 (p.68) to secure up to four LA-RAK II AVB on K1-BUMP.



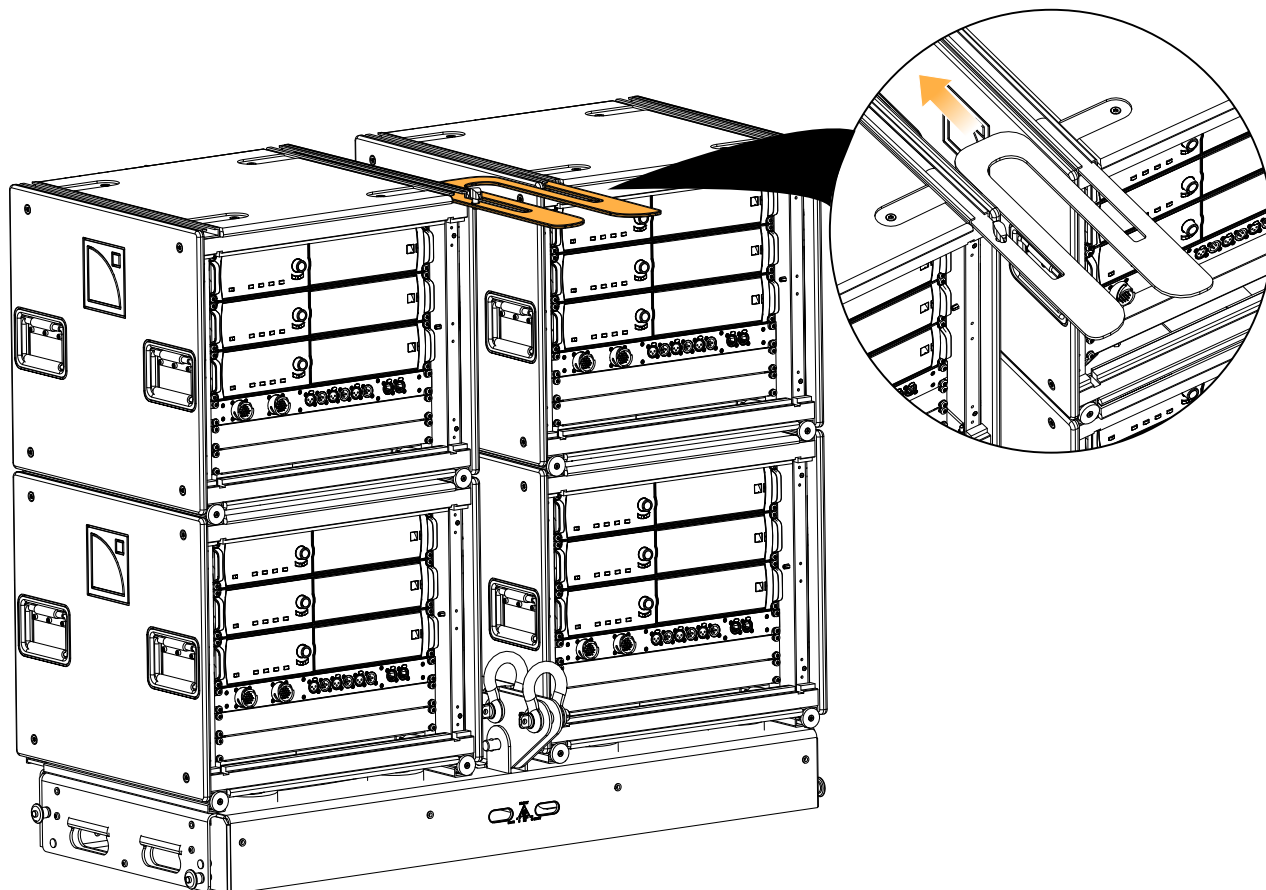
Risk of falling

If two racks are used, each one must be placed directly onto K1-BUMP.

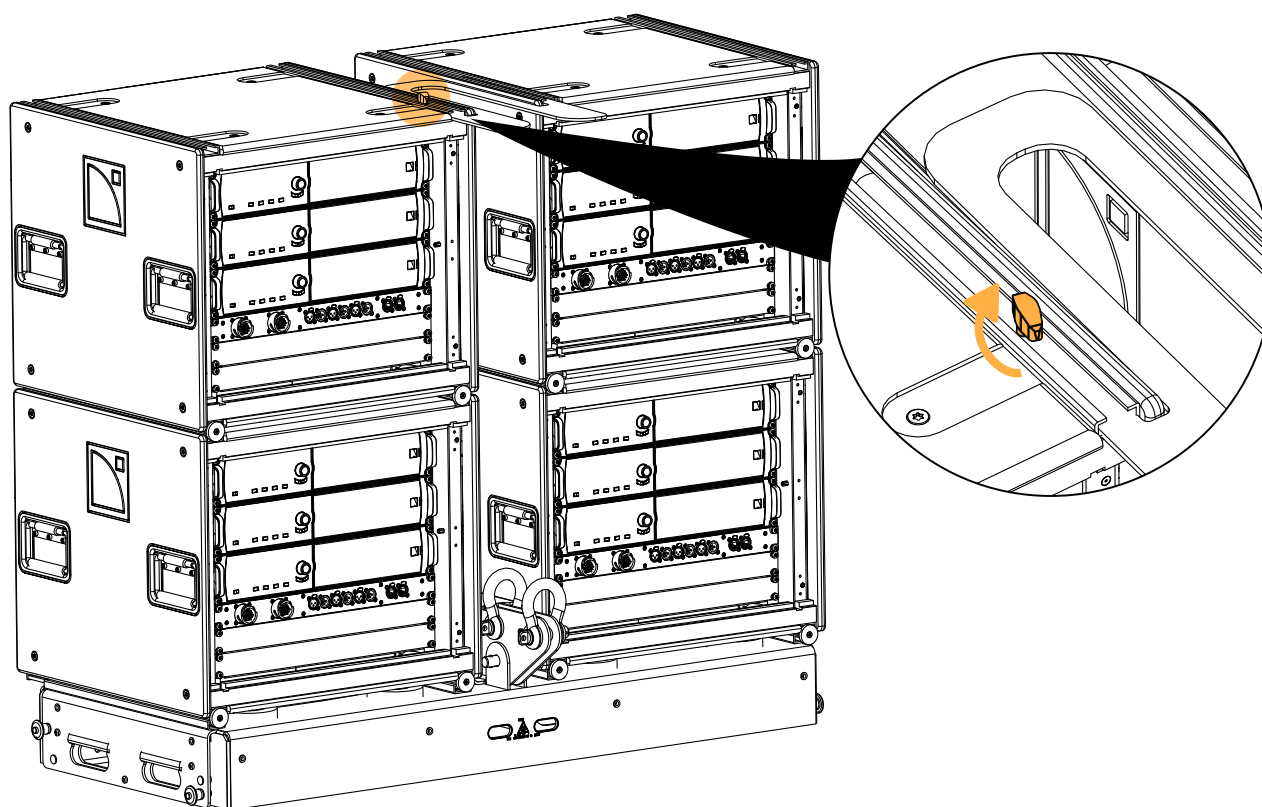
If four racks are used, they must be arranged in two dual-rack columns.



- 5.** Slide the rack stabilizer along the top LA-RAK II AVB rails until it locks. If necessary, unscrew the rack stabilizer safety screw.

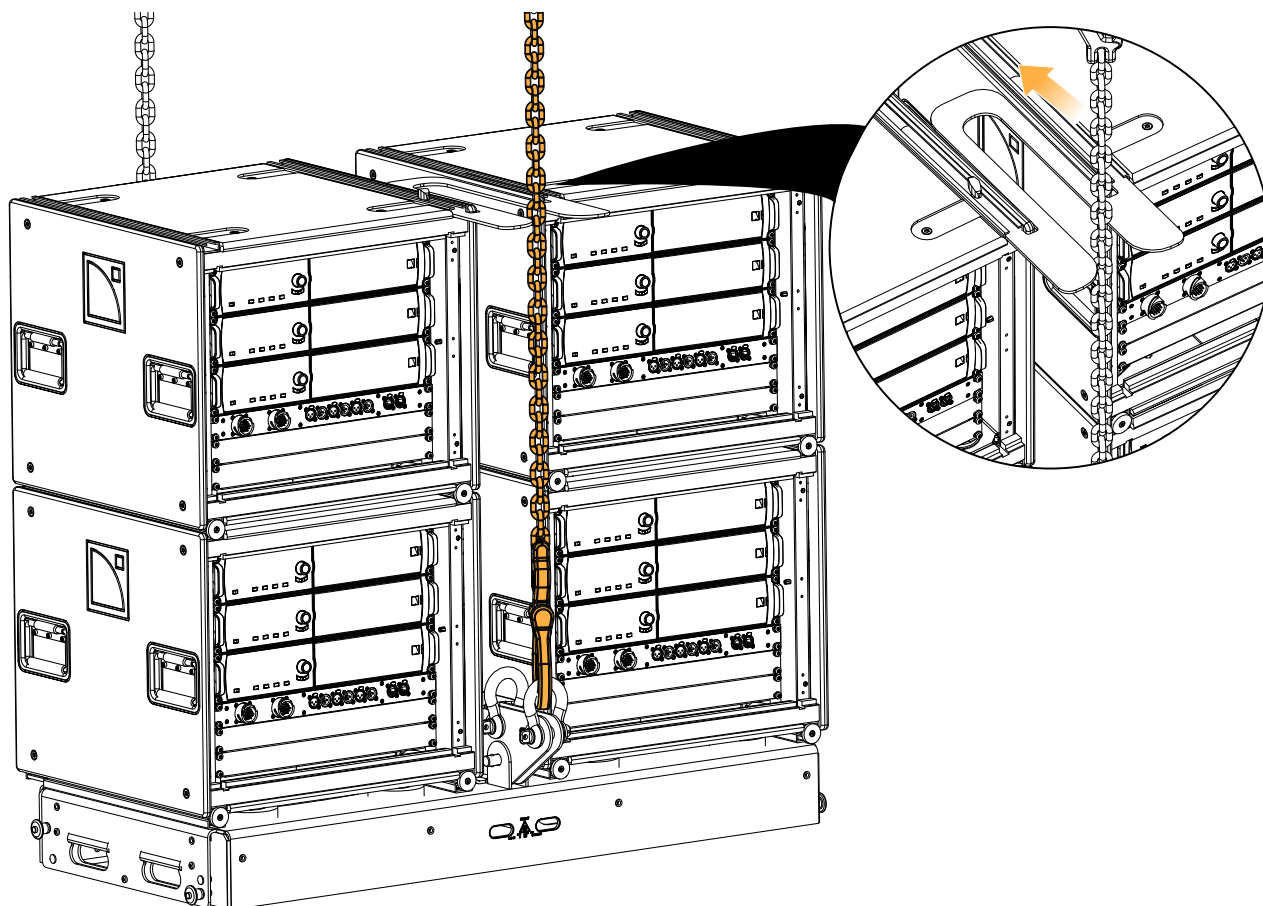


- 6.** Screw in the rack stabilizer safety screw.



7. Secure the motor hooks to the external shackles of the K1-BUMP rigging bar.

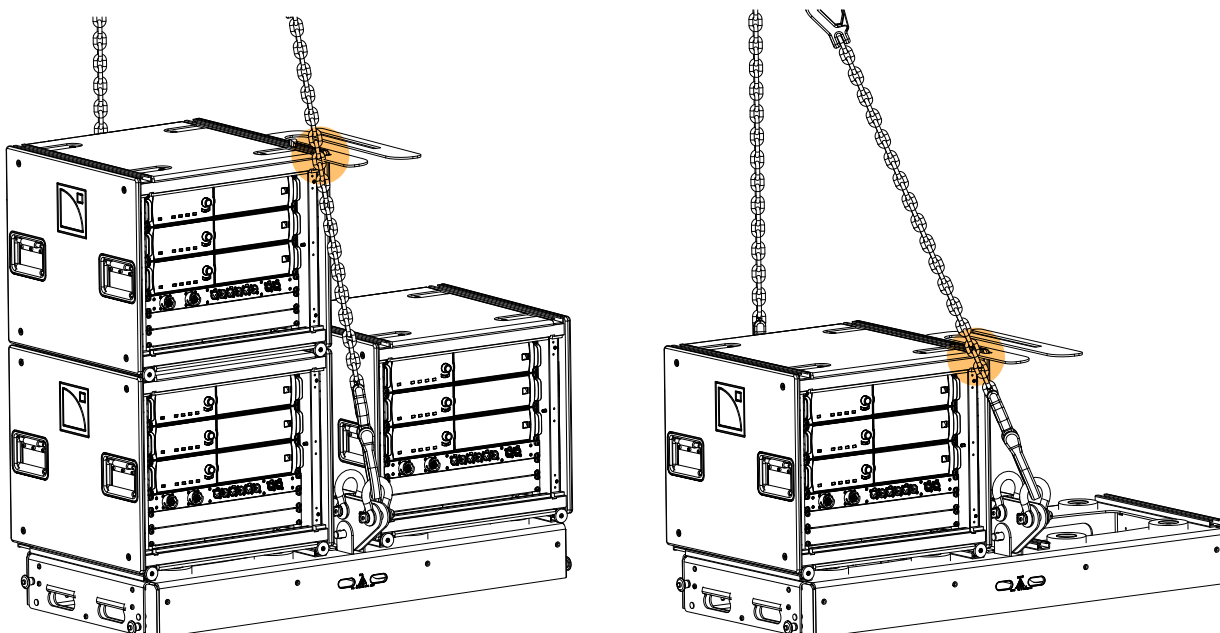
If two or four LA-RAK II AVB are installed, place the front motor chain into the groove of the rack stabilizer.



If one or three LA-RAK II AVB are installed, place the front motor chain on the external side of the rack stabilizer to prevent K1-BUMP from tilting during enclosure rigging.



The front motor chain must be slid back into the groove of the rack stabilizer after enclosures rigging.



Preparing a block of four K1/K1-SB

Min number of operators

2

Rigging accessories

K1-CHARIOT



All along the procedure:

Systematically ensure that each ball-locking pin is fully inserted by pulling on it.



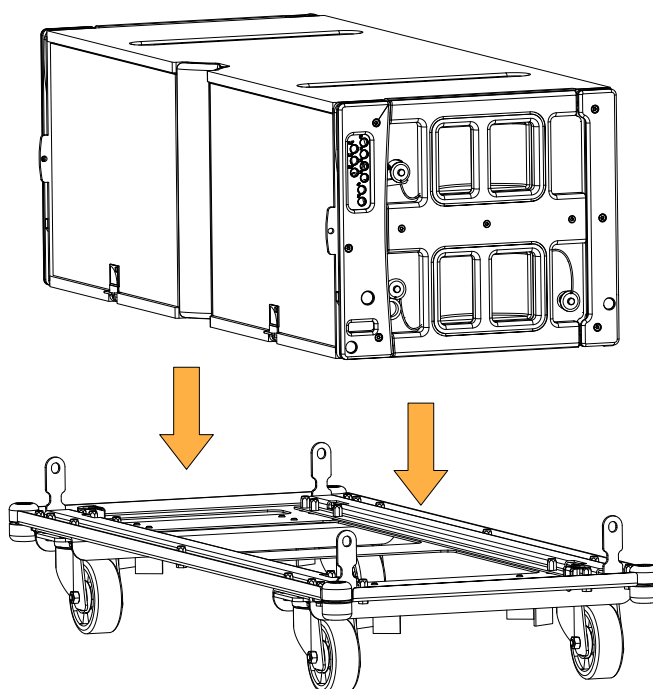
Follow the same procedure for K1-SB enclosures.

Procedure

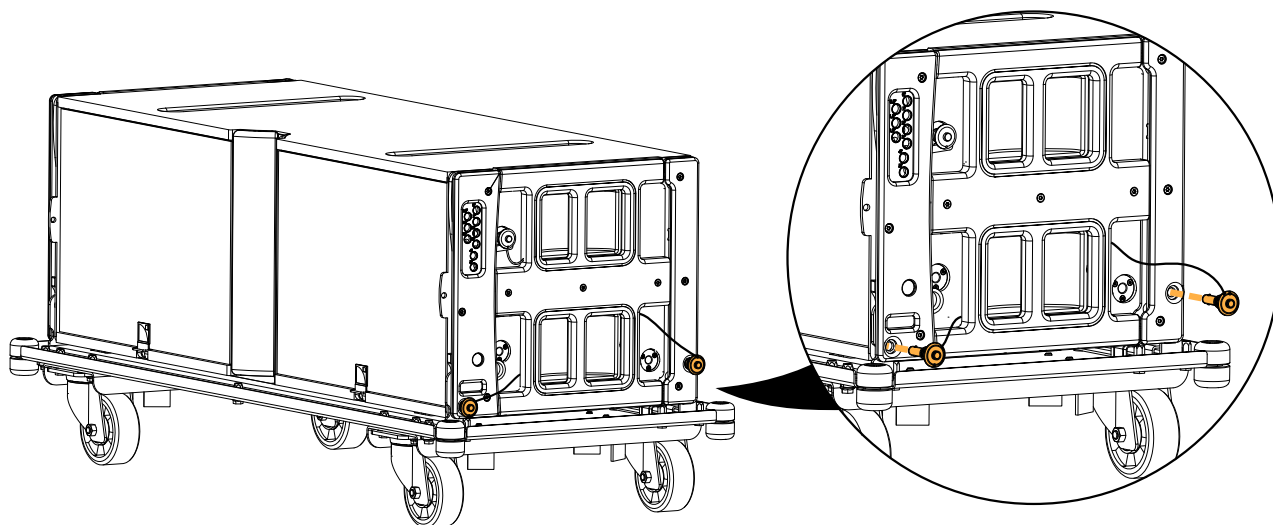
1. Position a first K1 on K1-CHARIOT.



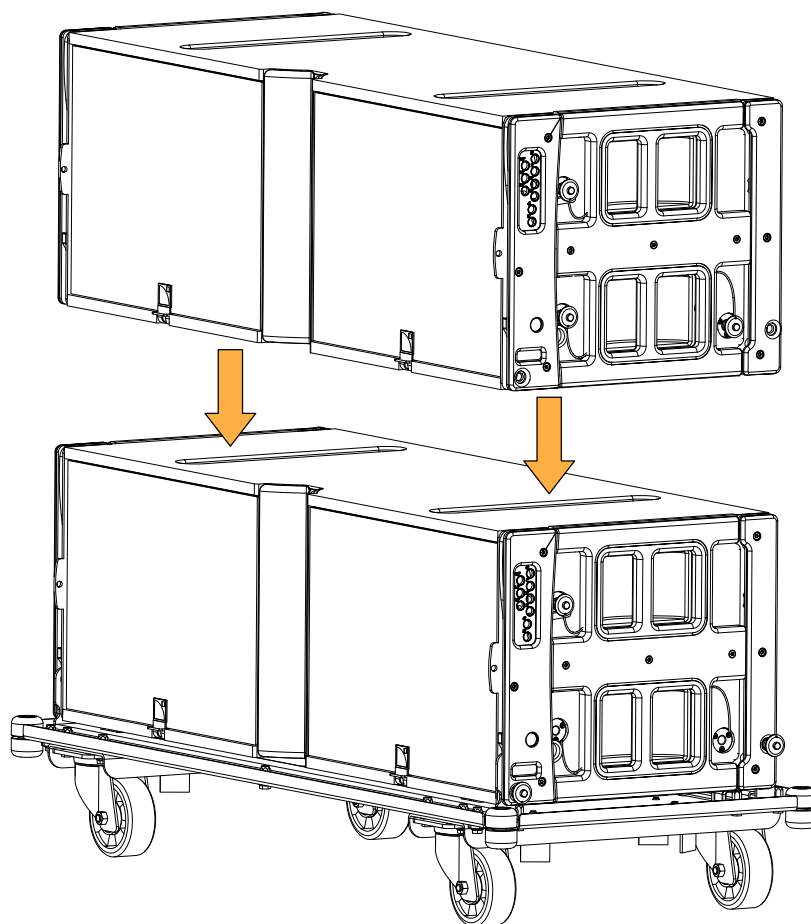
Enclosure can face either way.



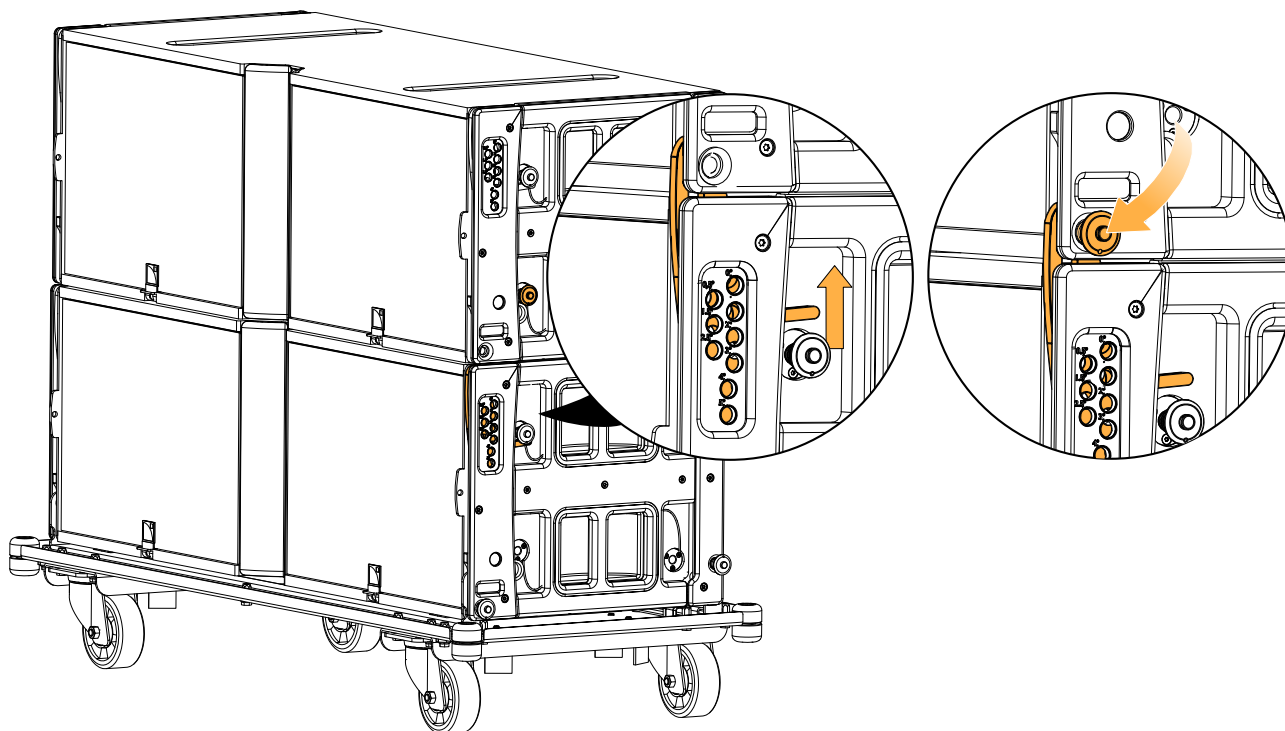
2. Secure the enclosure using the ball-locking pins.



3. Position another K1 on the first one.

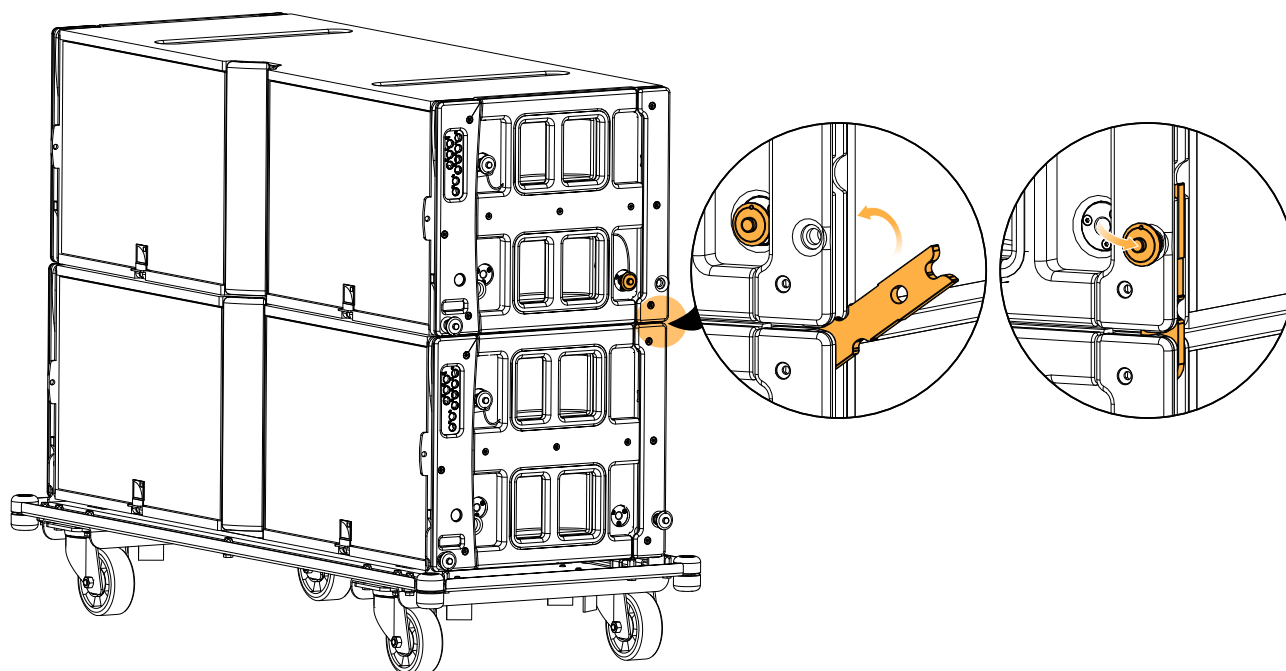


4. On each side of the bottom enclosure, secure the front rigging arm to the top enclosure:
a) Slide the front rigging arm.
b) Secure the front rigging arm using the ball-locking pin of the top enclosure.



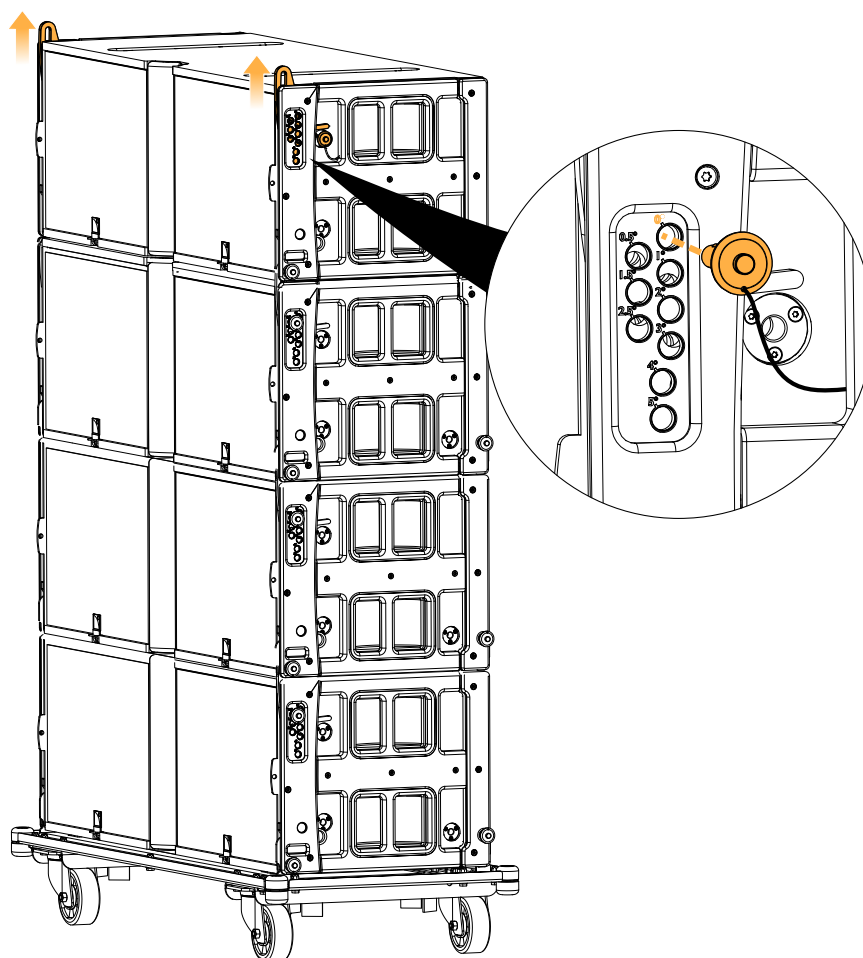
5. On each side of the bottom enclosure, secure the rear rigging arm to the top enclosure:

- a) Rotate the rear rigging arm.
- b) Secure the rear rigging arm using the ball-locking pin of the top enclosure.



6. Repeat steps 3 (p.72) to 5 (p.73) until the block of four K1 is complete.

7. On each side of the top enclosure, set the angle at 0° using the top ball-locking pin.



Presetting the inter-enclosure angles of K1/K1-SB



All along the procedure:

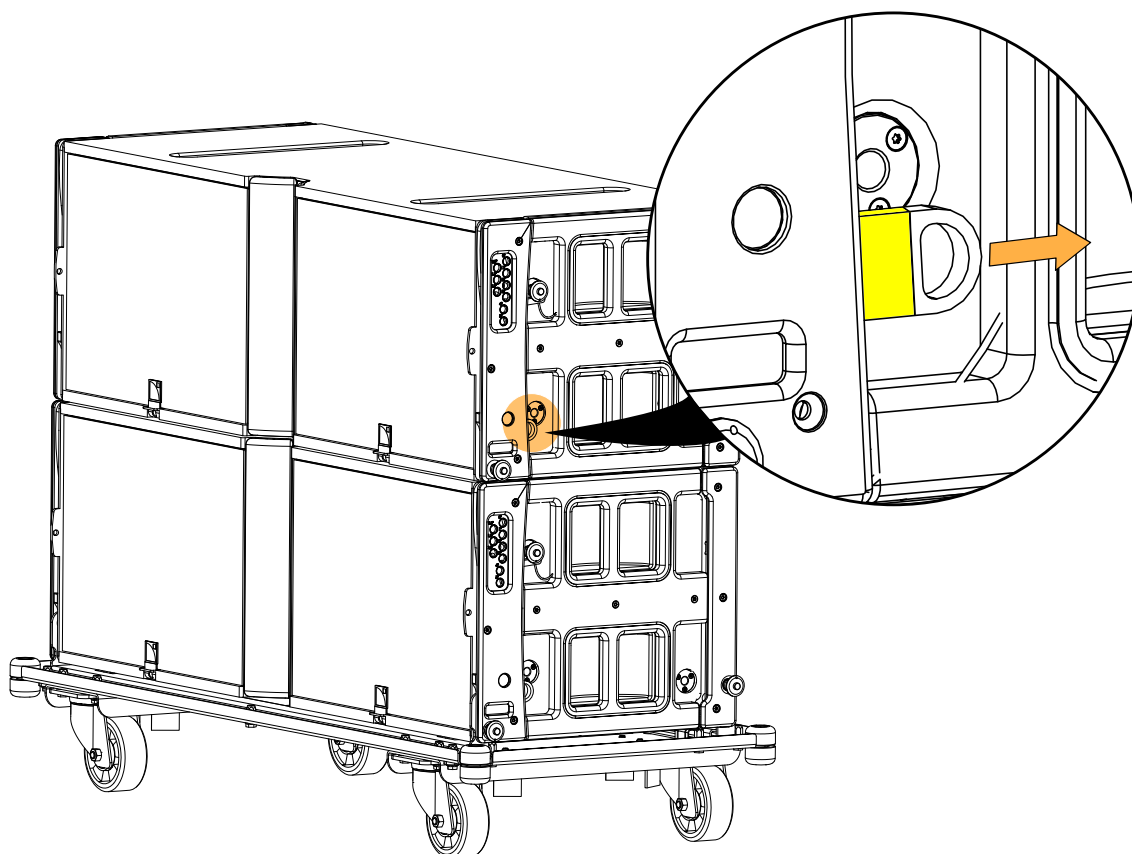
Systematically ensure that each ball-locking pin is fully inserted by pulling on it.



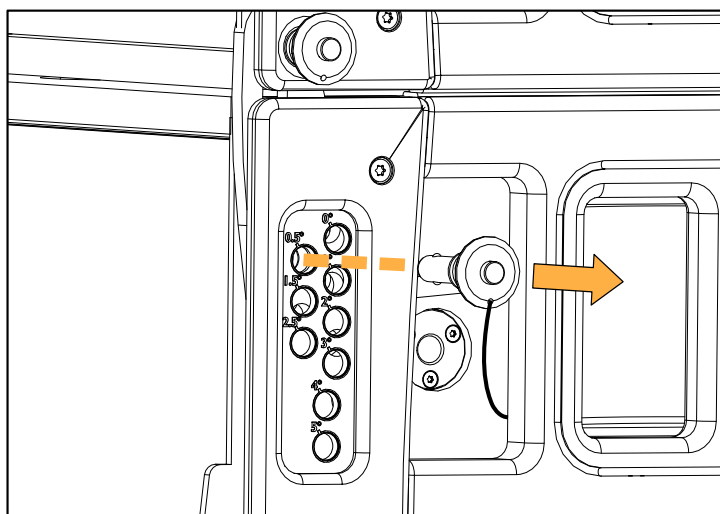
Follow the same procedure for K1-SB enclosures.

Procedure

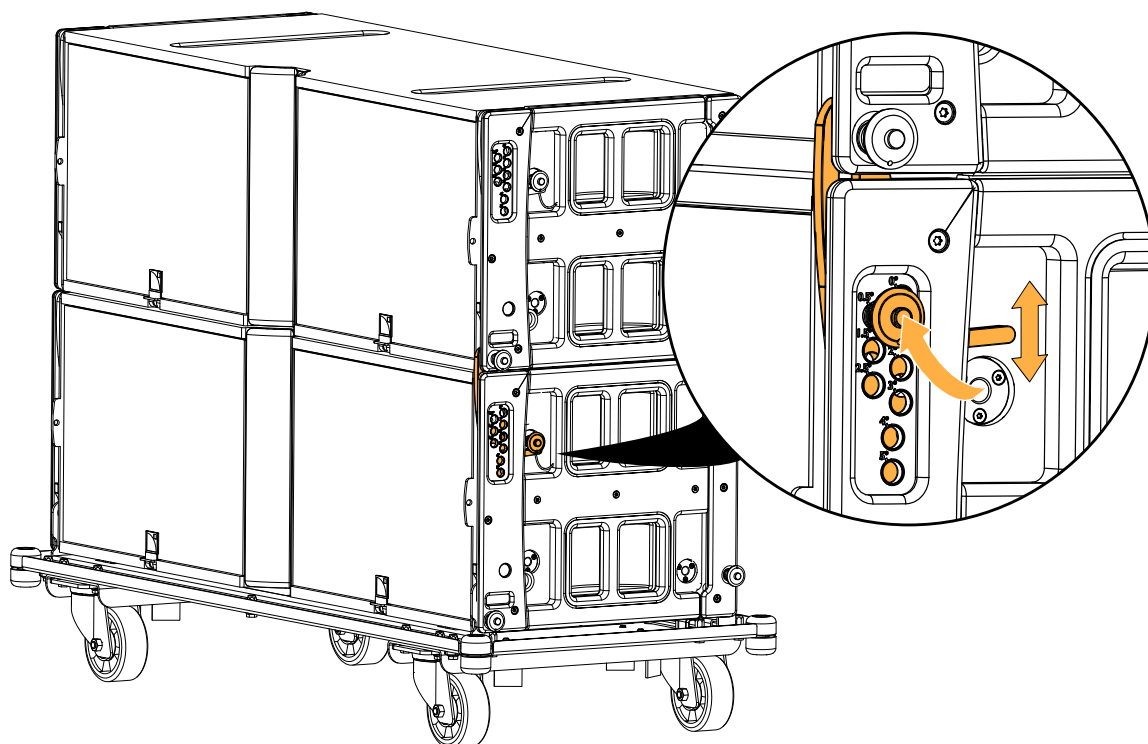
1. Make sure the automatic locking system is deactivated.
If the latch is engaged, pull it to deactivate the automatic locking system.



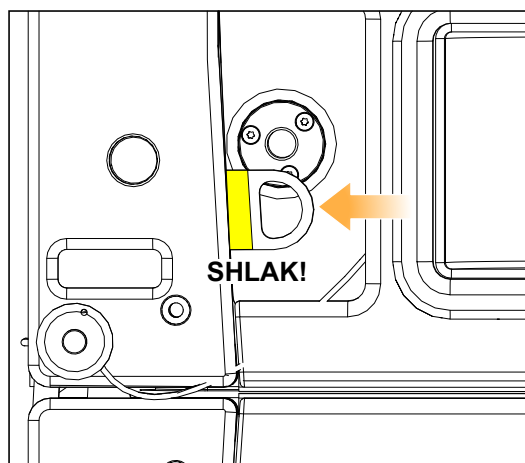
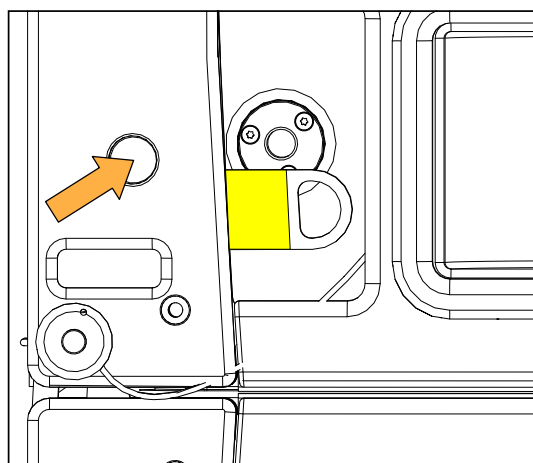
2. If necessary, on each side of the bottom enclosure, remove the ball-locking pin from its current angle hole.



- 3.** Slide out the front rigging arm to the desired angle hole and insert the ball-locking pin.



- 4.** On each side of the top enclosure, press the angle locking button to lock the inter-enclosure angle. The latch slightly retracts when the button is pushed (yellow label visible).



- 5.** Repeat the procedure for each inter-enclosure angle of the array.

Flying K1

Flying a K1/K1-SB array with K1-BUMP

Min number of operators	3
Rigging accessories	4 Ø 22 mm shackles WLL 3.25 t
	K1-BUMP

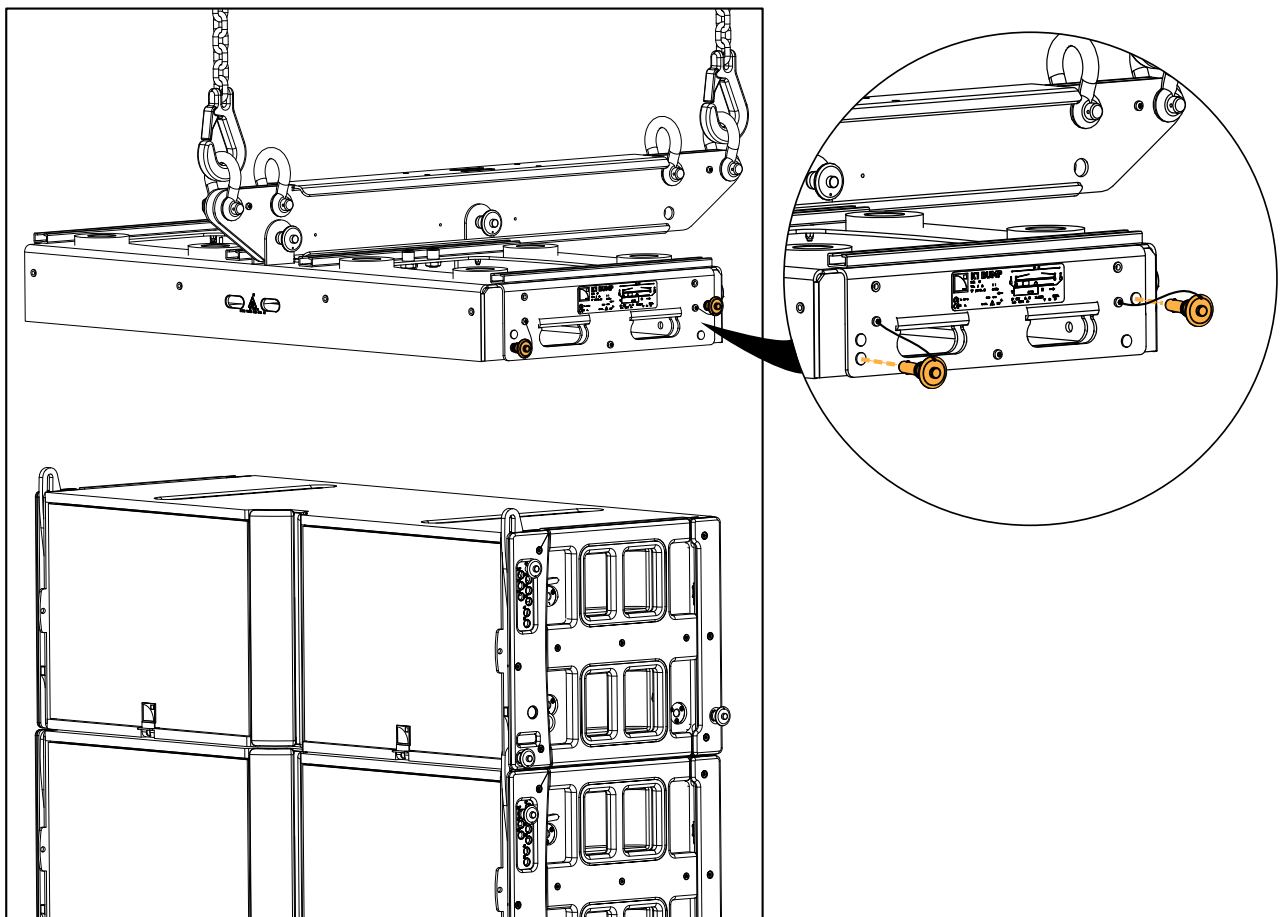
- ! All along the procedure:**
Systematically ensure that each ball-locking pin is fully inserted by pulling on it.
- ! For clarity purposes, the loudspeaker cable procedure is not described.**
The loudspeaker cables are not represented in the figures.
- i** Follow the same procedure for K1-SB enclosures.

Prerequisite

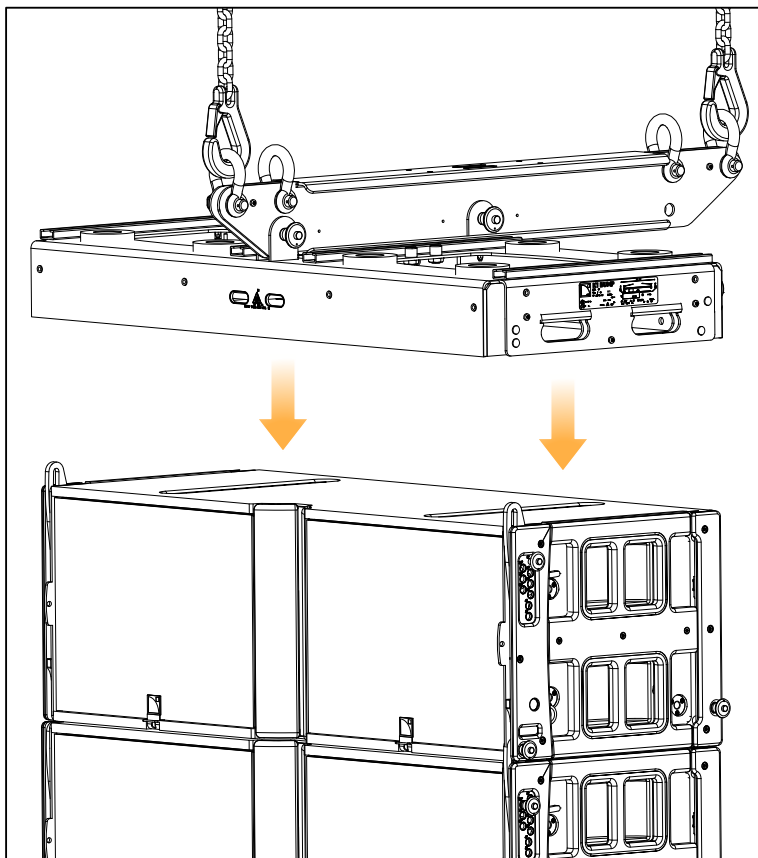
- Prepare K1-BUMP. Refer to [Preparing K1-BUMP](#) (p.62).
- If necessary, secure LA-RAK II AVB on K1-BUMP. Refer to [Securing LA-RAK II AVB on K1-BUMP](#) (p.67).
- Prepare a block of four K1. Refer to [Preparing a block of four K1/K1-SB](#) (p.71).
- Preset the inter-enclosure angles. Refer to [Presetting the inter-enclosure angles of K1/K1-SB](#) (p.74).

Procedure

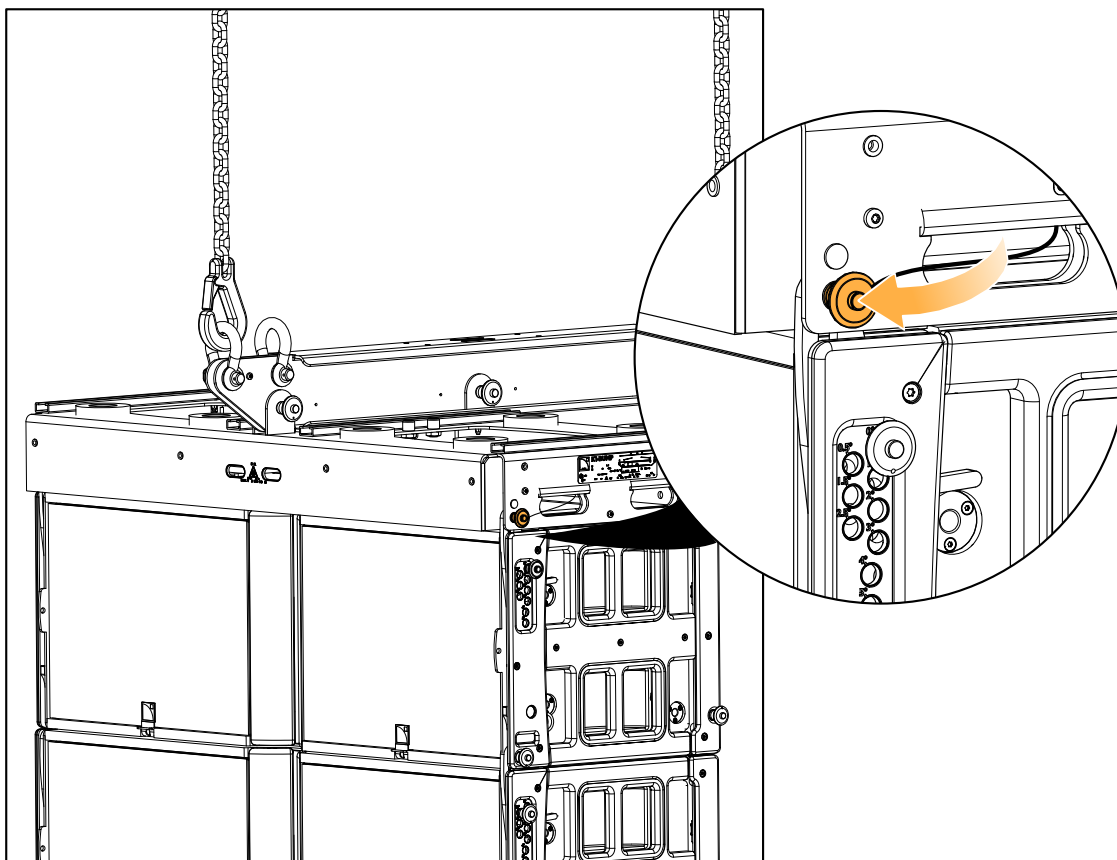
1. Lift K1-BUMP to more than 2 m / 6.5 ft height using the motor.
2. Position a block of four K1/K1-SB under K1-BUMP.
3. Secure K1-BUMP on top of the array:
 - a) On each side of the K1-BUMP rigging frame, remove the four ball-locking pins.



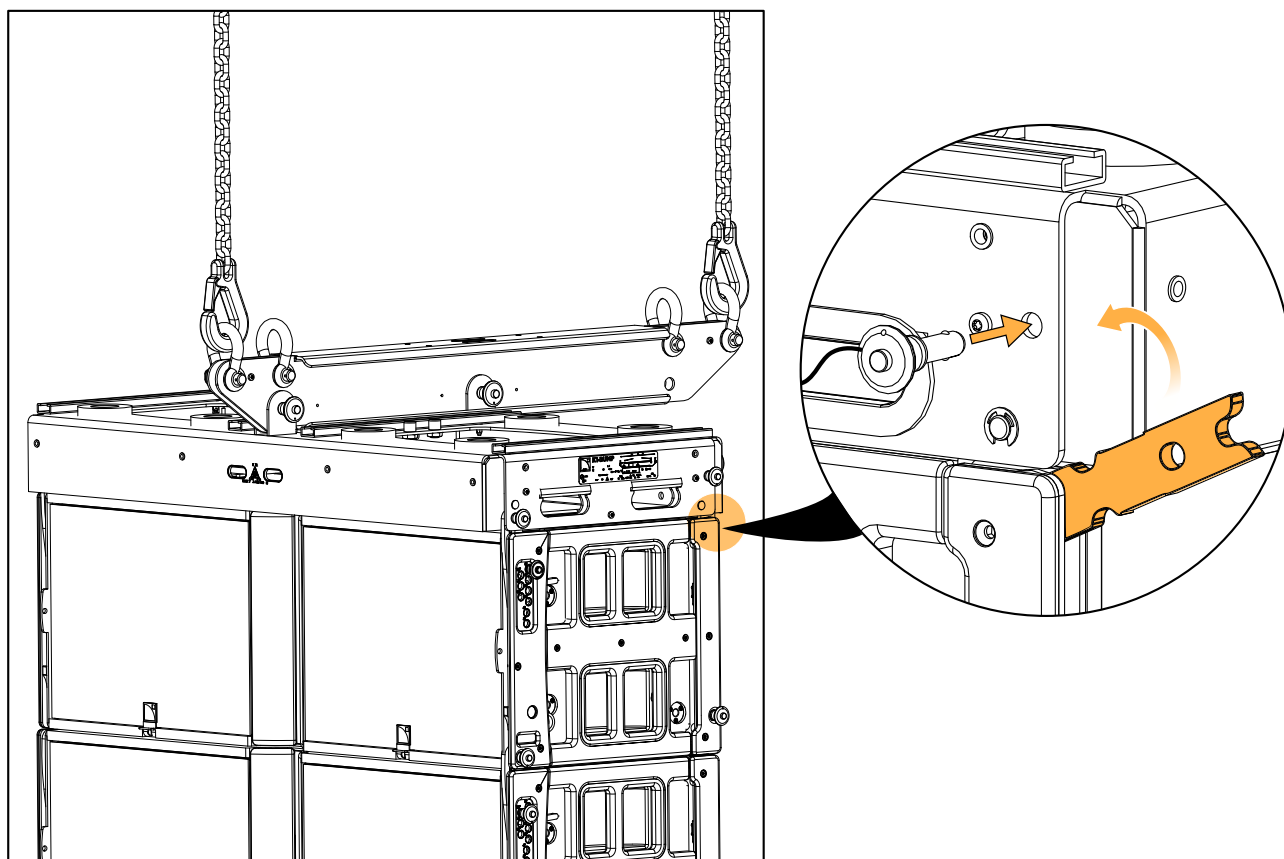
b) Lower K1-BUMP to facilitate the connection between the enclosures and the rigging frame.



c) On each side of the top enclosure, secure the front rigging arm using the K1-BUMP ball-locking pin.



- d) On each side of the top enclosure, rotate the rear rigging arm and secure it using the K1-BUMP ball-locking pin.

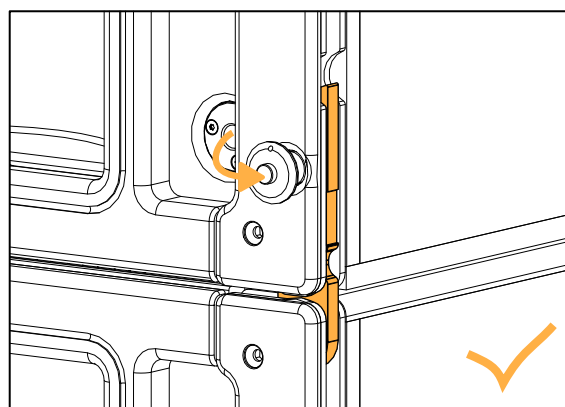
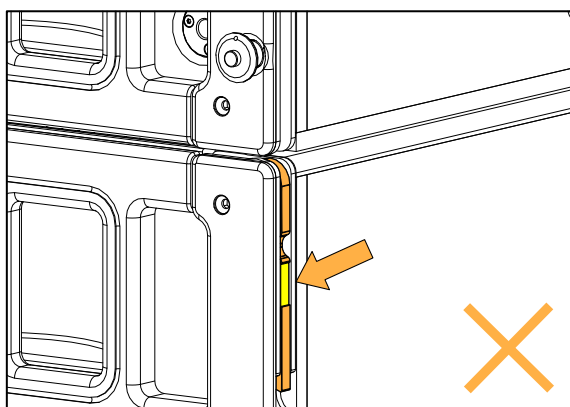


4. Before raising the array, check that all ball-locking pins and rigging arms are properly secured.



Check that all ball-locking pins are **fully inserted** in rigging position.

Check that all rear rigging arms are secured by making sure **no yellow sticker is visible**.



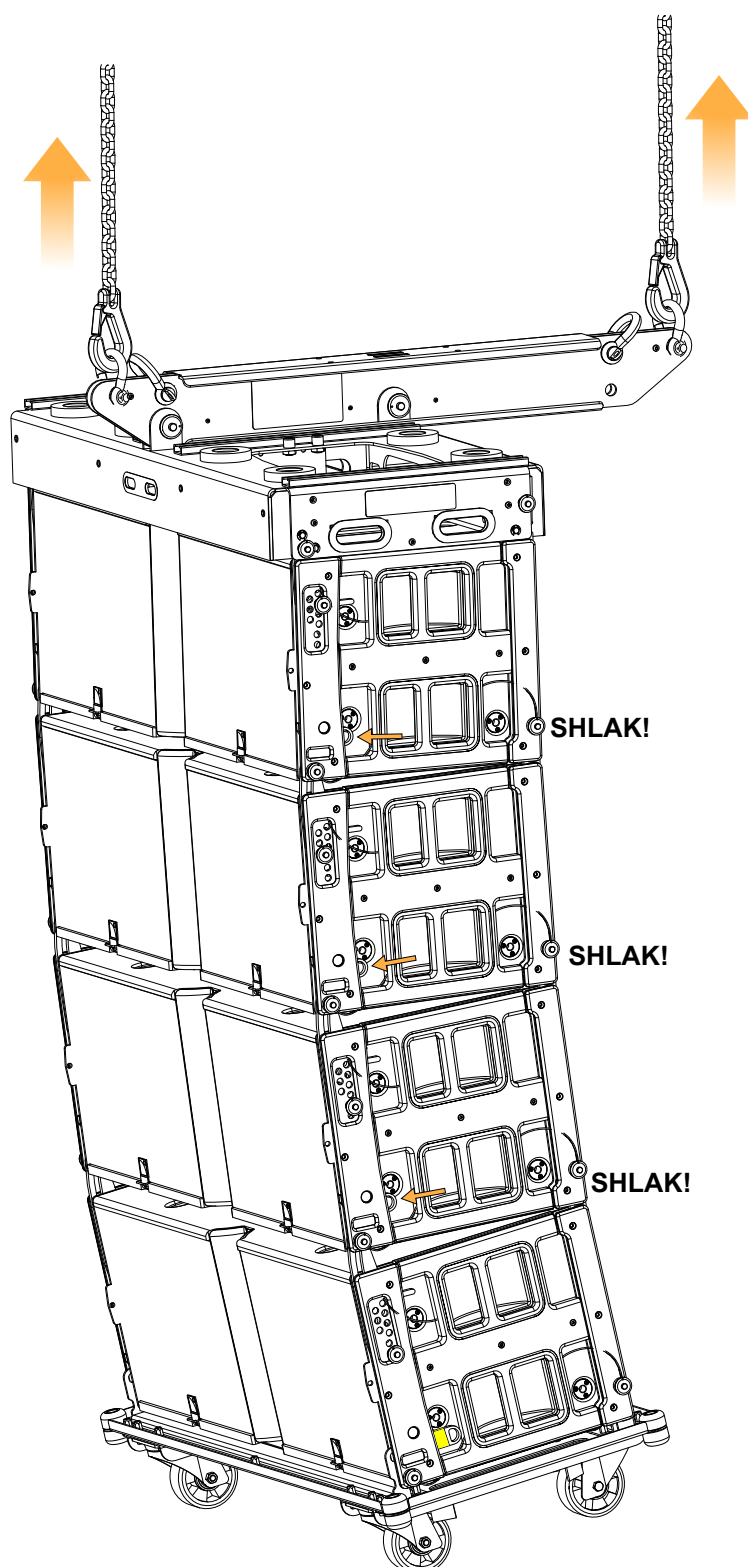
5. Slightly raise the array to lock the inter-enclosure angles.

The automatic system engages and the latches retract (except on the bottom enclosure).



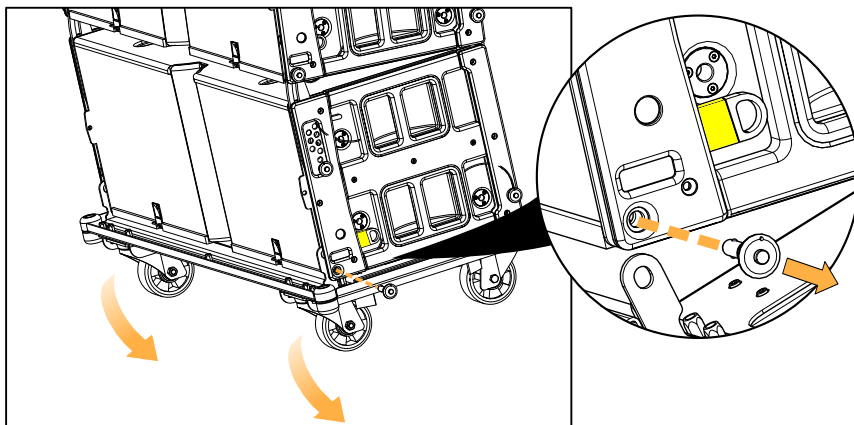
Check that the automatic locking system is **activated**.

When the automatic locking system is activated, **no yellow sticker is visible** on the latches.

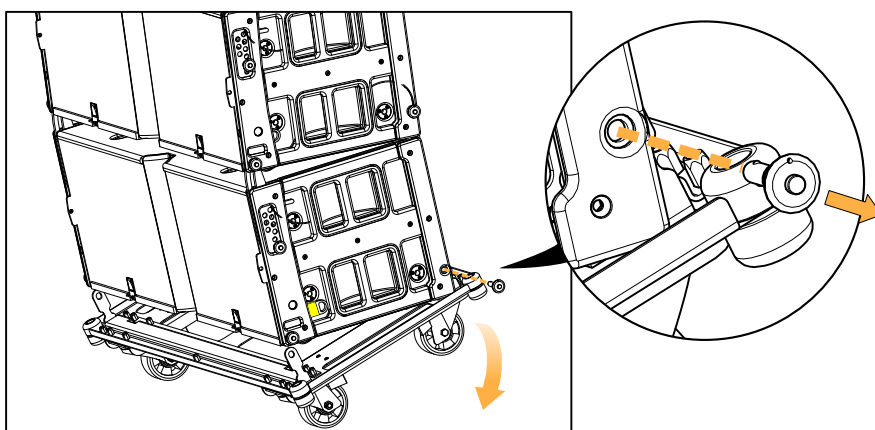


6. Remove K1-CHARIOT from the array:

a) Hold K1-CHARIOT and remove the two front ball-locking pins. Put them into their storage locations.



b) Hold K1-CHARIOT and remove the two rear ball-locking pins. Put them into their storage locations.



What to do next

- Attach a block of four K1 under the array. Refer to [Attaching a block of four K1 under a K1 array](#) (p.81).
- Add a Kara II downfill array. Refer to [Rigging a Kara II downfill array under a K1 array with KARA-DOWNK1](#) (p.92).

Attaching a block of four K1 under a K1 array

Min number of operators	3
Rigging accessories	2 Ø 22 mm shackles WLL 3.25 t K1-BUMP



All along the procedure:

Systematically ensure that each ball-locking pin is fully inserted by pulling on it.



Secondary safety

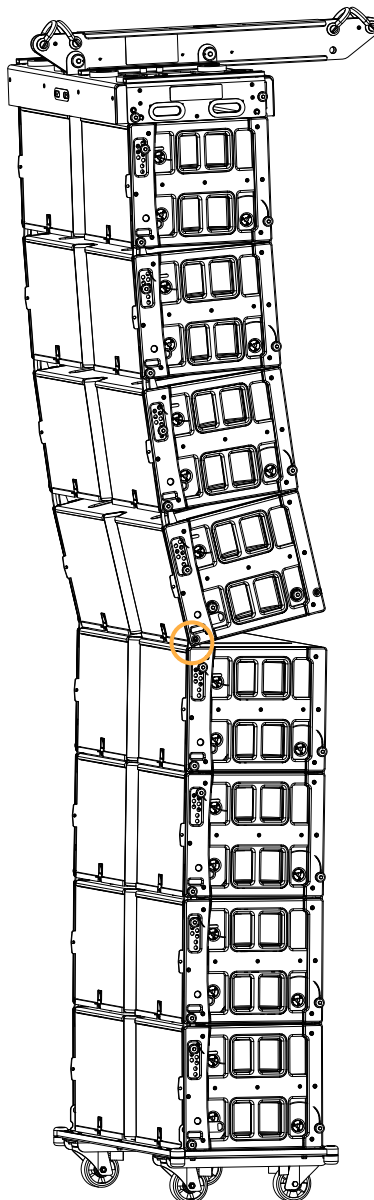
Use available holes on the rigging accessories to implement a secondary safety.

Prerequisite

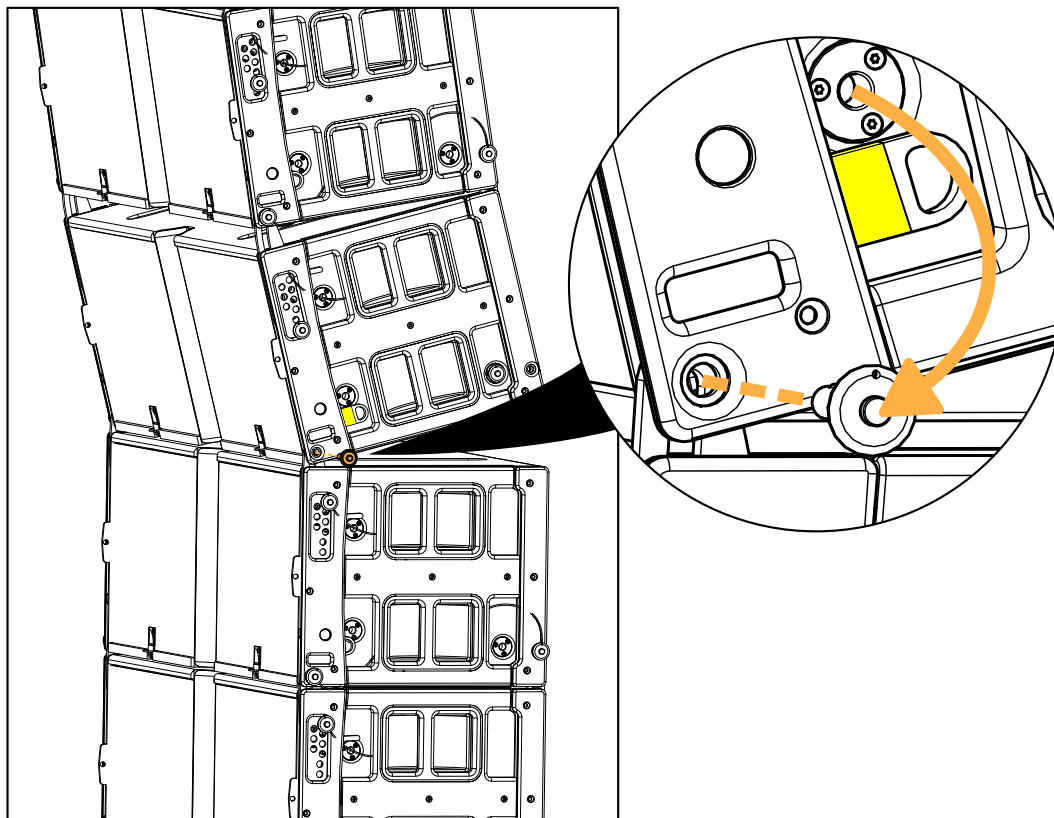
- Prepare a block of four K1. Refer to [Preparing a block of four K1/K1-SB](#) (p.71).
- Preset the inter-enclosure angles. Refer to [Presetting the inter-enclosure angles of K1/K1-SB](#) (p.74).

Procedure

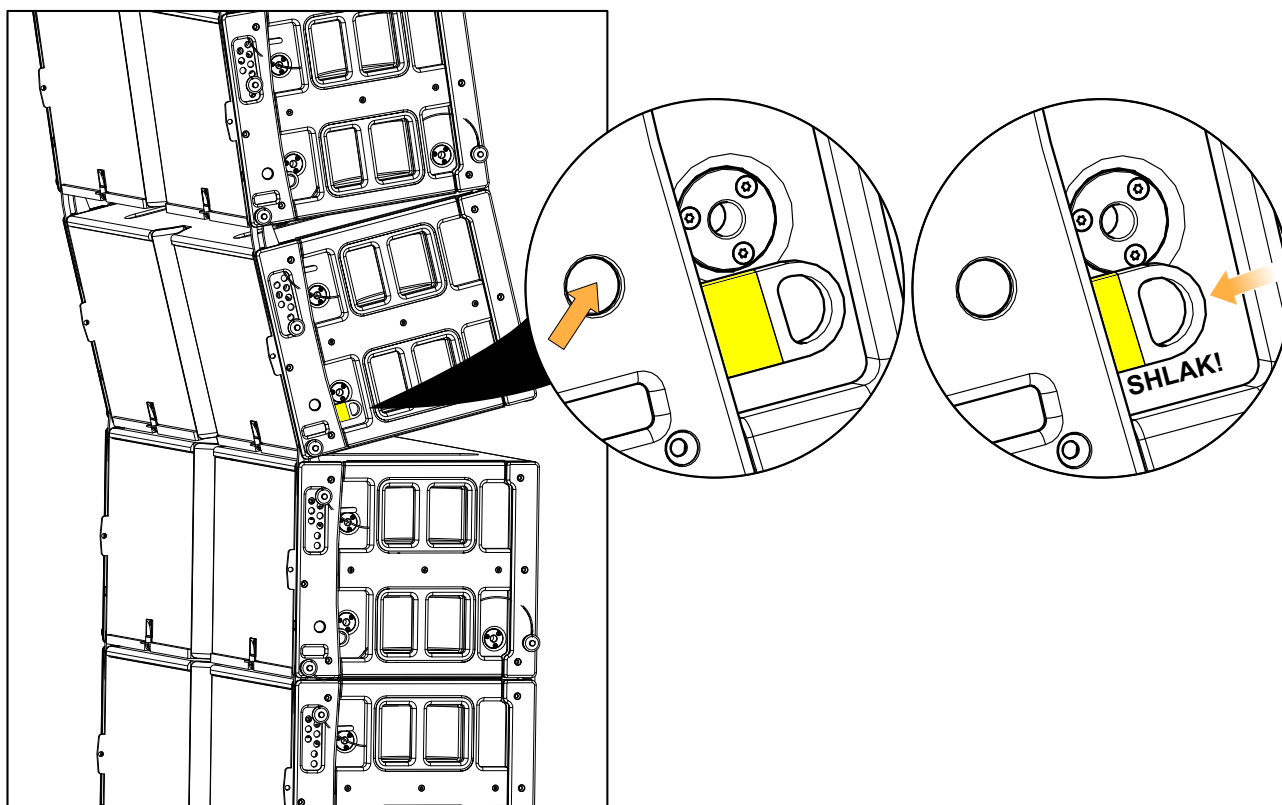
1. Position the block of four K1 under the array.
2. On each side, secure the front rigging arm:
 - a) Lower the array to align the front rigging points.



- b) Secure the front rigging arms using the ball-locking pins.
If necessary, slide the front rigging arms to align the holes.



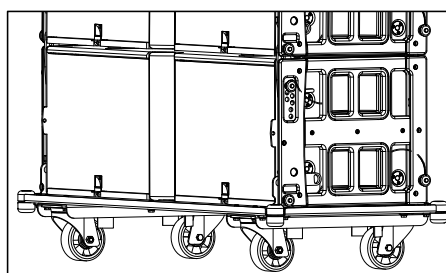
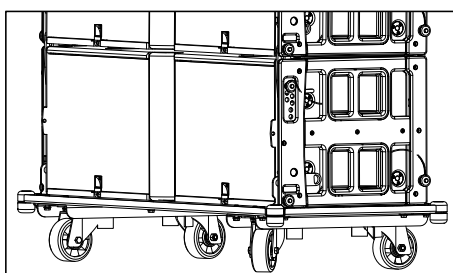
3. Press the angle locking buttons to lock the inter-enclosure angles.
The latches slightly retract when the buttons are pushed (yellow label visible).



4. Raise the array until the K1-CHARIOT wheels are off the ground.
The latches engage and **no yellow sticker is visible**, except on the bottom enclosure.

5. Turn the wheels inside the chariot.

! This operation avoids instability when the wheels are in contact with the ground.

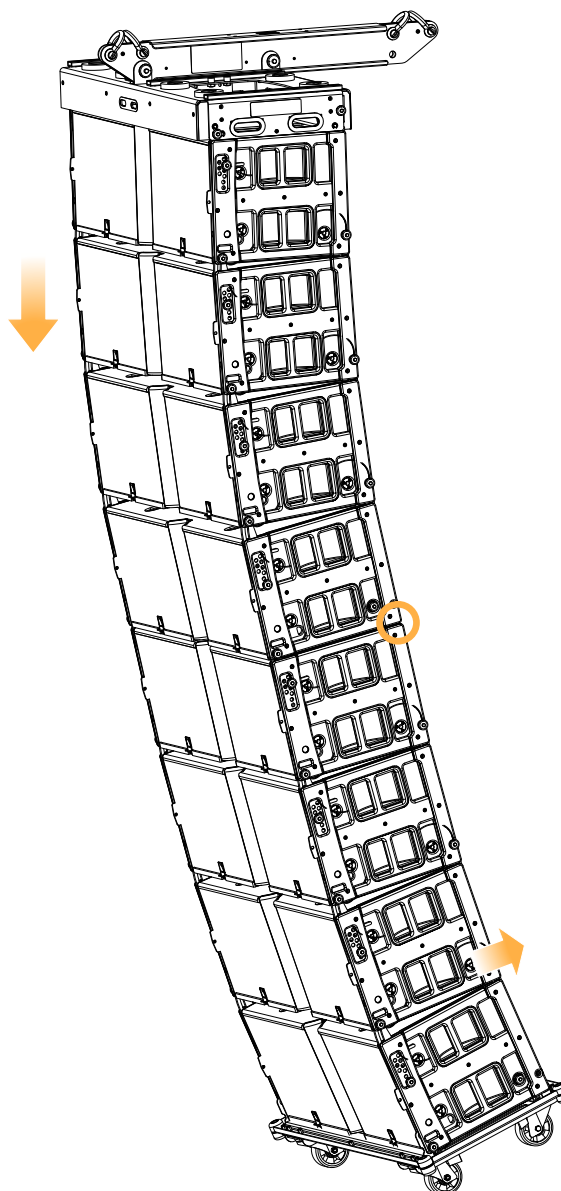
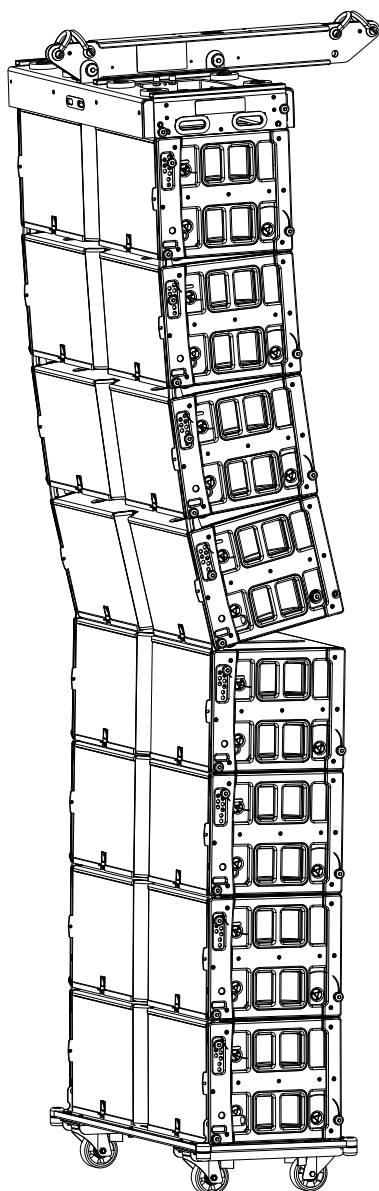


6. On each side, secure the rear rigging arms:

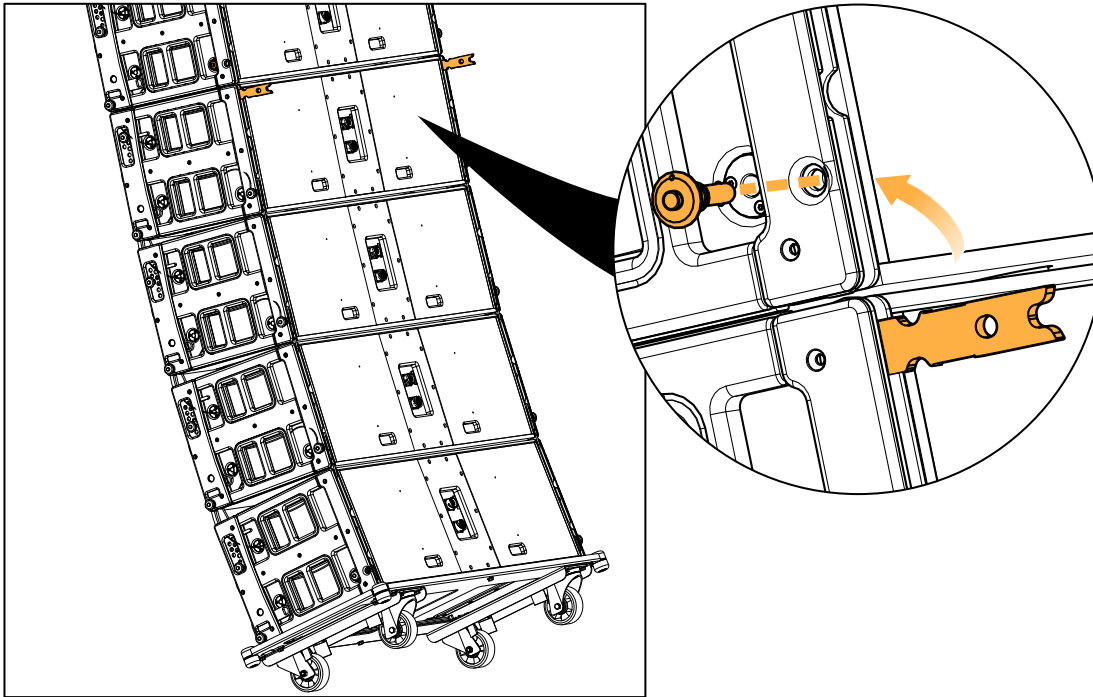
a) Pull the array towards the back while lowering it until the rear corners are in contact.

! Three people are needed for this operation:
 One manipulating the hoist controller,
 Two on both sides of the array, grabbing the handles of the bottom enclosure.

! Stop lowering the array when both rear corners are in contact.
 This operation avoids putting too much stress on the K1-CHARIOT wheels.



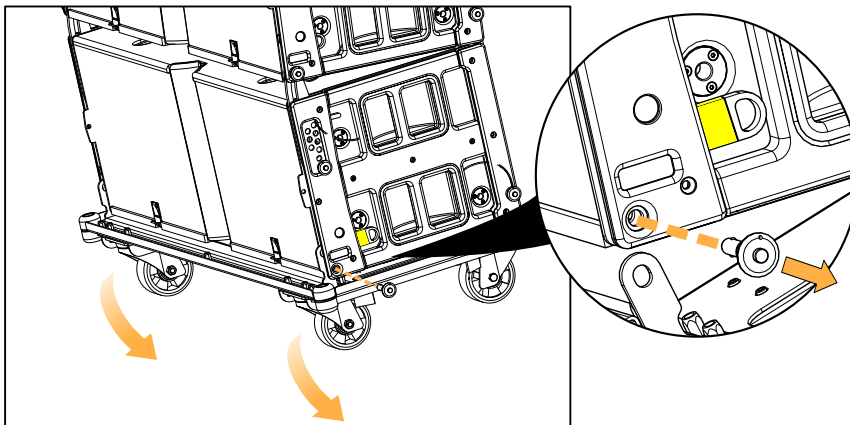
- b) Rotate the two rear rigging arms of the top enclosure and secure them to the array.



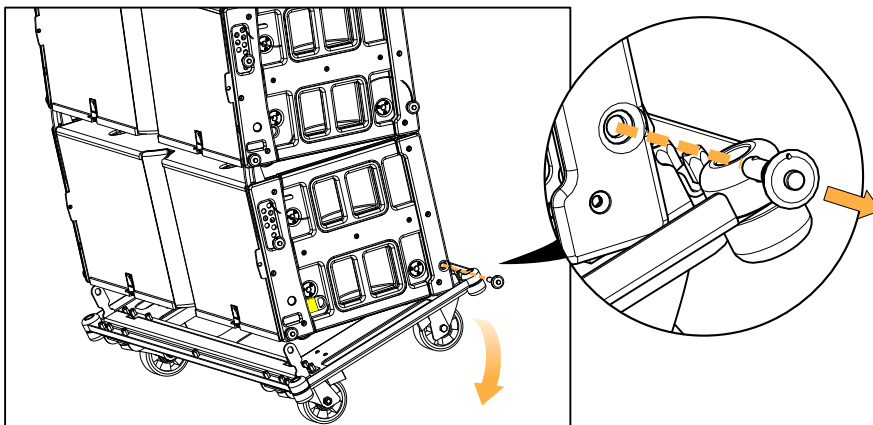
- 7.** Raise the array until the K1-CHARIOT wheels are off the ground.

- 8.** Remove K1-CHARIOT from the array:

- a) Hold K1-CHARIOT and remove the two front ball-locking pins. Put them back into their storage locations.



- b) Hold K1-CHARIOT and remove the two rear ball-locking pins. Put them back into their storage locations.



- 9.** Repeat the procedure until the array is complete.

- 10.** Fly the array at desired trim height and adjust the site and azimuth angles.

Attaching a block of four K2 under a K1/K1-SB array

Min number of operators	3
Rigging accessories	4 Ø 22 mm shackles WLL 3.25 t K2-LINK



All along the procedure:

Systematically ensure that each ball-locking pin is fully inserted by pulling on it.



K1 / K1-SB must always be on top of a K2 array.



K2 arrays can be flown with K1-BUMP or K2-BUMP.

Refer to the **K2 owner's manual** for the procedure with K2-BUMP.



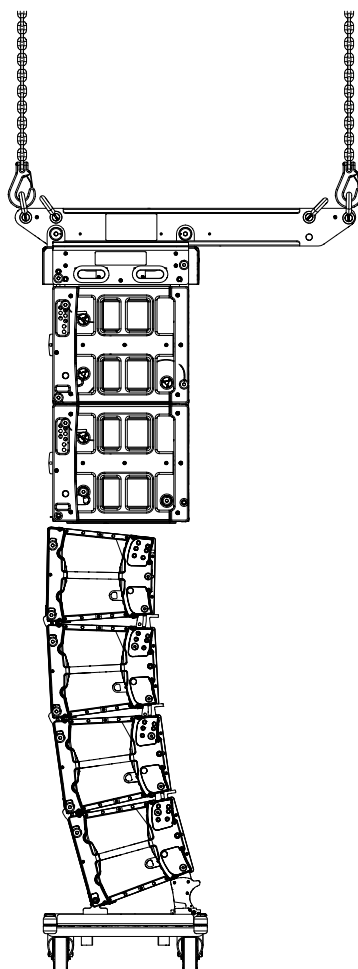
Follow the same procedure for K1-SB enclosures.

Prerequisite

- Prepare a block of four K1. Refer to [Preparing a block of four K1/K1-SB](#) (p.71).
- Preset the inter-enclosure angles. Refer to [Presetting the inter-enclosure angles of K1/K1-SB](#) (p.74).
- Prepare a block of four K2 and preset the inter-enclosure angles. Refer to the **K2 owner's manual**.

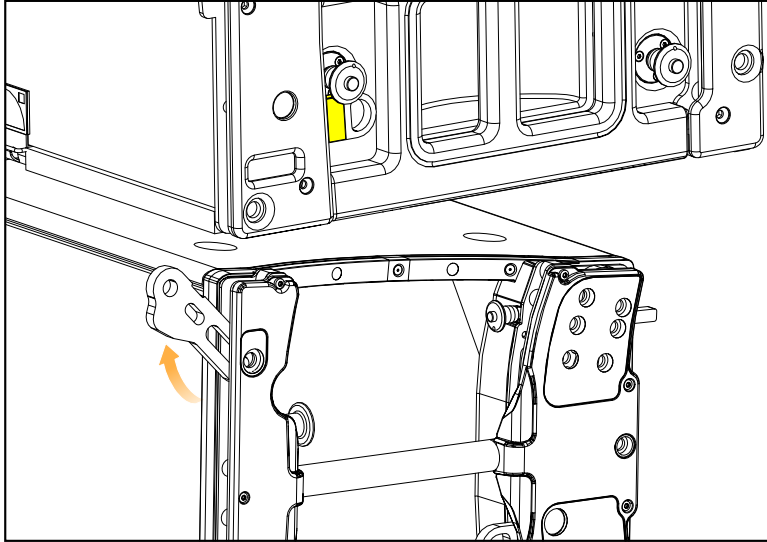
Procedure

1. Position the block of four K2 under the array.

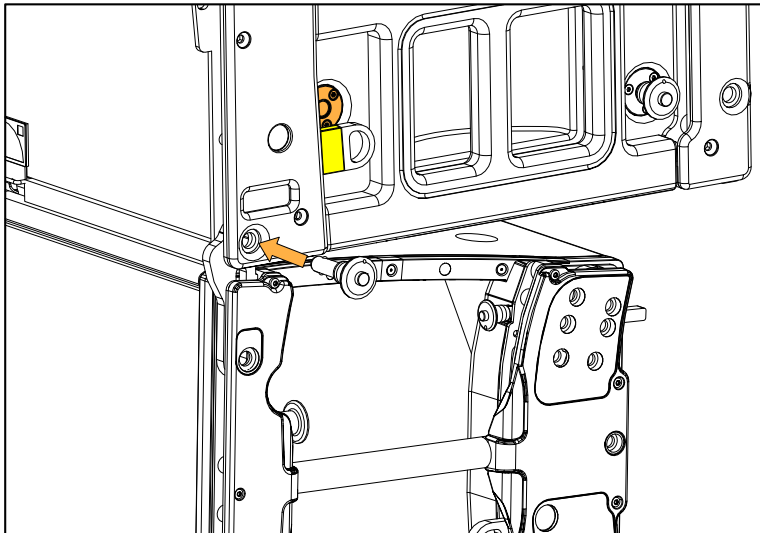


2. On each side, secure the front rigging arm:

- a) Rotate the front rigging arm to align its hole with the K1/K1-SB rigging hole.

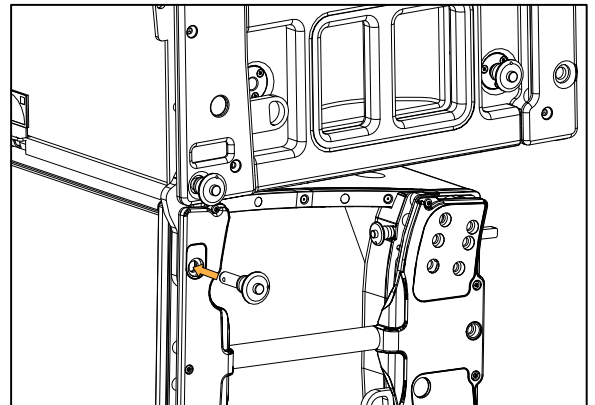
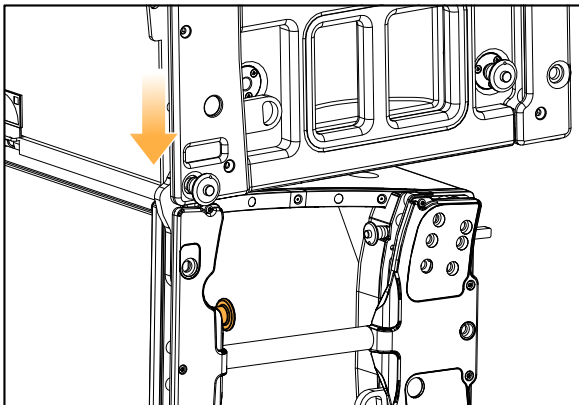


- b) Secure the front rigging arm using the ball-locking pin



- c) Lower the flown array and secure the assembly using the ball-locking pin.

i If necessary, move the flown array back and forth holding the enclosure handles to insert the ball-locking pin.



3. Before raising the array, check that all ball-locking pins and rigging arms are properly secured.



Check that all ball-locking pins are **fully inserted** in rigging position.

Check that all rear rigging arms are secured by making sure **no yellow sticker is visible**.

4. Slightly raise the array to lock the inter-enclosure angles.

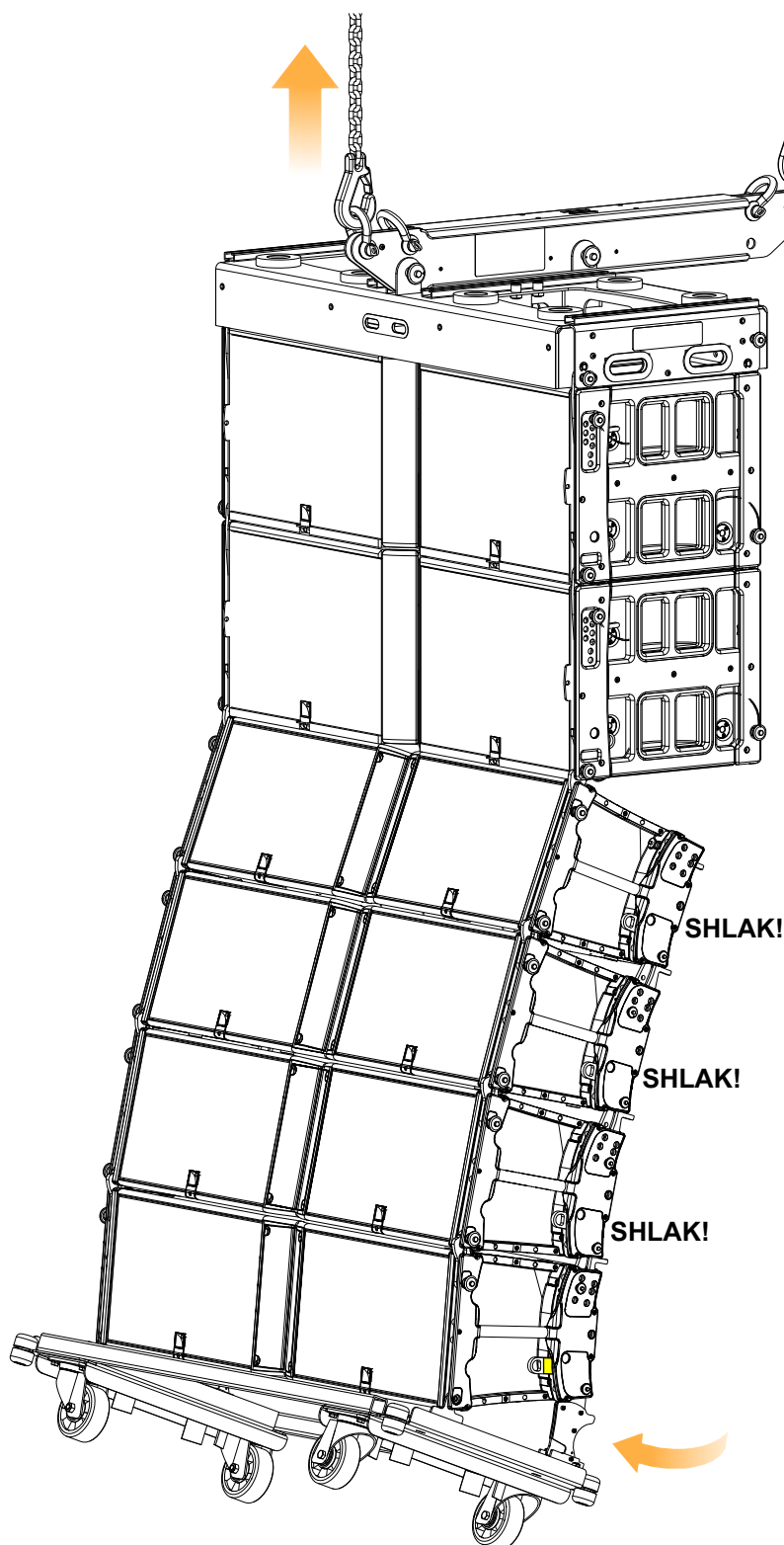
The array swings.

The automatic system engages and the latches retract (except on the bottom enclosure).

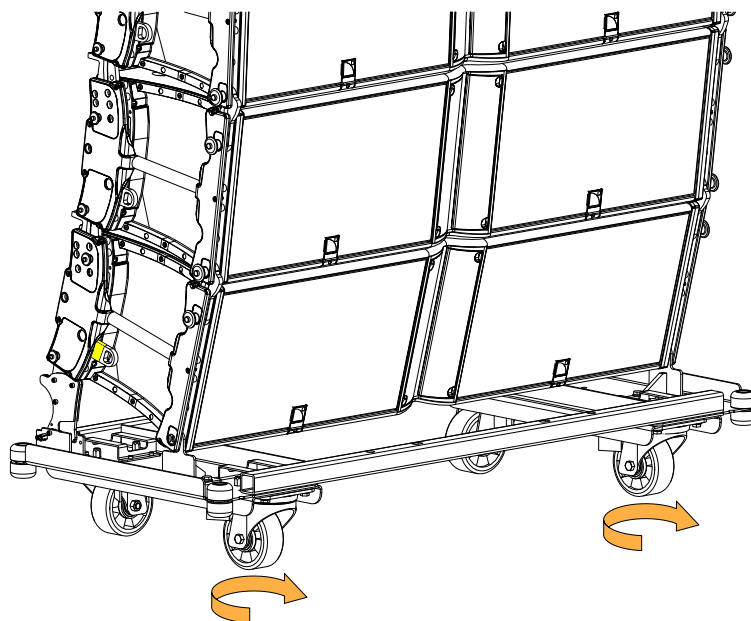


Check that the automatic locking system is **activated**.

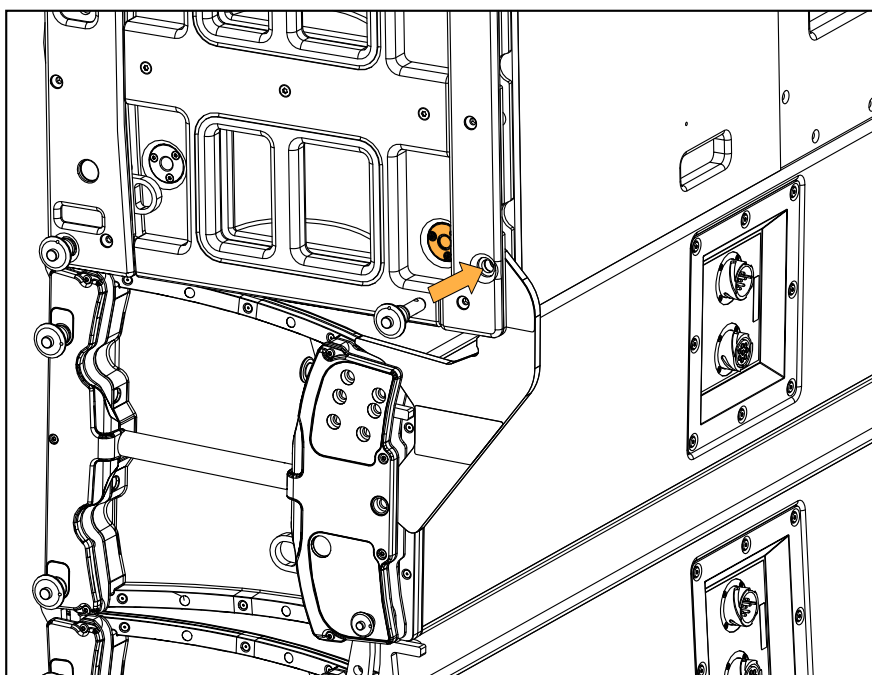
When the automatic locking system is activated, **no yellow sticker is visible** on the latches.



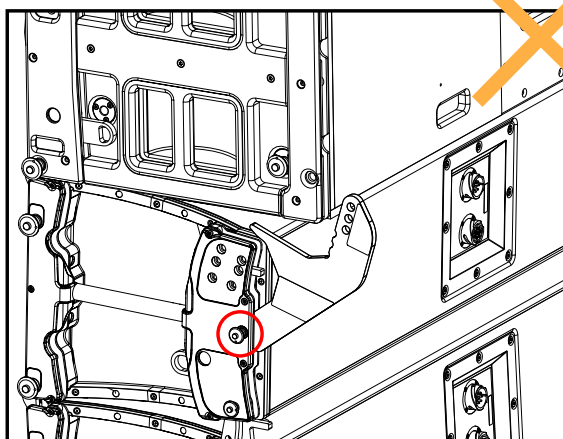
5. Turn the front wheels inside K2-CHARIOT.



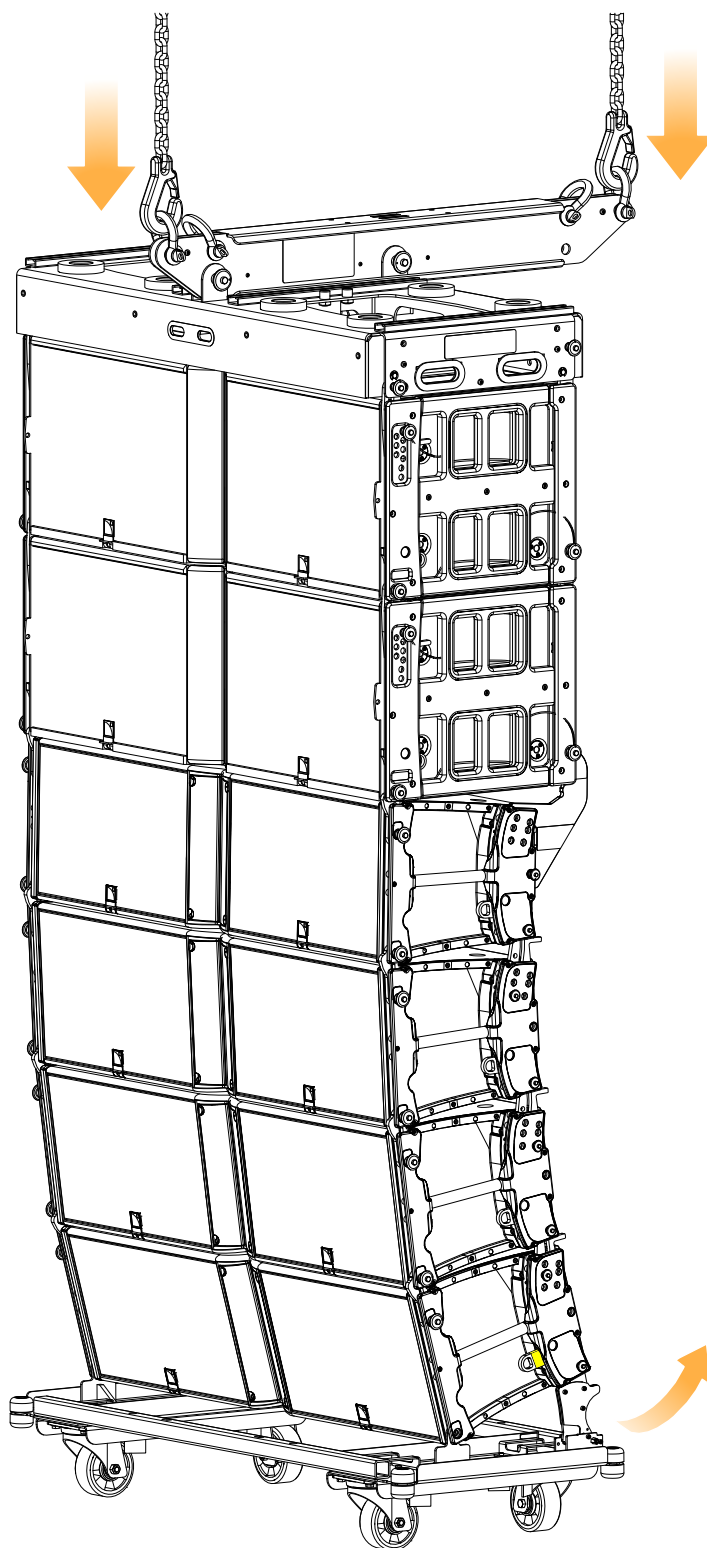
6. Connect the rear of the K1/K1-SB array to the rear of the K2 array with the K2-LINK interfaces:
a) On each side, attach a K2-LINK at the back of the K1/K1-SB enclosure.



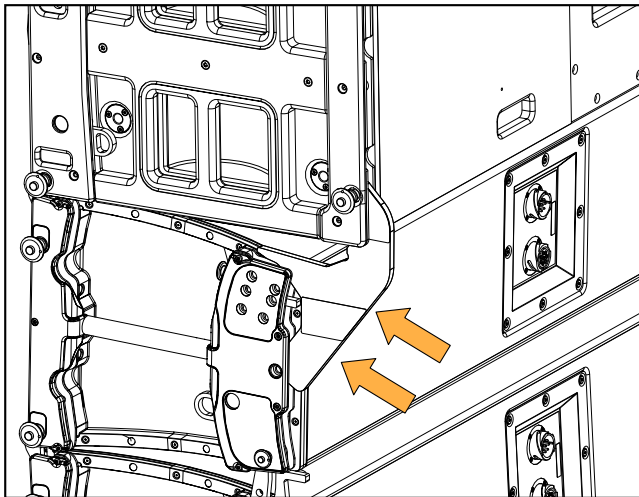
i Do not pin K2-LINK on K2 yet.



- b) Pull back the bottom enclosures while lowering the array until only the front wheels touch the ground.



c) Push the K2-LINK into the K2 rear rigging.

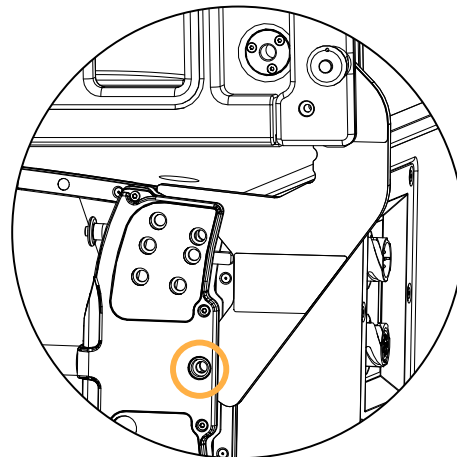
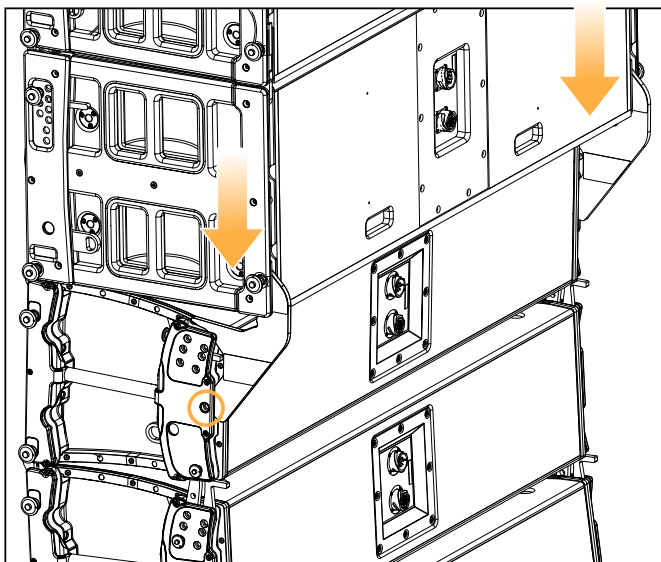


7. Lower the array until the hole of K2-LINK matches the K2 rigging middle hole.

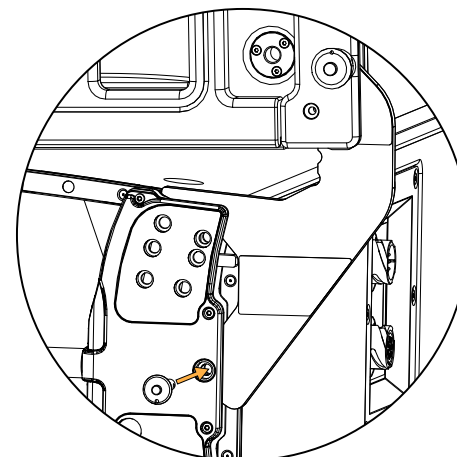
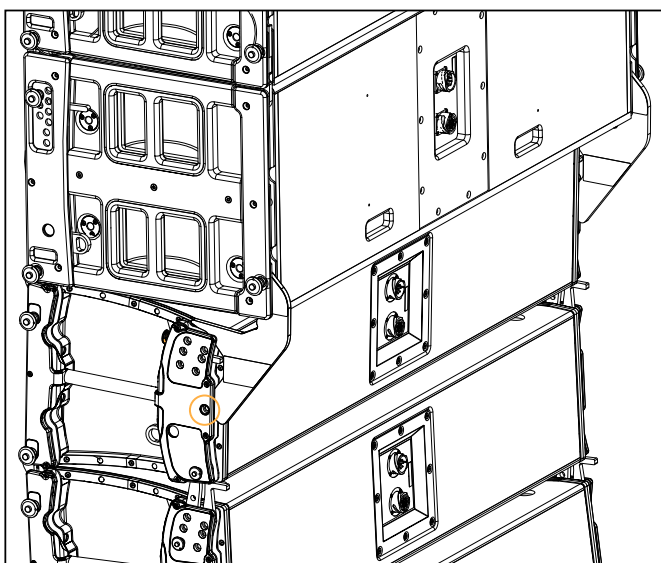


Risk of pinching

Do not touch K2-LINK while lowering the array.



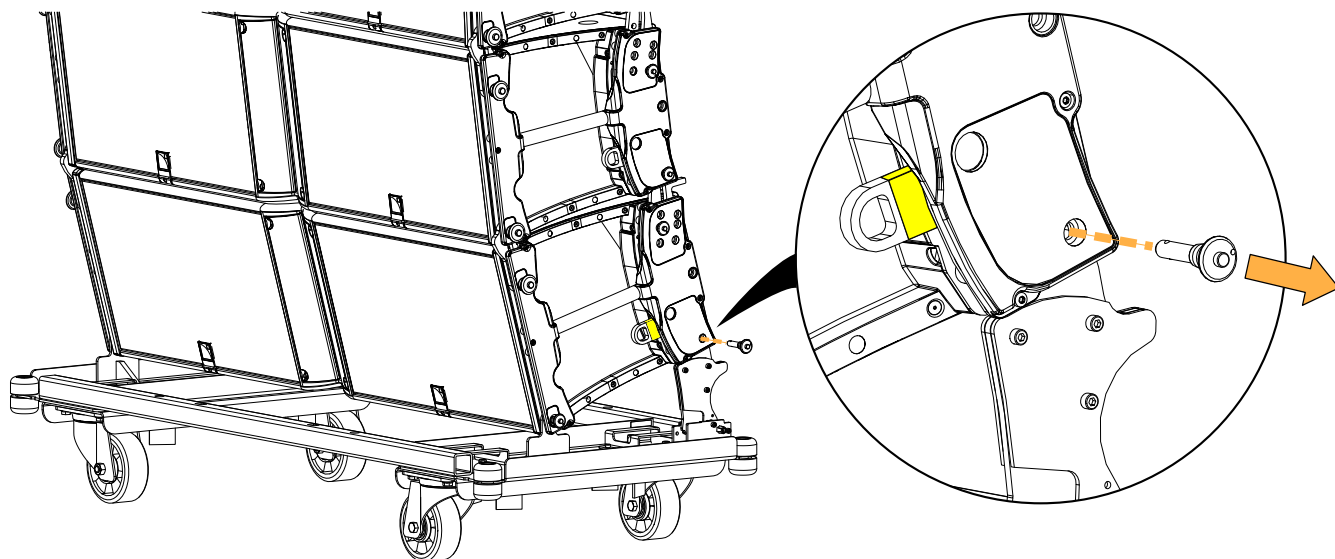
8. Secure K2-LINK on the K2 rigging middle hole.



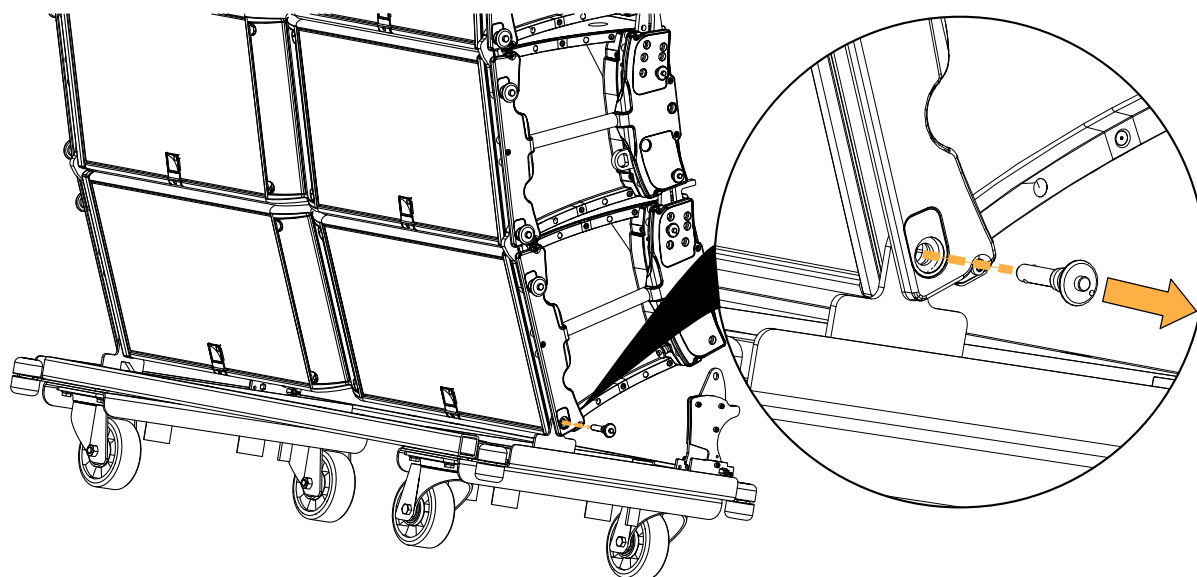
9. Raise the array until the K2-CHARIOT wheels are off the ground.

10. Remove K2-CHARIOT from the array:

- a) Hold K2-CHARIOT and remove the two rear ball-locking pins. Put them back into their storage locations .



- b) Hold K2-CHARIOT and remove the two front ball-locking pins. Put them back into their storage locations .

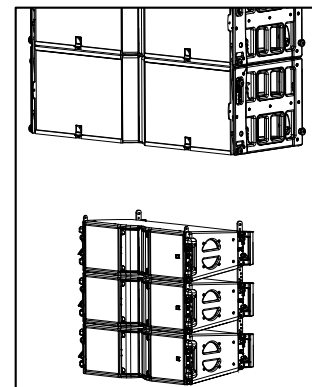


Rigging a Kara II downfill array under a K1 array with KARA-DOWNK1

Min number of operators	2
Rigging accessories	KARA-DOWNK1

Prerequisite

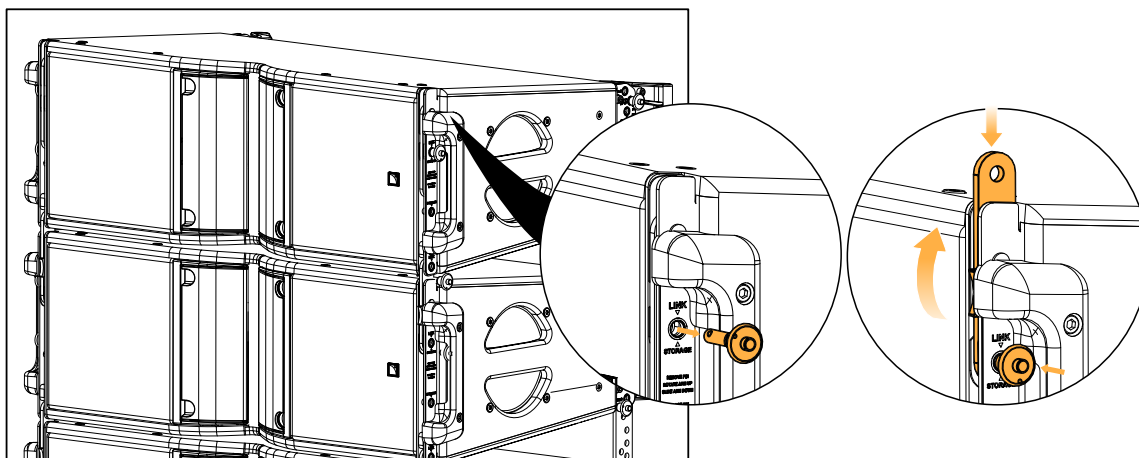
- Prepare and raise a K1 array. Refer to [Flying K1](#) (p.76).
- Prepare an array of three Kara II with 0° inter-enclosure angles. Refer to the **Kara II owner's manual**.
- Position the Kara II array under the K1 array.



Procedure

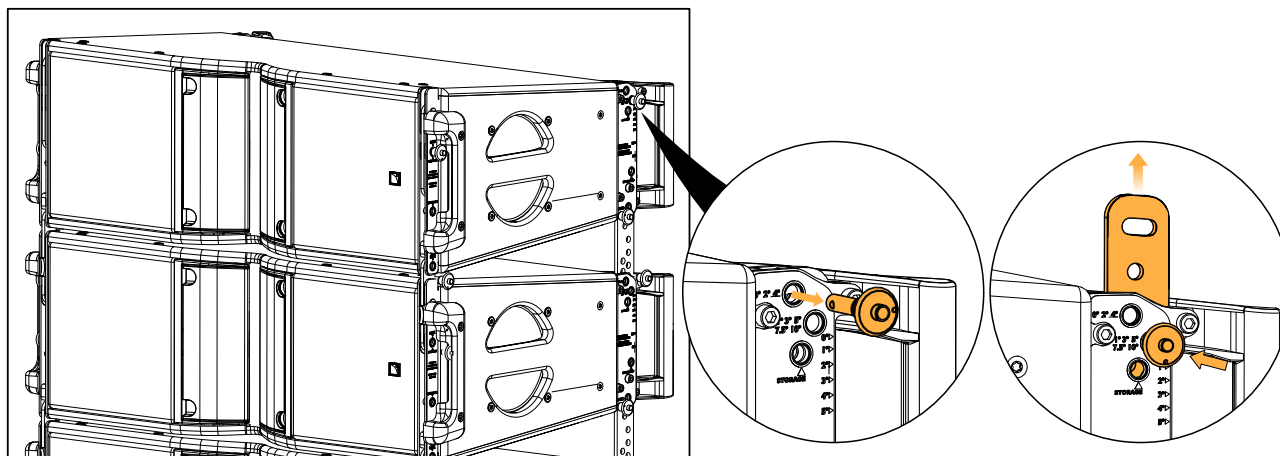
1. On each side of the top Kara II enclosure, set the front rigging arm into rigging position:

- a) Rotate and slide down the front rigging arm.
- b) Secure the front rigging arm with the ball-locking pin.



2. On each side of the top Kara II enclosure, set the rear rigging arm into rigging position:

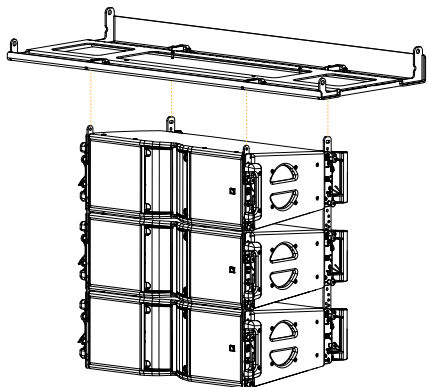
- a) Slide up the rear rigging arms and select the same inter-element angle, from 0° to 10°.
- b) Secure the rear rigging arm with the ball-locking pin.



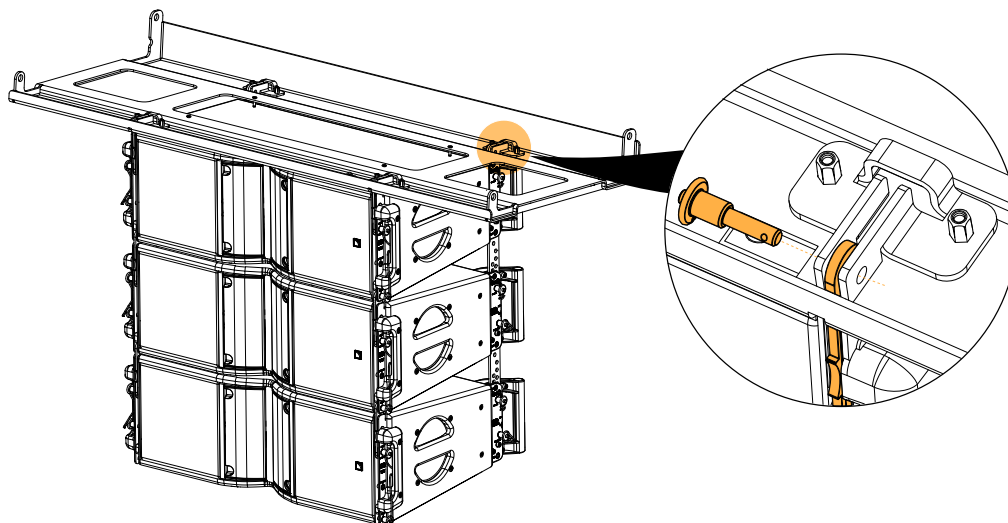
3. Secure KARA-DOWNK1 on top of the Kara II array.



Position the higher ledge at the rear.



Secure the four Kara II rigging arms inside the slits using the ball-locking pins.



4. Lower the K1 array as close as possible to the assembly without resting on it.
5. Disconnect the rear ball-locking pins between the two top Kara II enclosures and place them in storage position.



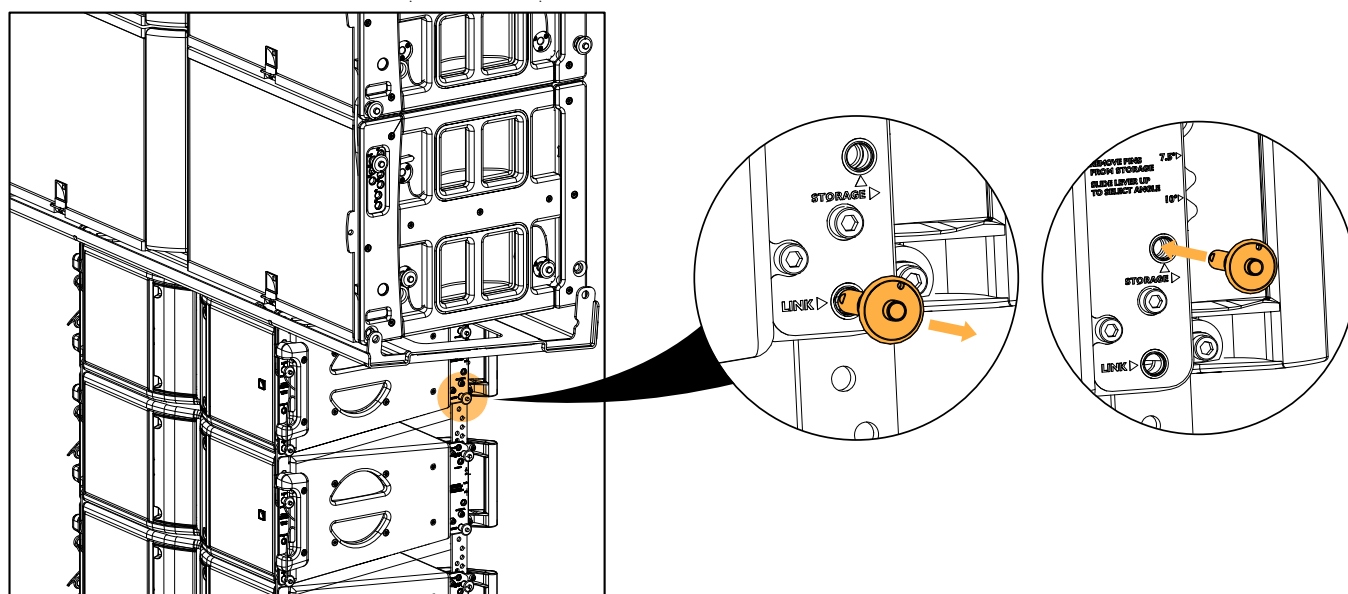
Risk of crushing injury.

Keep fingers away from the contact area between the ledge and the cabinet.



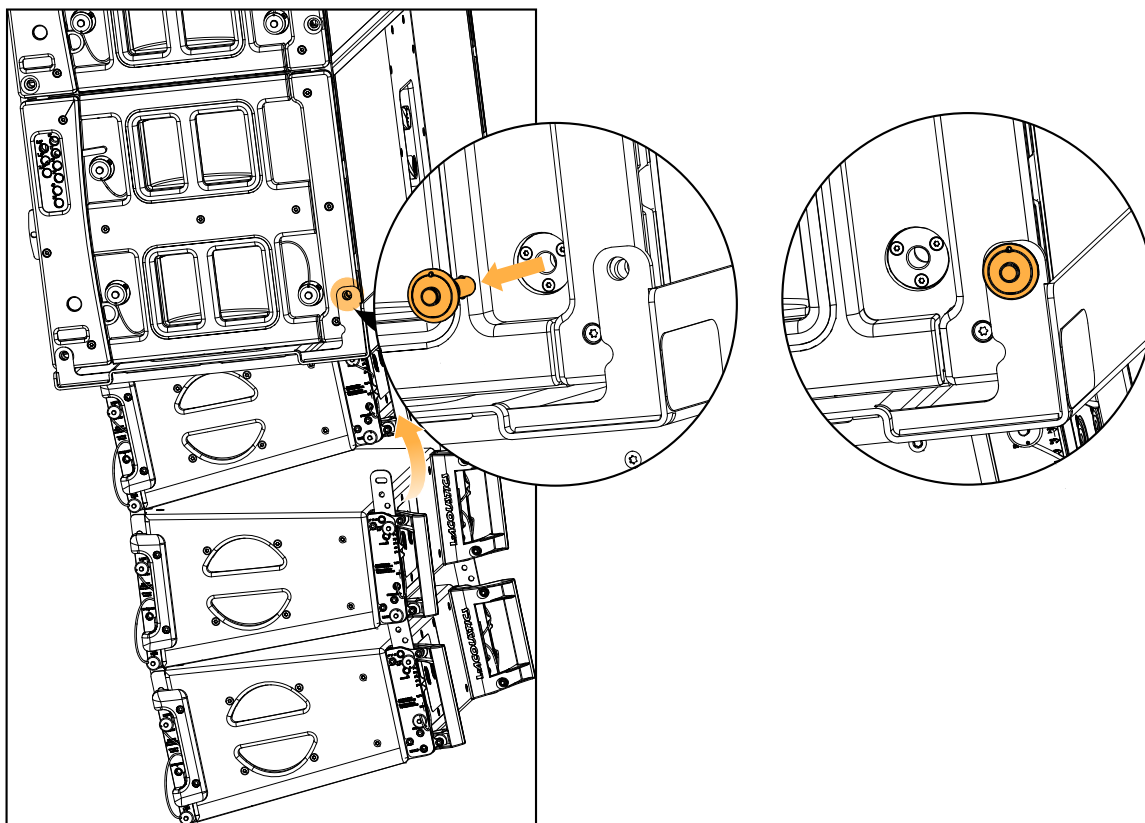
Risk of cable damage.

Disconnect any cable from the K1 bottom connector.



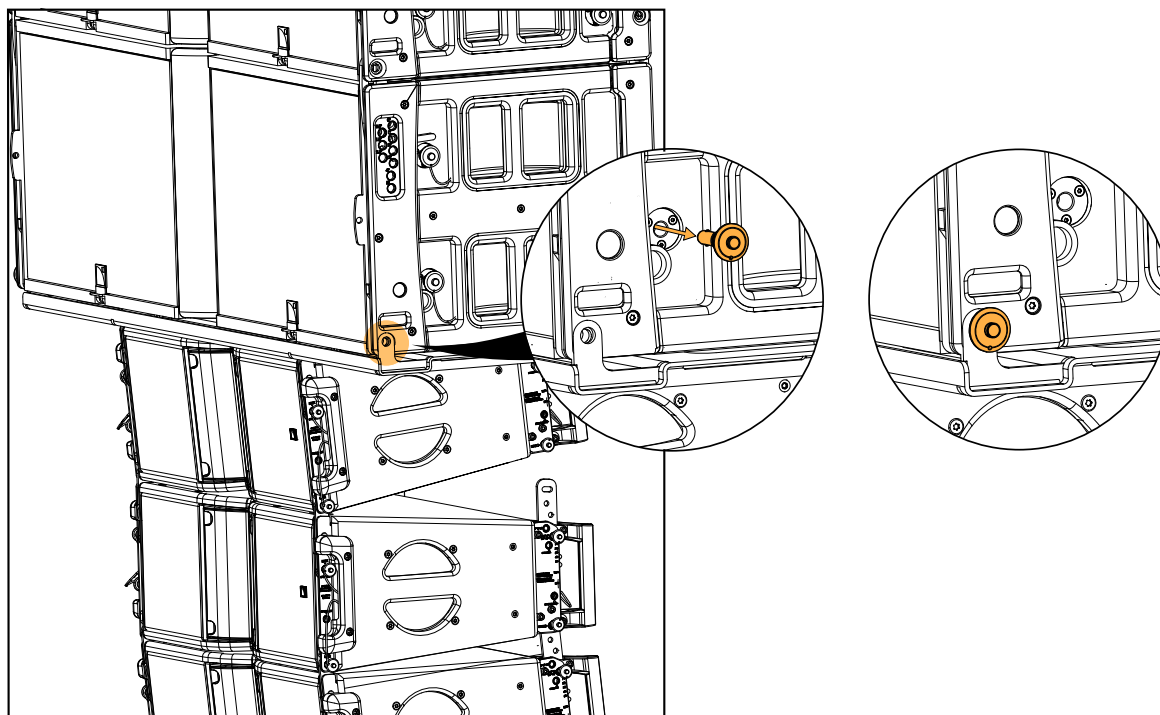
6. Connect KARA-DOWNK1 to the bottom of the K1 array:

- a) Rotate the top Kara II upwards and secure the rear KARA-DOWNK1 tabs to the K1 rigging.



- b) Secure the front KARA-DOWNK1 tabs to the K1 rigging.

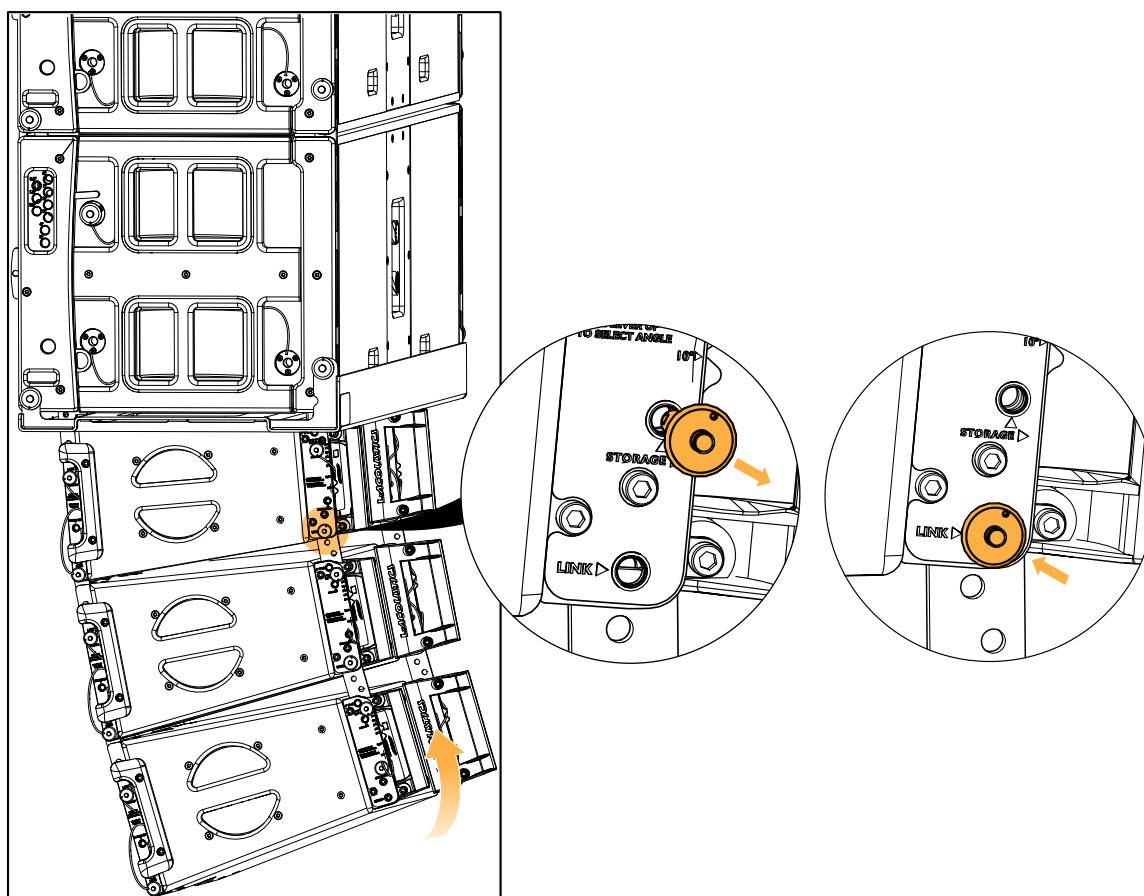
If necessary, slightly lower the K1 array.



c) Reconnect the top Kara II at the rear to the assembly.



Use the handles of bottom Kara II.



7. Raise the array.
8. Set the inter-enclosure angles.
9. Repeat steps 1 (p.92) to 8 (p.95) the Kara II array is complete.



Refer to the **Kara II owner's manual**.

Flying a K1 array with K1-BUMP using the CATERPILLAR method

The following procedure describes how to use the CATERPILLAR method to fly a vertical array of K1.

The CATERPILLAR method consists in preparing the array on the ground by adding the enclosures one by one. Each enclosure ships onto a K1-PLA dolly board.

Min number of operators	3
Rigging accessories	4 Ø 22 mm shackles WLL 3.25 t
	K1-BUMP
	K1-PLA



All along the procedure:

Systematically ensure that each ball-locking pin is fully inserted by pulling on it.



Secondary safety

Use available holes on the rigging accessories to implement a secondary safety.



For clarity purposes the loudspeaker cabling procedure is not described.

The loudspeaker cables are not represented in the figures.

Use a strain relief to avoid mechanical stress at the connector locations due to cable weight.



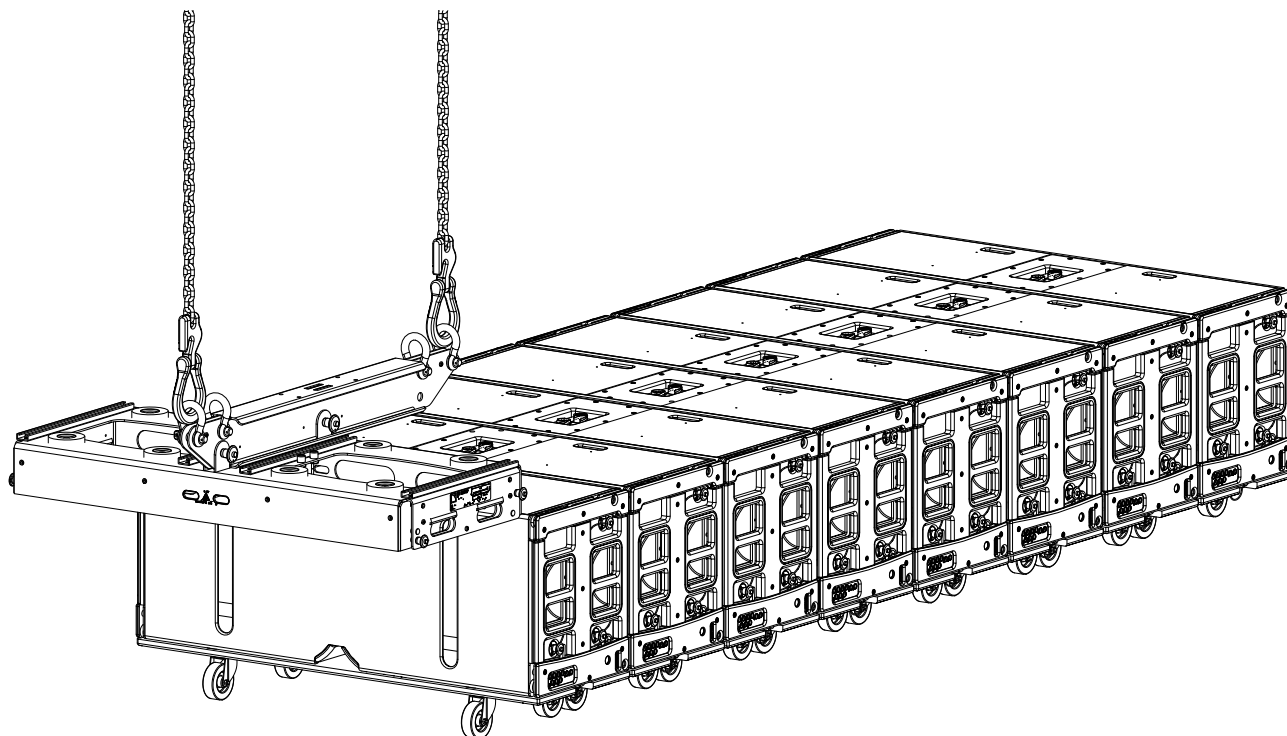
Follow the same procedure for K1-SB enclosures.

Prerequisite

- Prepare K1-BUMP. Refer to [Preparing K1-BUMP](#) (p.62).
- If necessary, secure LA-RAK II AVB on K1-BUMP. Refer to [Securing LA-RAK II AVB on K1-BUMP](#) (p.67).

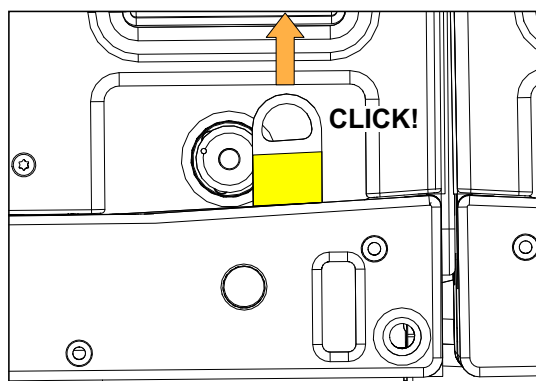
Procedure

1. Line up K1-BUMP and all enclosures with dolly boards at the rigging location.
2. Raise K1-BUMP in front of the first enclosure to more than 0.7 m / 2.3 ft height.

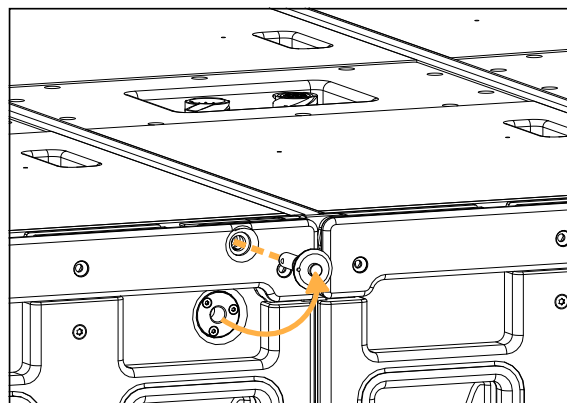
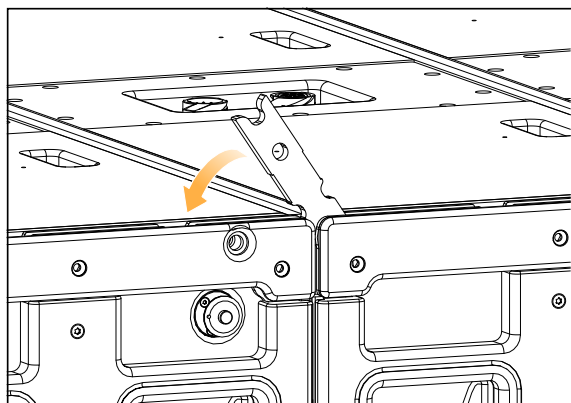


3. On both sides of each enclosure, make sure the automatic locking system button is deactivated. If the button is pressed, pull to disengage the latch.

! When the automatic locking system is deactivated, **a yellow sticker is visible** on the latch.

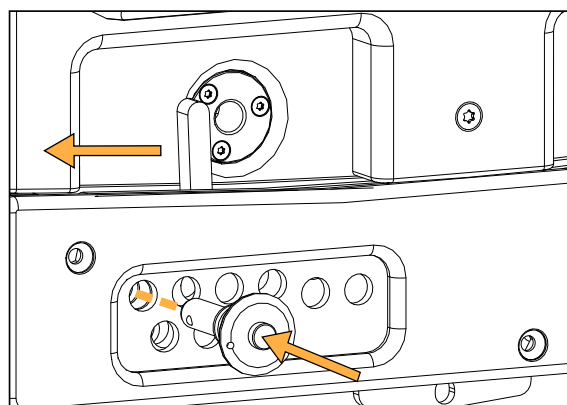
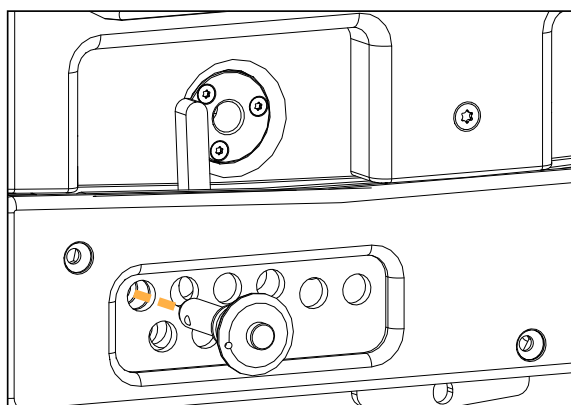


4. On both sides of each enclosure **except for the top one**, lift out the rear rigging arm and secure it to the adjacent enclosure.

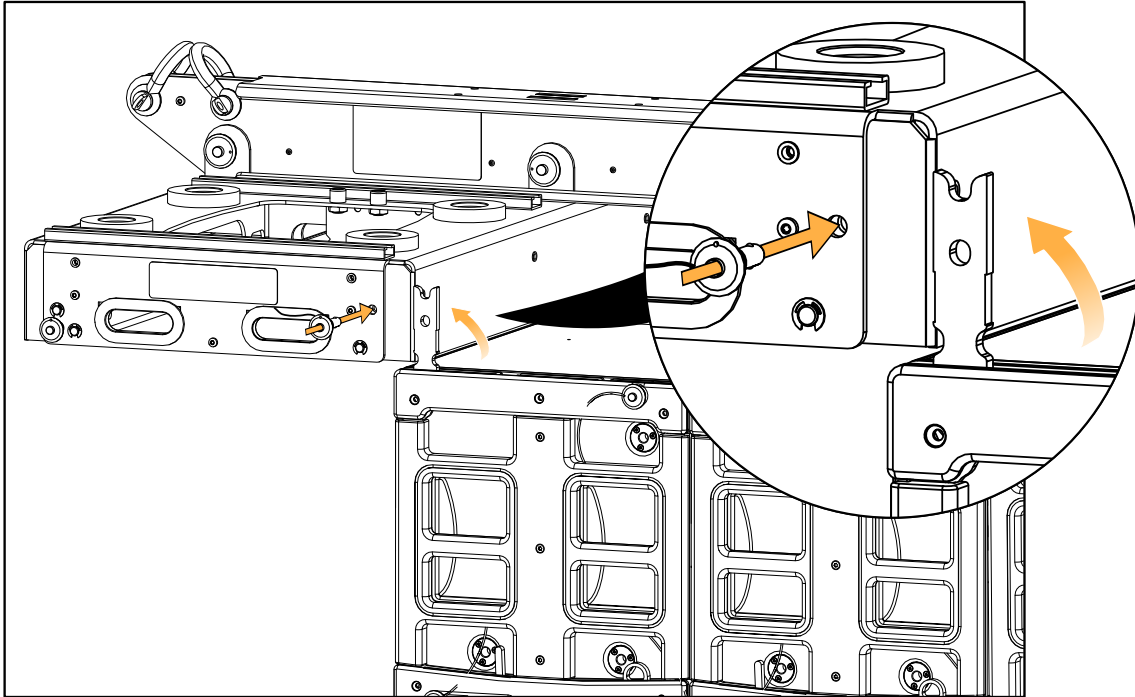


5. On both sides of each enclosure, preset the inter-enclosure angles. Slide out the angle arm to the desired angle hole and insert the ball-locking pin.

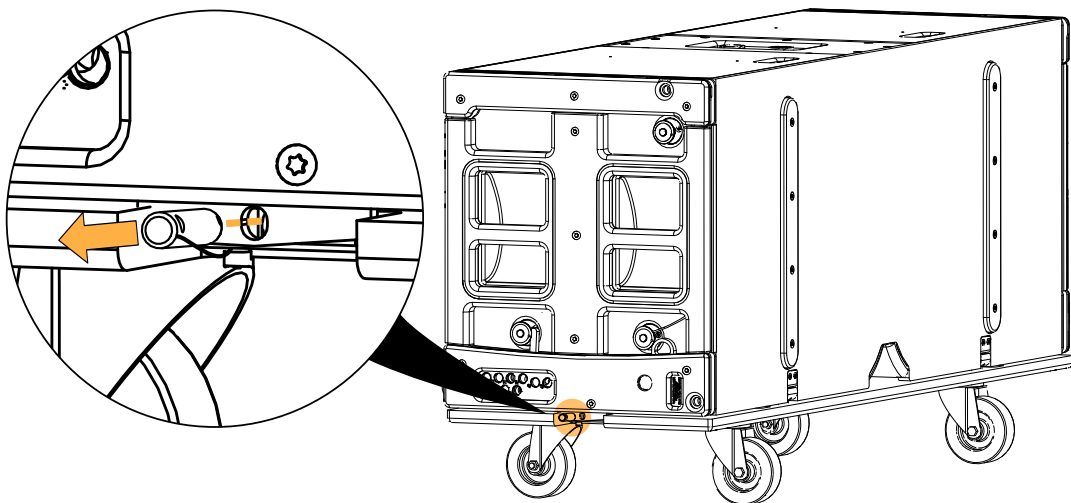
i Select the 0° angle for the top enclosure to be aligned with with the K1-BUMP site angle.



6. On the top enclosure, lift out both rear arms. Secure them to the K1-BUMP rear rigging points using the K1-BUMP ball-locking pins.



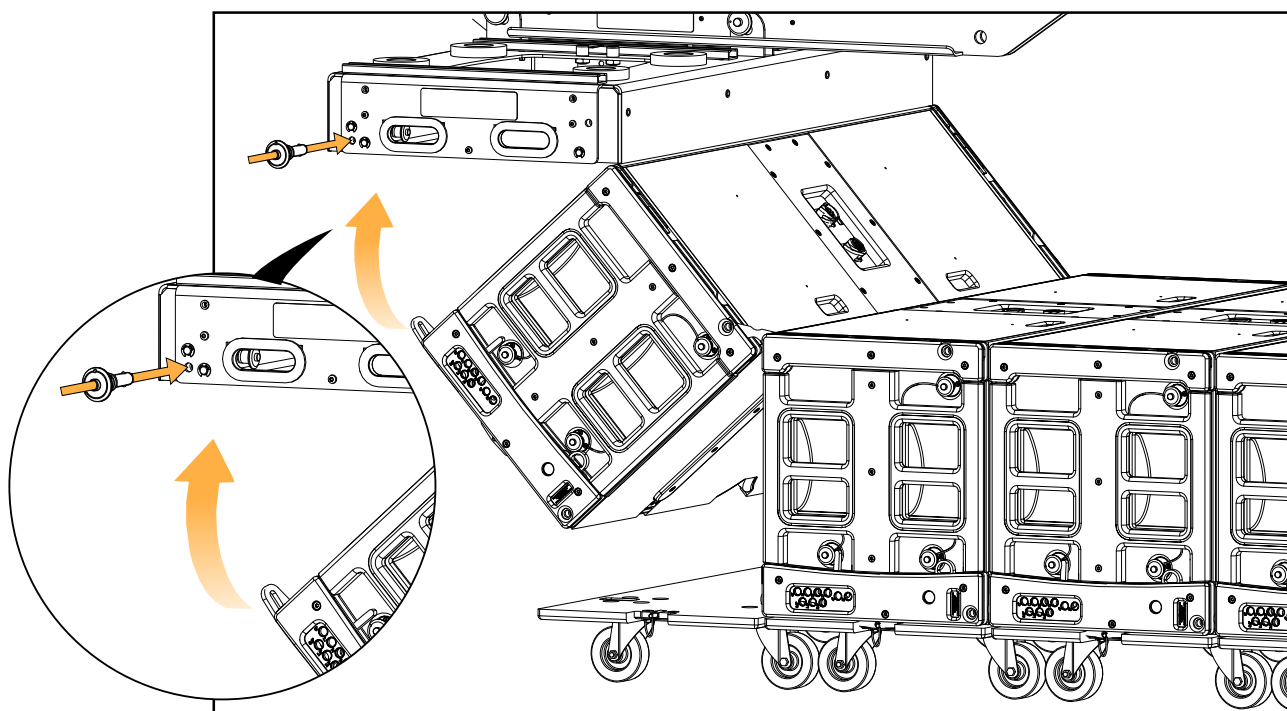
7. On both sides of each K1-PLA, pull on the lanyards to remove the ball-locking pins.



8. Raise the array until the top enclosure is flown and K1-BUMP is in the horizontal position. K1-PLA separates from the top enclosure.

9. Lift the front of the top enclosure and secure the two front rigging arms to K1-BUMP using K1-BUMP ball-locking pins.

! Two people are needed for this operation: one on each side of the enclosure.

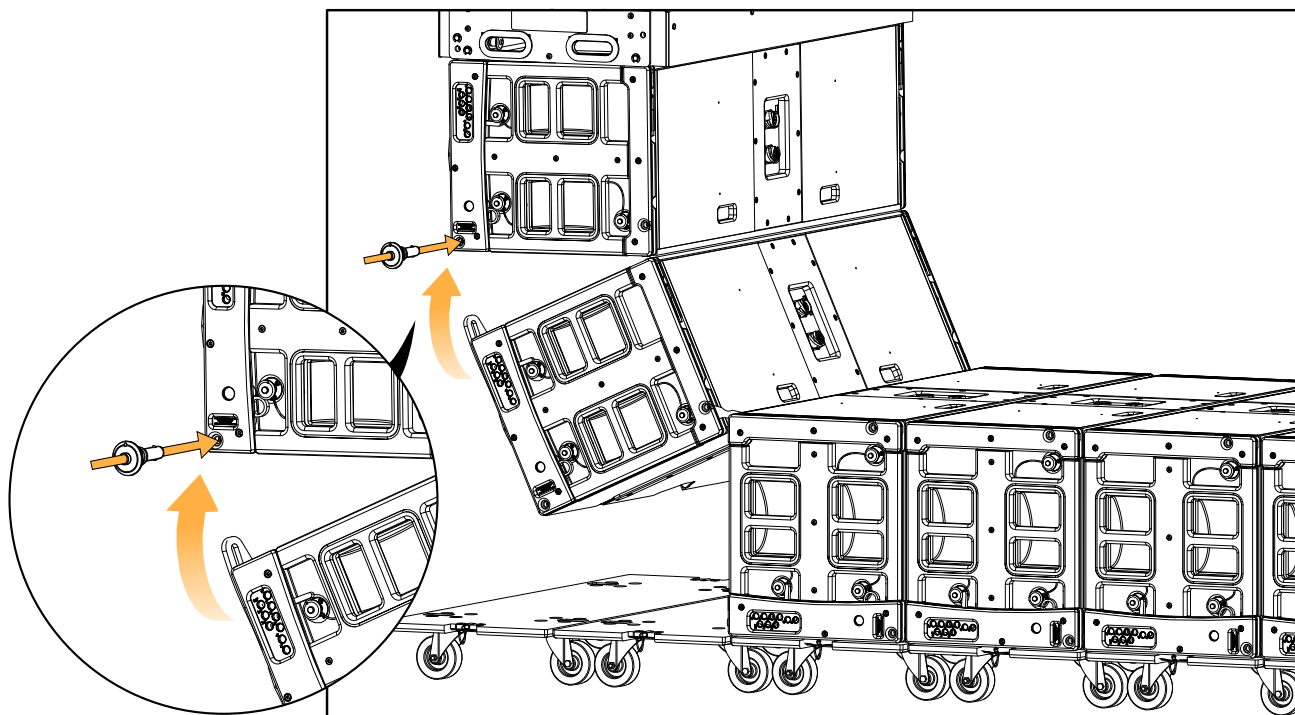


10. Raise the array until the second enclosure is flown.

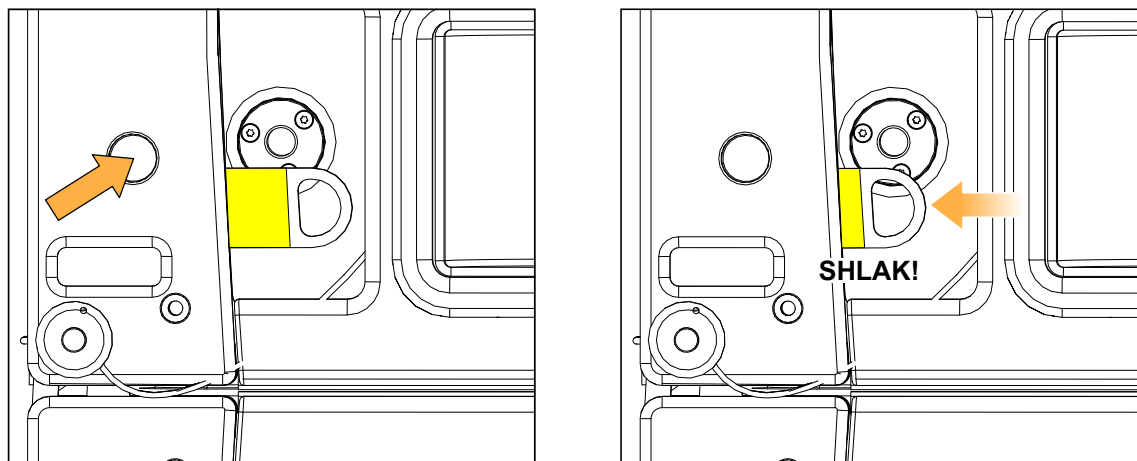
! Have one person holding the last enclosure to prevent array rough movements.

11. Lift the front of the second enclosure and secure both front rigging arms to the top enclosure.

! Two people are needed for this operation: one on each side of the enclosure.



- 12.** On each side of the top enclosure, press the angle locking button to lock the inter-enclosure angle.
The latch slightly retracts when the button is pushed (yellow label visible).

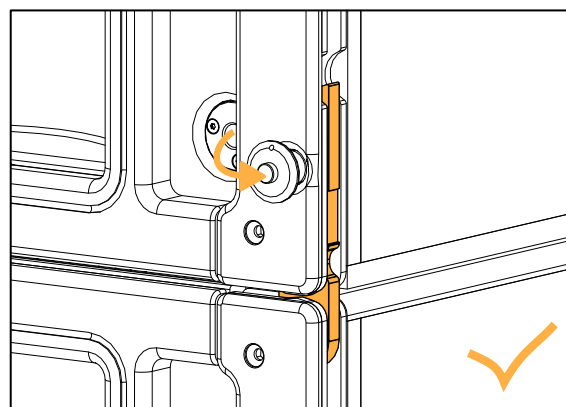
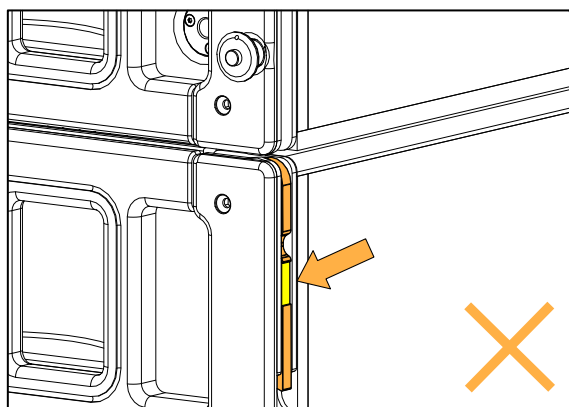


- 13.** Repeat steps 10 (p.99) to 12 (p.100) until all enclosures composing the array are flown.



Check that all ball-locking pins are **fully inserted** in rigging position.

Check that all rear rigging arms are secured by making sure **no yellow sticker is visible**.



- 14.** Fly the array at desired trim height and adjust the site and azimuth angles.



Always check that the automatic locking system is activated (except on the bottom enclosure). When the automatic locking system is activated, **no yellow stickers is visible** on the latches.

What to do next

Add a Kara II downfill array. Refer to [Rigging a Kara II downfill array under a K1 array with KARA-DOWNK1](#) (p.92).

Removing a K1 array using the CATERPILLAR method

The following procedure describes how to use the CATERPILLAR method to remove an array of K1/K1-SB.

The CATERPILLAR disassembly method consists in landing each enclosure on a K1-PLA dolly board for individual shipping.

Min number of operators	3
Rigging accessories	K1-PLA

! All along the procedure:

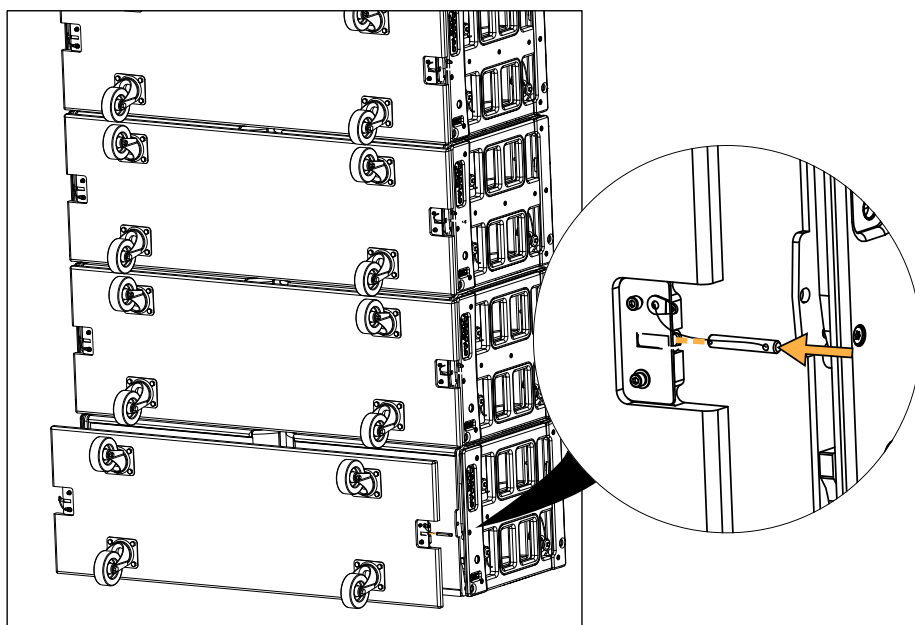
Systematically ensure that each ball-locking pin is fully inserted by pulling on it.

Prerequisite

To remove K2 or Kara II downfill arrays, refer to the **K2** or **Kara II owner's manual**.

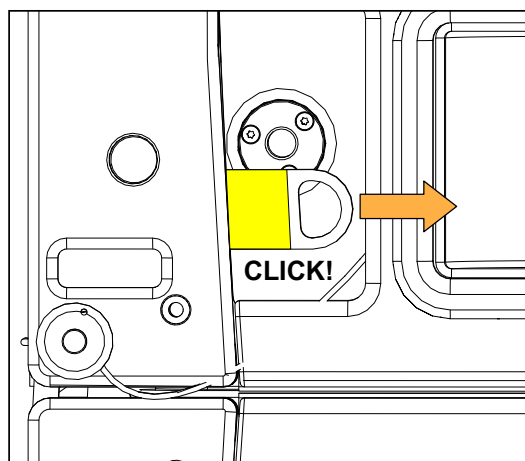
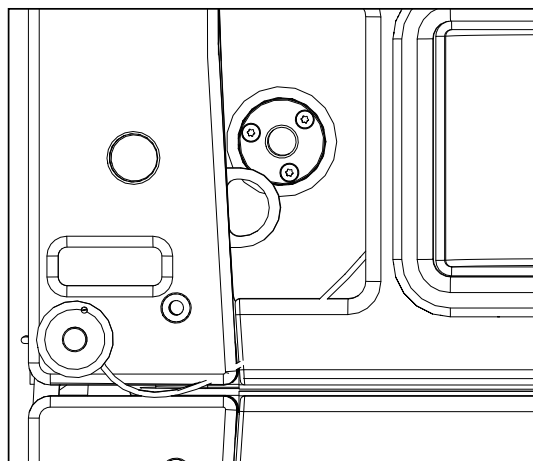
Procedure

1. Lower the array at 0.3 m / 1 ft height.
2. Secure a K1-PLA to the **four bottom enclosures** of the array.
Insert both K1-PLA ball-locking pins on each K1-PLA.



3. On each side of the four enclosures, pull the latch until a click is heard.

! A yellow sticker must be visible on each latch.



4. Land the four enclosures:

- a) Firmly hold the bottom enclosure by its handles.



Two people are needed for this operation: one on each side of the enclosure.

Use the ergonomic handles to lift the enclosure.

- b) Remove the two ball-locking pins securing the bottom enclosure at the front. Put them in their storage locations.



The rear rigging points must remain attached to prevent the bottom enclosure from falling.

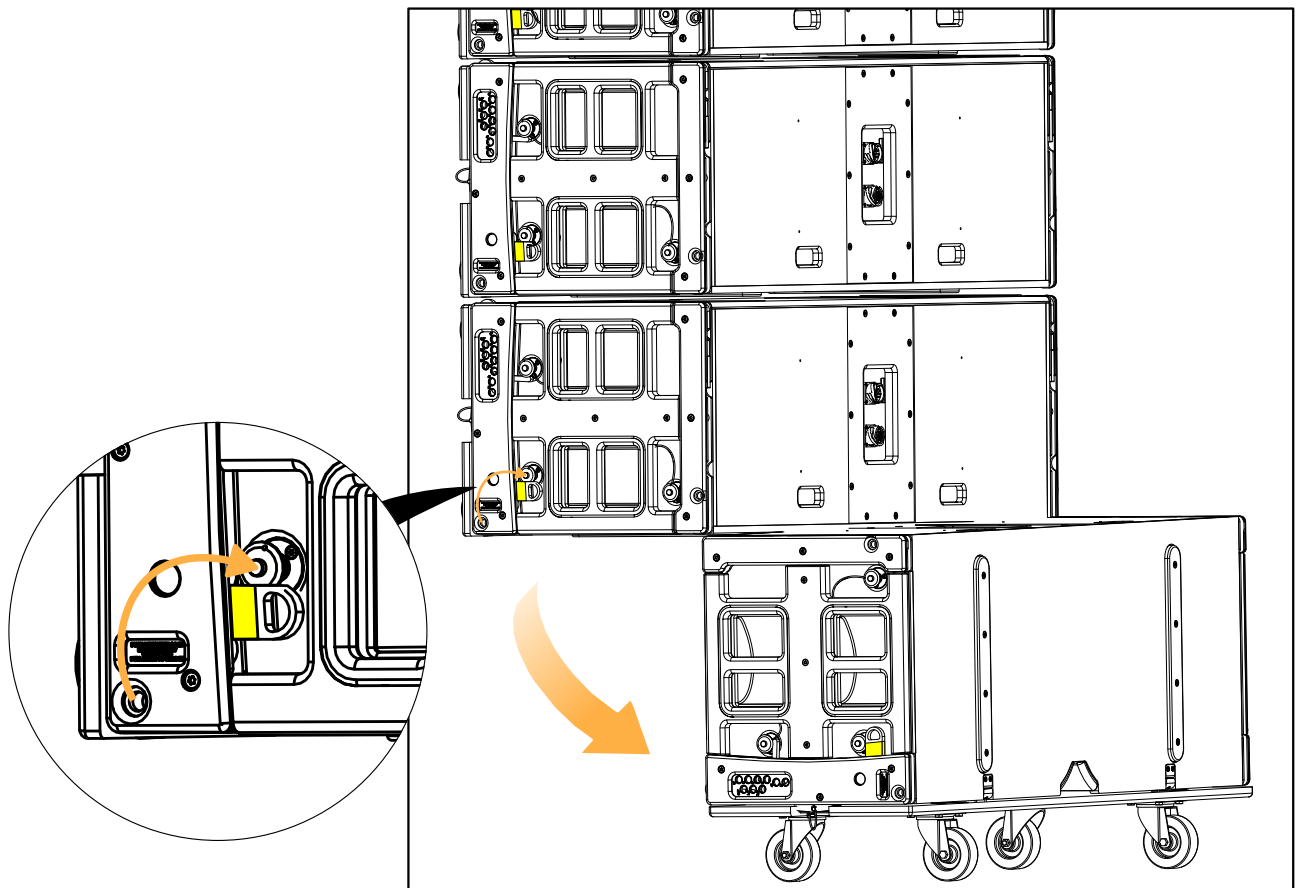
- c) Pull back the bottom enclosure while lowering the array until the four K1-PLA wheels touch the ground.



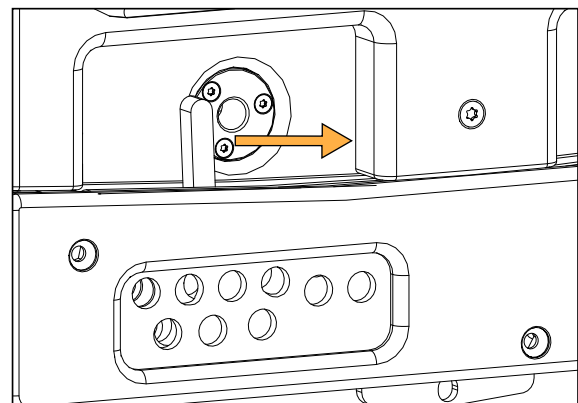
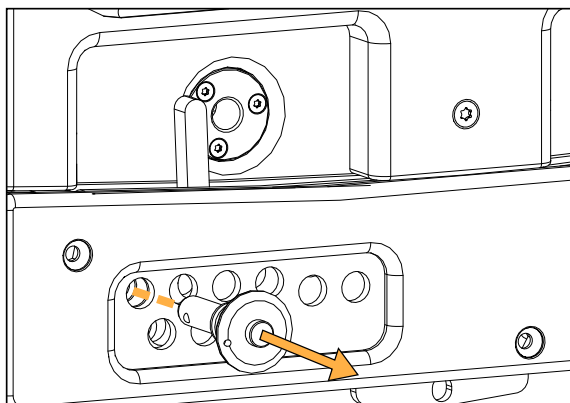
Two people are needed for this operation: one on each side of the enclosure.

Use the ergonomic handles to pull back the enclosure.

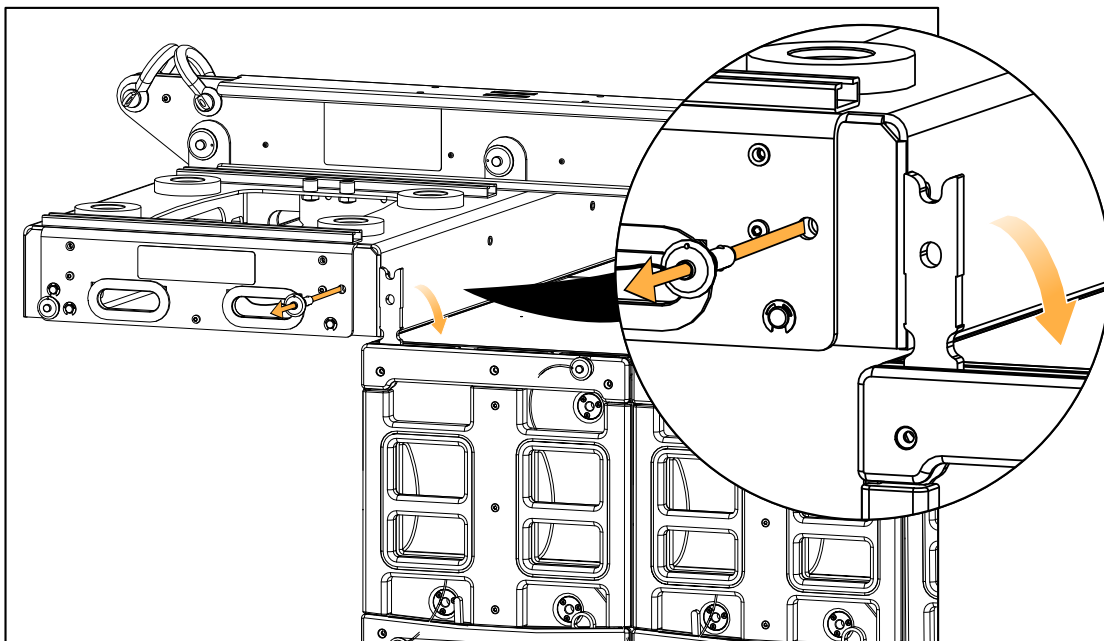
- d) Repeat step 4 (p.102) until the four enclosures are on the ground, attached by the rear rigging arms.

**5.** On both sides of the four landed enclosures:

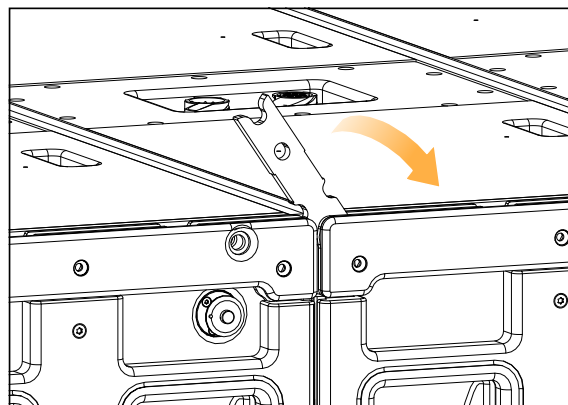
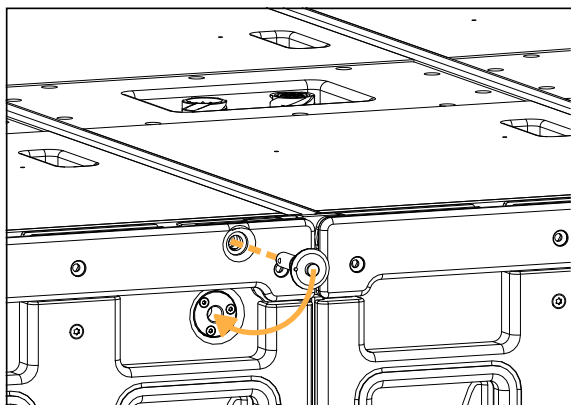
- a) Remove the angle setting ball-locking pins. Put them in their storage locations.
b) Slide in the front rigging arm.

**6.** If necessary, repeat steps 1 (p.101) to 7 (p.102) until all enclosures composing the array are on the ground.

- 7.** On the K1-BUMP, remove the two rear ball-locking pins.
Pull back the two rear rigging arms of the top enclosure into their storage locations.



- 8.** On all other enclosures, remove the rear ball-locking pins. Pull back the rear rigging arms into their storage locations.



- 9.** If necessary, lower K1-BUMP until it touches the ground and remove the LA-RAK II AVB touring racks.
10. Place K1-BUMP in K-BUMPFLIGHT using the motor.

Connection to LA amplified controllers

! Refer to the **Amplification reference** technical bulletin for the latest information on compatibility with amplified controllers and cabling schemes for all enclosure types.

Enclosure drive capacity per amplified controller

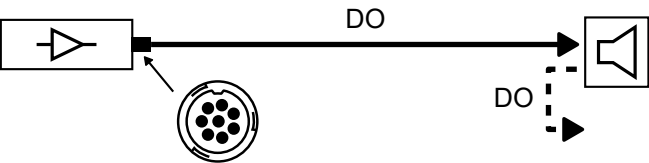
Make sure the total number of connected enclosures does not exceed the maximum number of enclosures per controller (refer to the footnotes).

	LA12X	
	per output* / total	
K1	2 / 2	
K1-SB	1 / 4	
K2	3 / 3	
Kara II	3 / 6	

Cabling scheme for K1 or K2

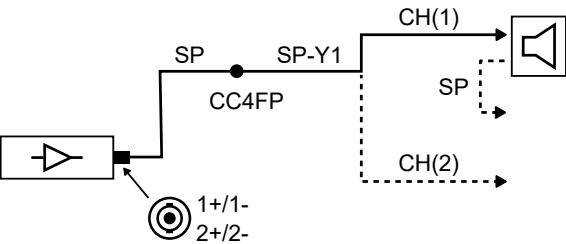
Refer to the cabling schemes to connect the enclosures to different types of output configurations.

Four-channel CA-COM output

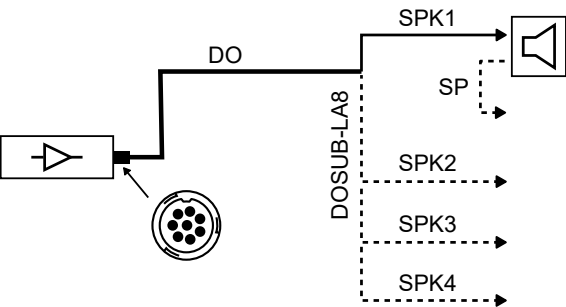


Cabling schemes for K1-SB

Two-channel speakON output



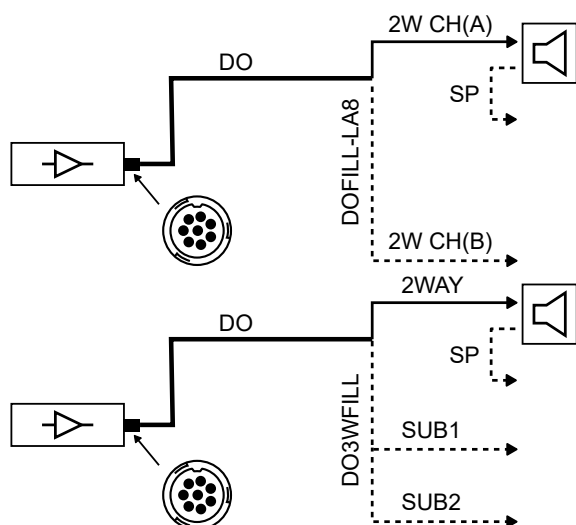
Four-channel CA-COM output



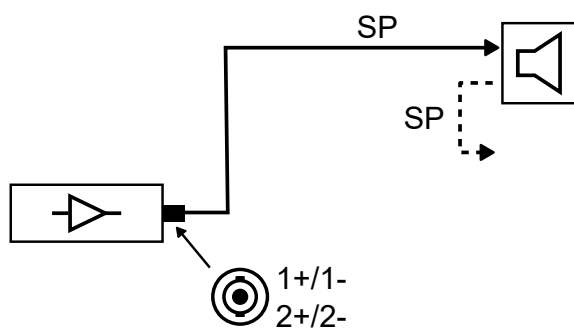
* For passive loudspeakers, the value corresponds to the number of enclosures in parallel on the output. For active loudspeakers, the value corresponds to the number of sections in parallel on the output.

Cabling scheme for Kara II

Four-channel CA-COM output



Two-channel speakON output



Corrective maintenance

Tools and consumables

Before performing maintenance on this product, make sure all the tools listed are available. References are given for FACOM® products in this table. Other manufacturers can be used.

name	reference	distributor
set of 6-point 1/4" sockets *	RL.NANO1 / R.360NANO	-
electric screwdriver with torque selector	-	-
Long nosed pliers	-	-
Electrical soldering material	-	-
combination pliers *	-	-
blue threadlocker	-	-



* included in the L-Acoustics Maintenance Toolcase.

Maintenance Toolcase

The Maintenance Toolcase is a carry-on suitcase that includes all the tools required to perform maintenance on L-Acoustics products. This toolcase is aimed at Certified Providers.

The Maintenance Toolcase uses a Peli™ 1510 Protector case that features three pre-cut layers of foam to safely fit the tools. The Maintenance Toolcase includes tools manufactured by FACOM®, Fluke®, Tohnichi, ABUS, and Würth.



All third-party trademarks, registered trademarks, or product names are the property of their respective owners.

K1

Introduction

This section contains the following maintenance procedures:

- [D/R - Grill](#) (p.109)
- [D/R - LF speaker](#) (p.110)
- [D/R - Fins](#) (p.111)
- [D/R - MF speaker](#) (p.112)
- [D/R - Connector plate](#) (p.113)
- [D/R - HF driver](#) (p.114)
- [D/R - HF diaphragm](#) (p.115)

For advanced maintenance, contact your L-Acoustics representative.

K1 Exploded views

In order to operate, follow the order outlined here. Each assembly refers to the corresponding Disassembly/Reassembly (D/R) procedure and the necessary repair kit (KR).

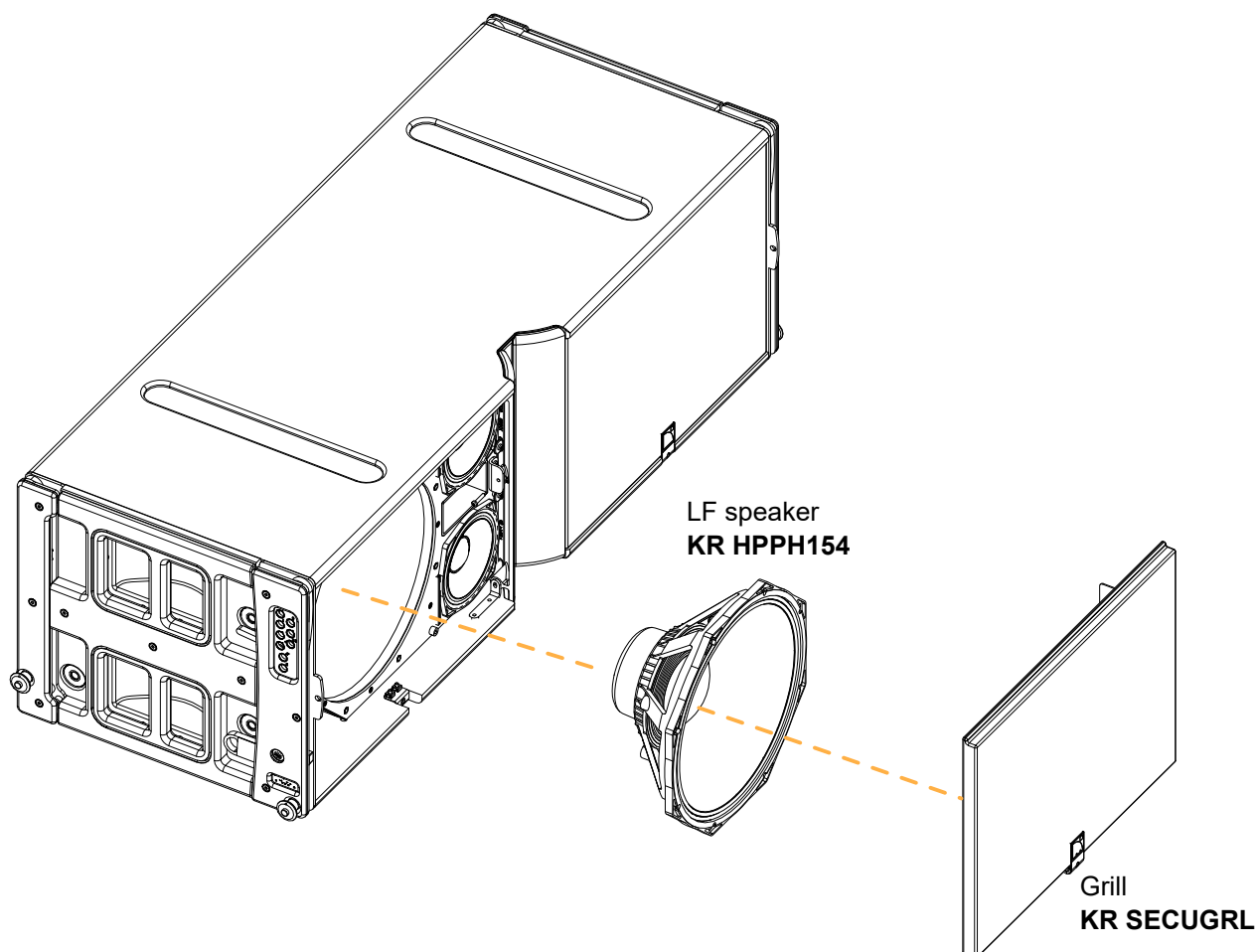


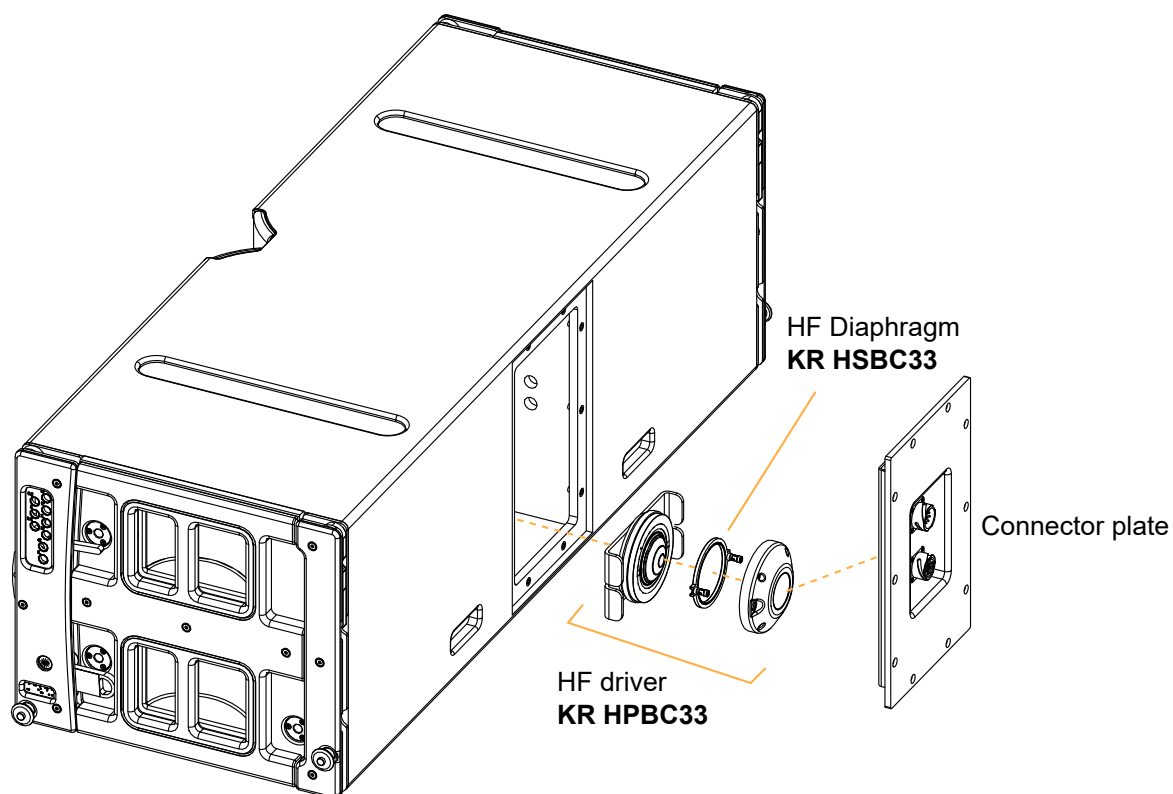
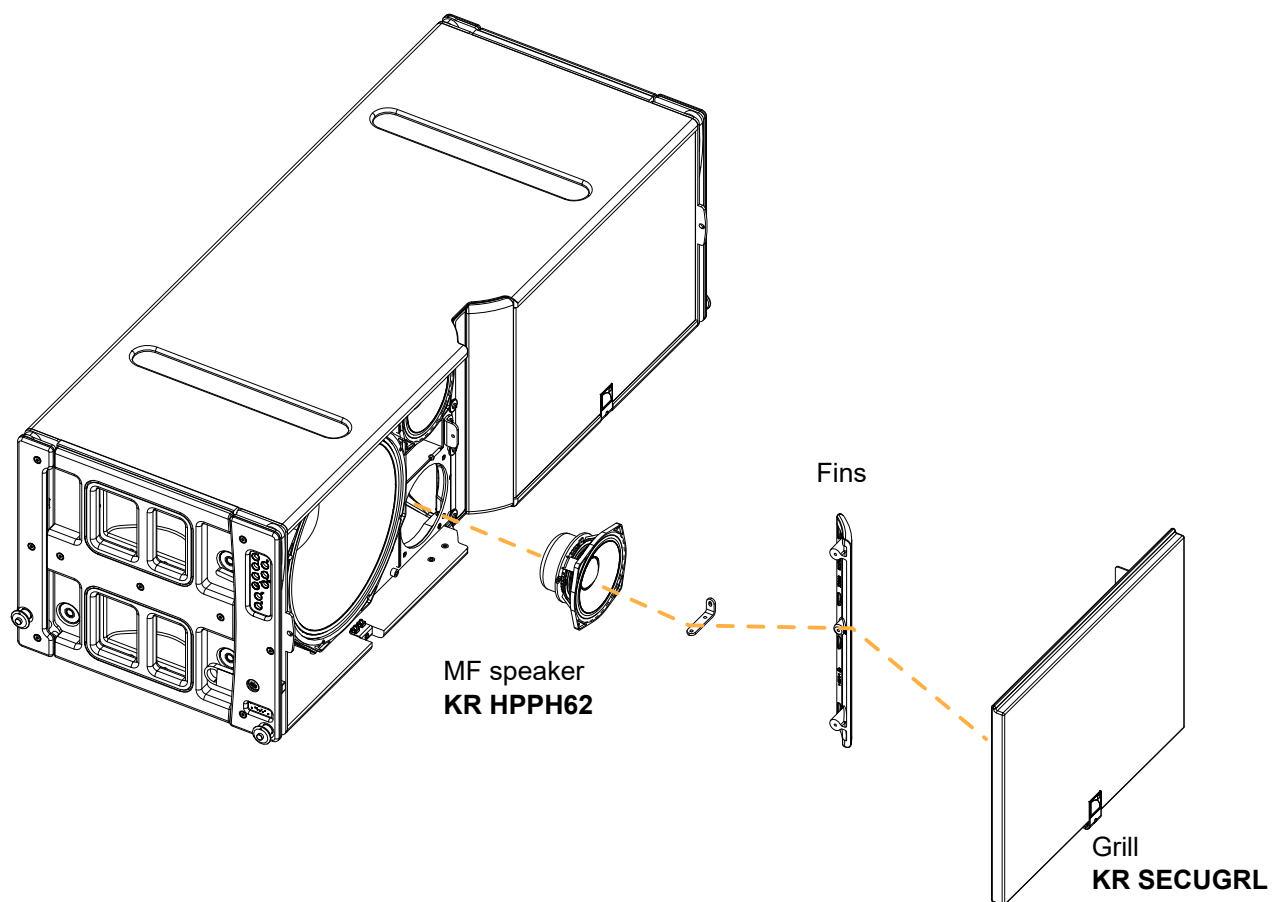
Using the KR HSPH154 reconing kit

Follow the instructions to remove the LF speaker from the enclosure.

Perform the reconing procedure according to the information provided by the manufacturer.

Use the screws and fasteners provided in KR HSPH154 for reassembly.





Disassembly and Reassembly procedures

D/R - Grill

Tools

- torque screwdriver
- 4 mm hex bit
- grill safety tool (provided)
- flat tool

Consumables

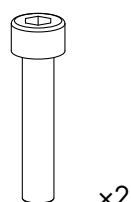
- blue threadlocker

Repair kit

KR HPPH154* - Kit HP PH154 Speaker 15" - 8 ohms

or KR HPPH62 - Kit HP PH62 Speaker 6.5" - 8 ohms

KR SECUGRL - Kit 100 safety O-rings for front grill



x2

S100155

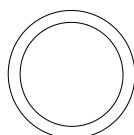
M5x25 hex



x2

S158

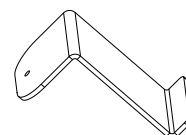
plain washer Ø 5 mm



x100

100557

grill safety O-ring



x1

100226

grill safety tool



*The screws and fasteners are also provided in the KR HSPH154 (Kit reconing 15").

Prerequisite

The enclosure is placed on its top.

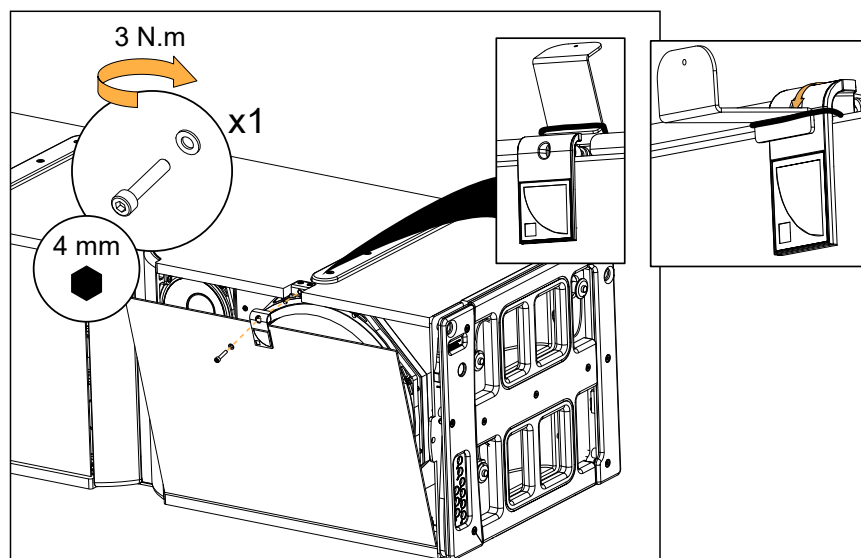
Exploded view



For safety reasons, always use the new screws and spare parts provided in the KR.

If no new screws are available, use blue threadlocker.

Use a flat tool to pull the O-ring over the logo.



D/R - LF speaker

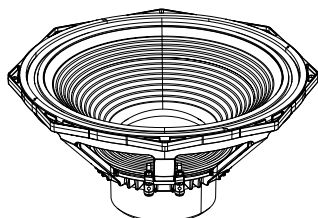
Tools

- torque screwdriver
- 5 mm hex bit

Repair kit

KR HPPH154*

Kit HP PH154 Speaker 15" - 8 ohms



x1

18463

15" speaker - 8 ohms



x8

S100054

M6x30 hex



*The screws and fasteners are also provided in the KR HSPH154 (Kit reconing 15").

Prerequisite

Grill removed.

See [D/R - Grill](#) (p.109).

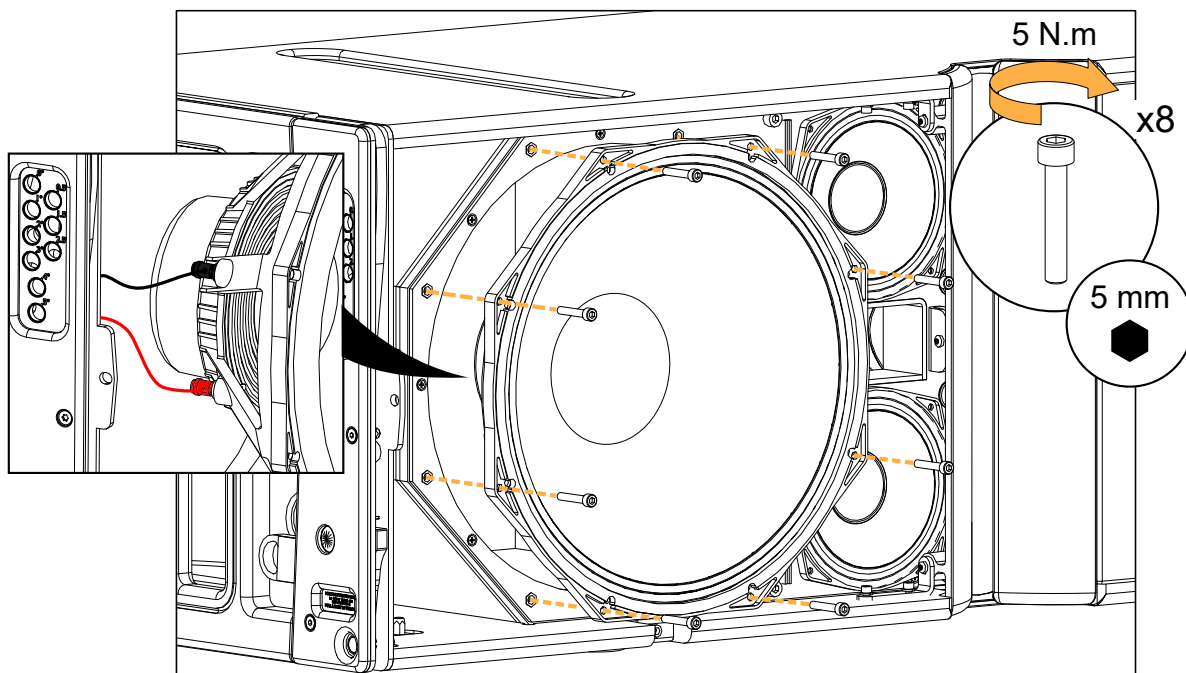
Exploded view



For safety reasons, always use the new screws and spare parts provided in the KR.
If no new screws are available, use blue threadlocker.



Gradually tighten the screws following a star pattern.



What to do next

Perform the [Acoustical check](#) (p.58) procedures.

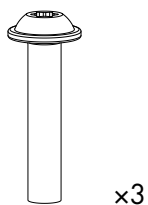
D/R - Fins

Tools

- torque screwdriver
- T20 Torx bit

KR HPPH62

Kit HP PH62 Speaker 6.5" - 8 ohms



S237

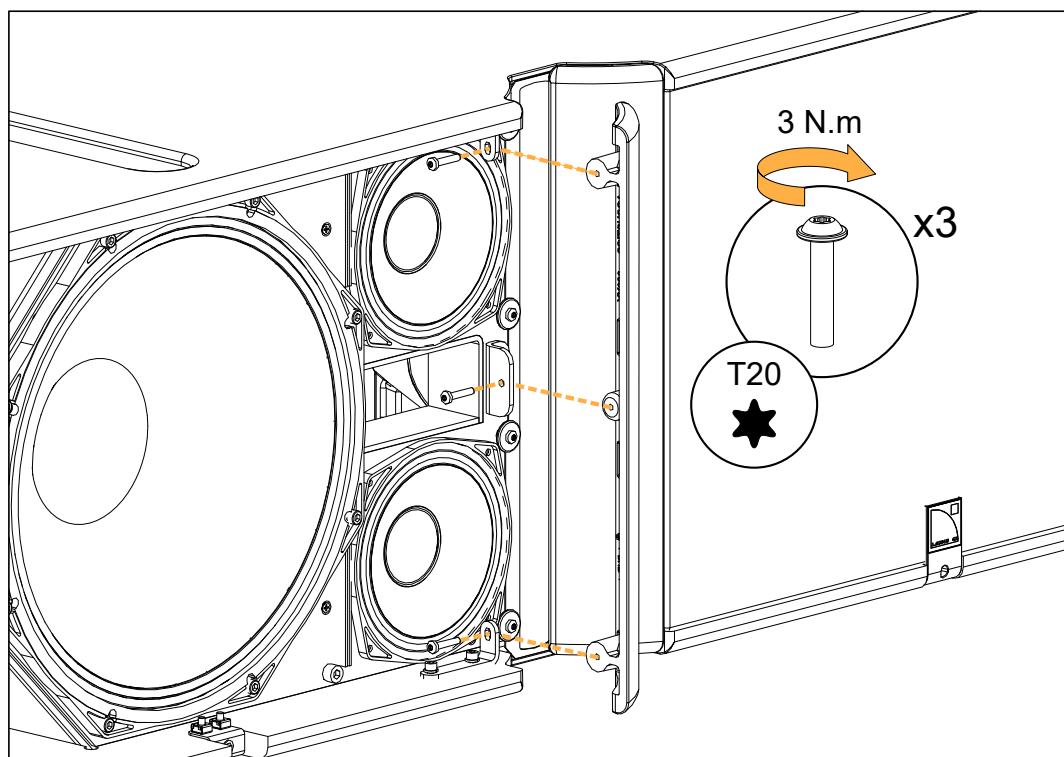
M5x16 hex

Prerequisite

Grill removed.

See [D/R - Grill](#) (p.109).

Exploded view



D/R - MF speaker

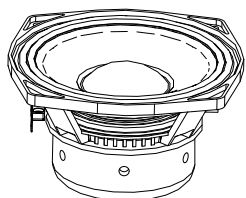
Tools

- torque screwdriver
- 4 mm hex bit

Repair kit

KR HPPH62

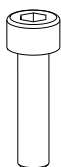
Kit HP PH62 Speaker 6.5" - 8 ohms



x1

18614

6.5" LF/MF speaker - 8 Ω



x4

S342

M5x20 hex



x2

S100113

M5x16 hex

Prerequisite

Grill removed.

See [D/R - Grill](#) (p.109).

Fin removed.

See [D/R - Fins](#) (p.111).

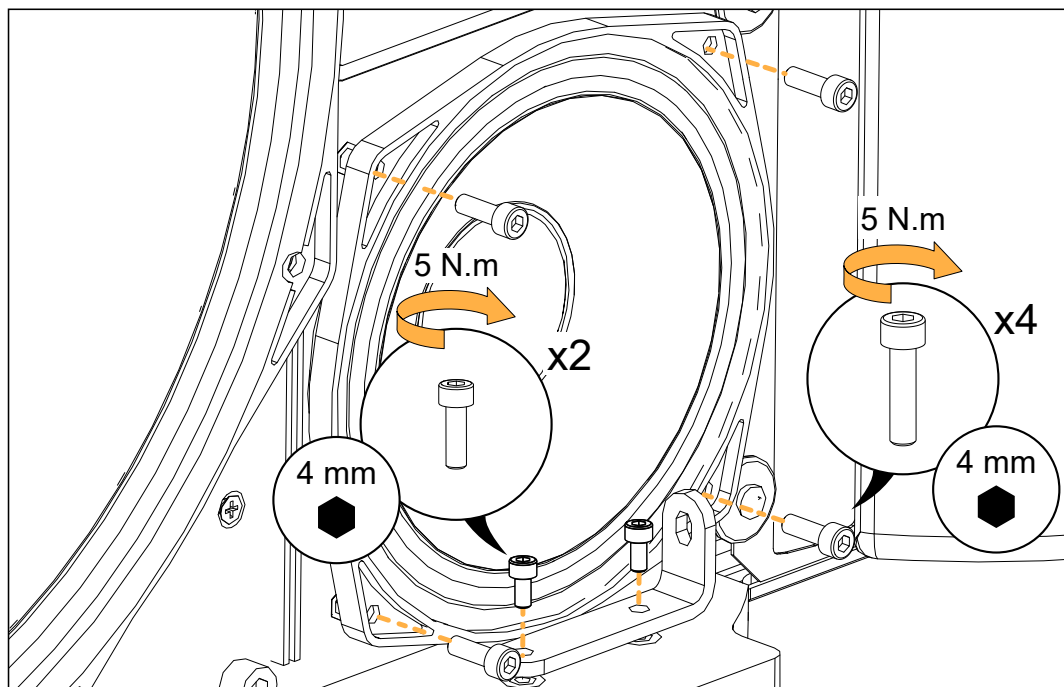
Exploded view



For safety reasons, always use the new screws and spare parts provided in the KR.



Gradually tighten the screws following a star pattern.



What to do next

Perform the [Acoustical check](#) (p.58) procedures.

D/R - Connector plate

Tools

- torque screwdriver
- T30 Torx bit

Repair kit

KR HPBC33 - Kit HP BC33 Driver 3" - 20 ohms

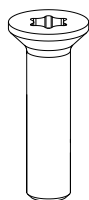
or KR HSBC33 - Kit diaphragm for 3" driver



x1

100585_2

adhesive gasket kit



x10

S100110

M6x25 Torx

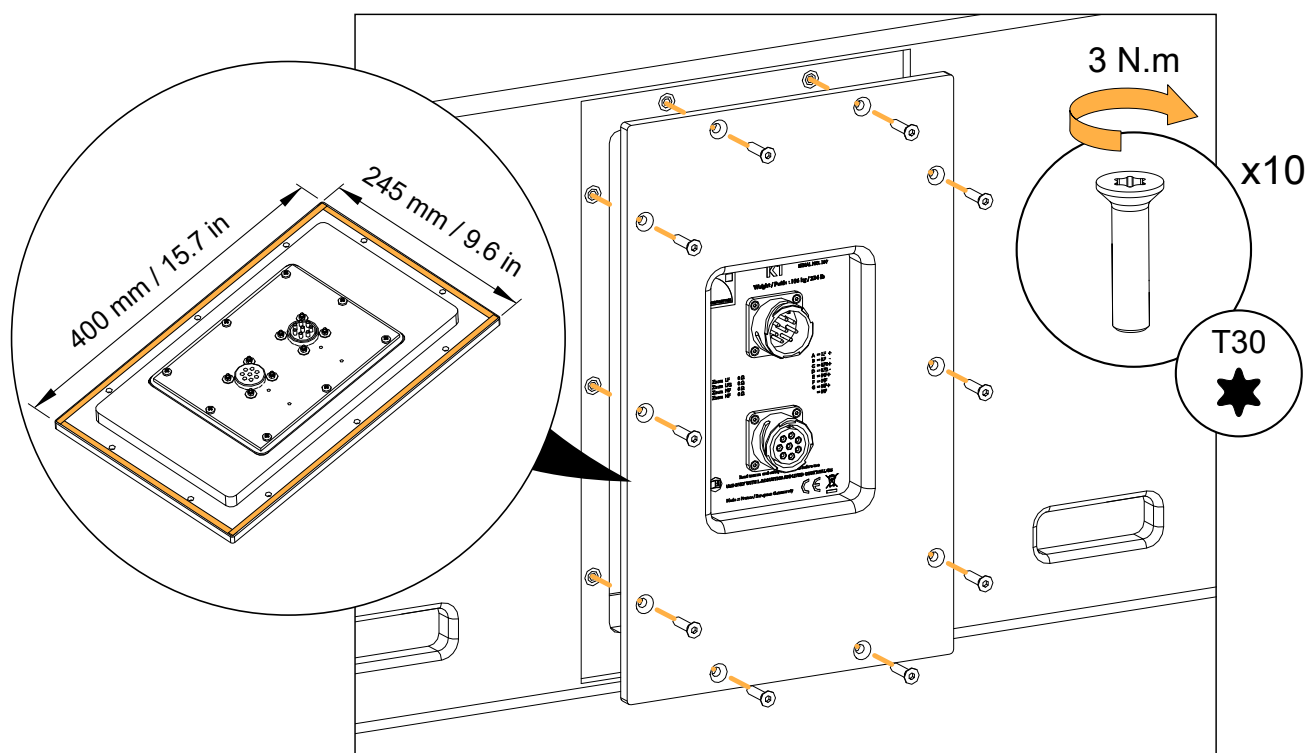
Exploded view



For safety reasons, always use the new screws and spare parts provided in the KR.
If no new screws are available, use blue threadlocker.



If the gasket is damaged, remove and replace it.



D/R - HF driver**Tools**

- torque screwdriver
- 5 mm hex bit

Repair kit**KR HPBC33**

Kit HP BC33 Driver 3" - 20 ohms



i * The gasket and the screws are also provided in the KR HSBC33 (Kit diaphragm for 3" driver).

Prerequisite

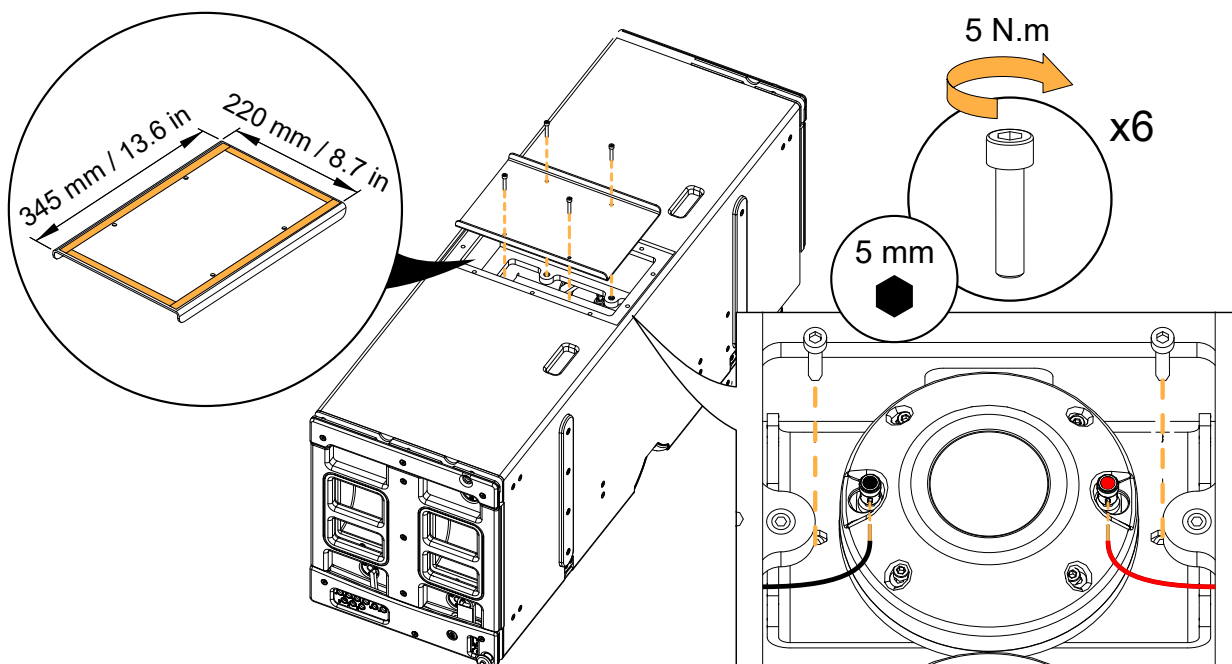
Connector plate removed.

See [D/R - Connector plate](#) (p.113).

Exploded view

! For safety reasons, always use the new screws and spare parts provided in the KR.
If no new screws are available, use blue threadlocker.

i If the gasket is damaged, remove and replace it.

**What to do next**

Perform the [Acoustical check](#) (p.58) procedures.

D/R - HF diaphragm

Tools

- torque screwdriver
- 3 mm hex bit
- 3 mm hex wrench
- compressed air blower

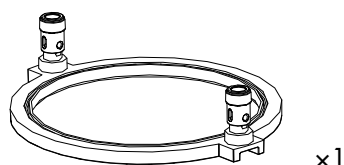
Consumables

- double face adhesive tape

Repair kit

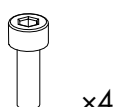
KR HSBC33

Kit diaphragm for 3" driver



03024

diaphragm for 1.4" driver - 20 ohms (with shims)



S03024

M4x14 hex

Prerequisite

Connector plate removed.

HF driver removed.

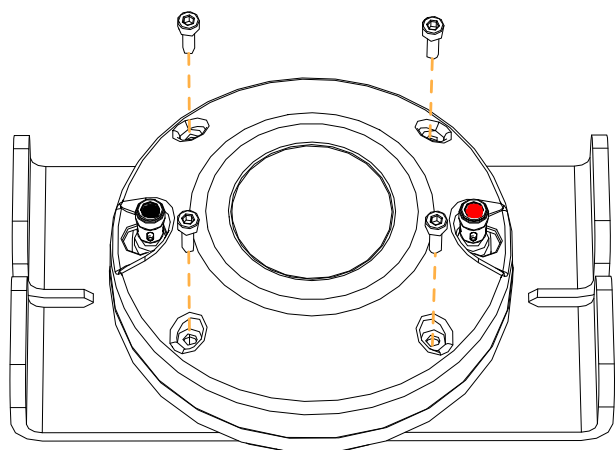
See [D/R - Connector plate](#) (p.113).

See [D/R - HF driver](#) (p.114)

Exploded view



Use a blower or double face adhesive tape to clean the dome and the air gap.



K1-SB

Introduction

This section contains the following maintenance procedures:

- [D/R - Grill](#) (p.117)
- [D/R - LF speaker](#) (p.118)

For advanced maintenance, contact your L-Acoustics representative.

Exploded views

In order to operate, follow the order outlined here. Each assembly refers to the corresponding Disassembly/Reassembly (D/R) procedure and the necessary repair kit (KR).

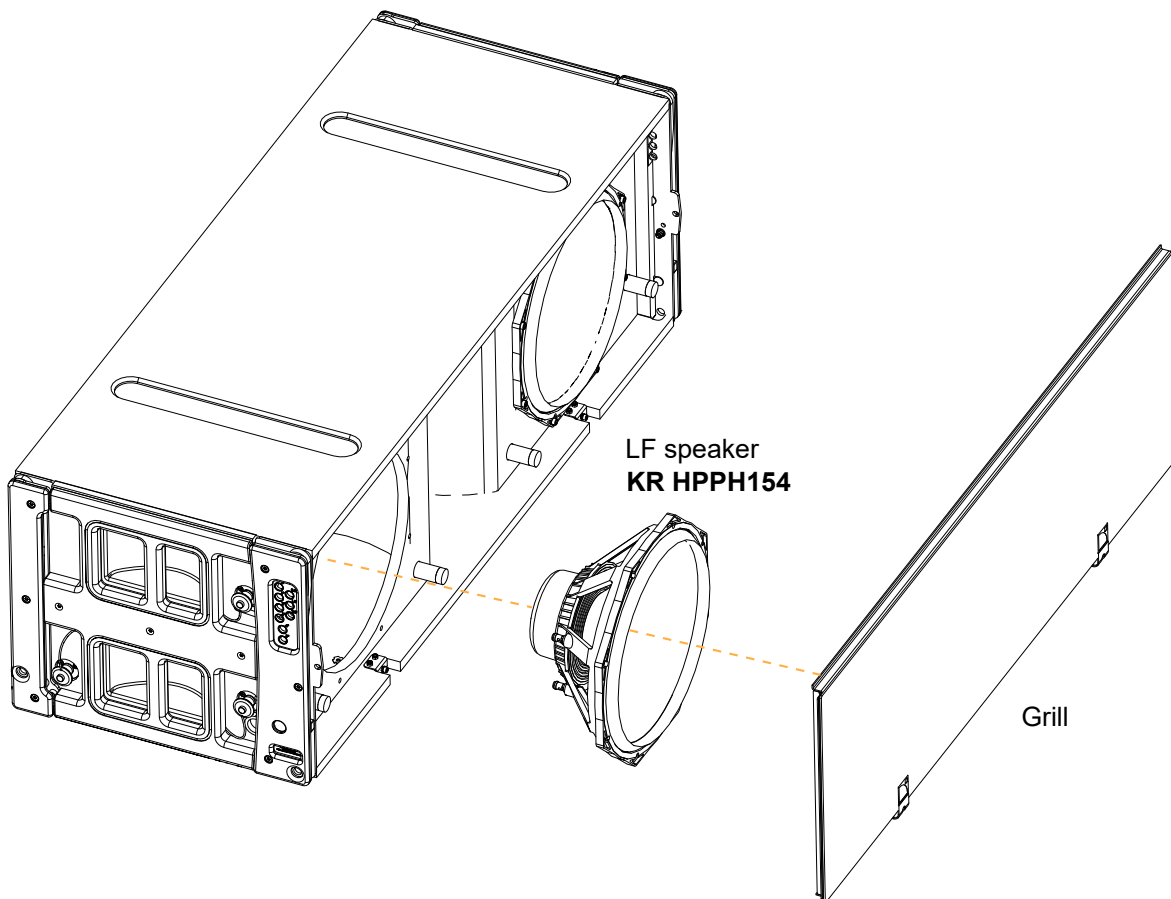


Using the KR HSPH154 reconing kit

Follow the instructions to remove the LF speaker from the enclosure.

Perform the reconing procedure according to the information provided by the manufacturer.

Use the screws and fasteners provided in KR HSPH154 for reassembly.



Disassembly and Reassembly procedures

D/R - Grill

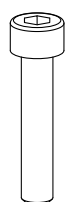
Tools

- torque screwdriver
- 4 mm hex bit

Repair kit

KR HPPH154*

Kit HP PH154 Speaker 15" - 8 ohms



x2

S100155

M5x25 hex



x2

S158

plain washer Ø 5 mm



*The screws and fasteners are also provided in the KR HSPH154 (Kit reconing 15").

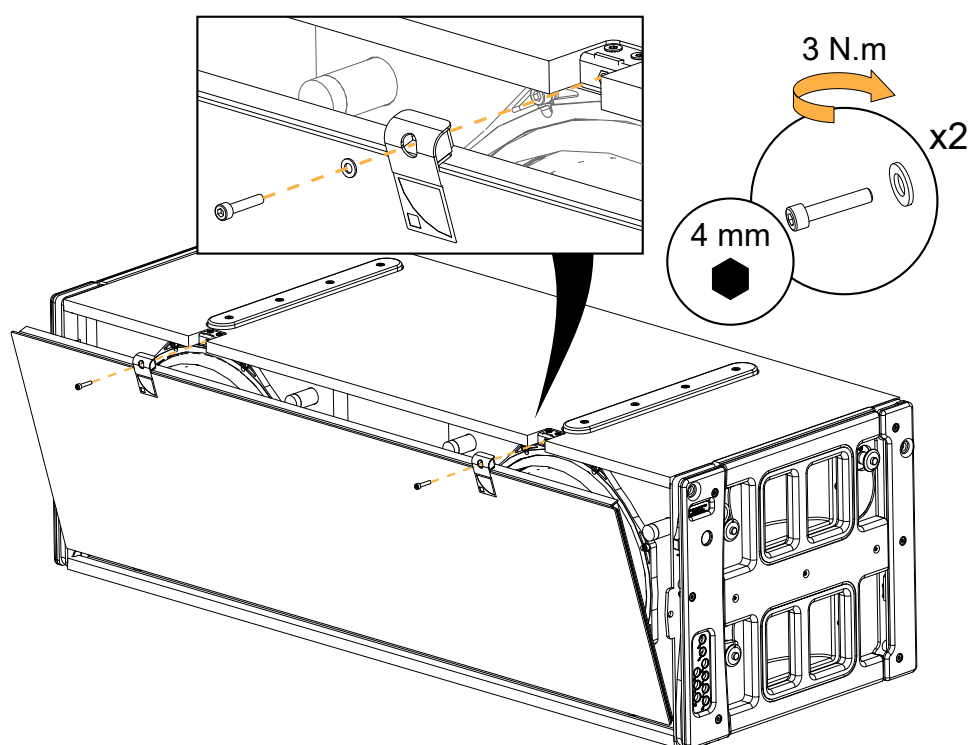
Prerequisite

The enclosure is placed on its top.

Exploded view



For safety reasons, always use the new screws and spare parts provided in the KR.
If no new screws are available, use blue threadlocker.

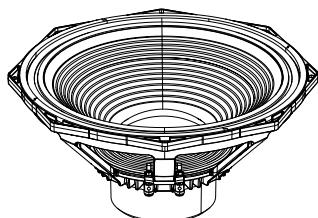


D/R - LF speaker**Tools**

- torque screwdriver
- 5 mm hex bit

Repair kit**KR HPPH154***

Kit HP PH154 Speaker 15" - 8 ohms



x1

18463

15" speaker - 8 ohms



x8

S100054

M6x30 hex



*The screws and fasteners are also provided in the KR HSPH154 (Kit reconing 15").

Prerequisite

Grill removed.

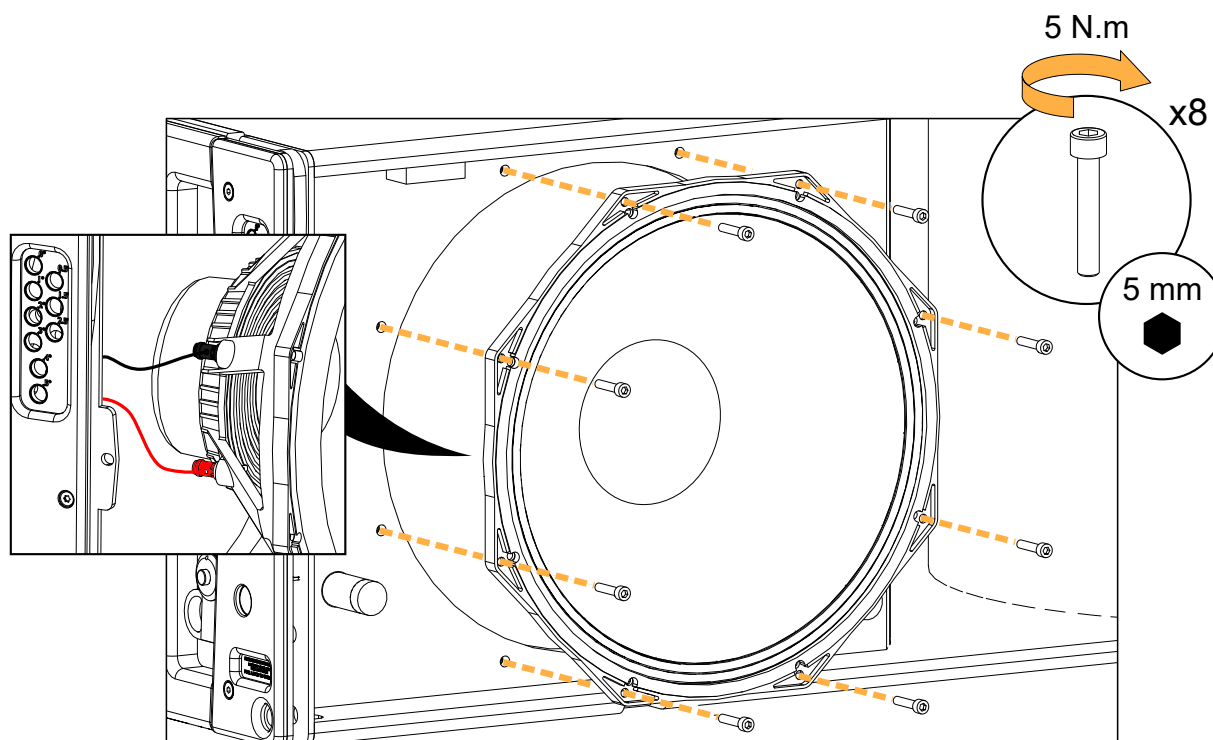
See [D/R - Grill](#) (p.117).

Exploded view

For safety reasons, always use the new screws and spare parts provided in the KR.
If no new screws are available, use blue threadlocker.



Gradually tighten the screws following a star pattern.



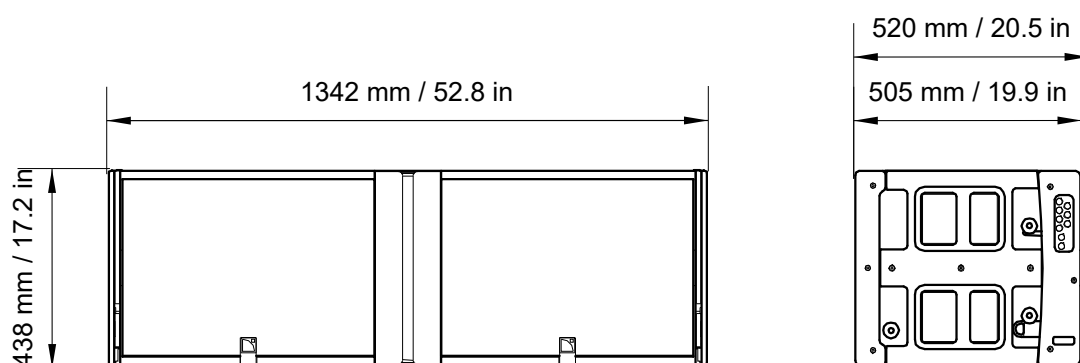
Specifications

K1 specifications

Description	3-way full-range active WST® enclosure, quad-amplified by LA12X
Usable bandwidth (-10 dB)	35 Hz - 20 kHz ([K1])
Maximum SPL¹	149 dB ([K1])
Nominal directivity (-6 dB)	horizontal: 90° symmetric vertical: dependent upon the number of elements and the line source curvature
Transducers	LF: 2 × 15" neodymium MF: 4 × 6.5" neodymium, direct-radiating HF: 3 × 3" diaphragm compression drivers
Acoustical load	LF: bass-reflex MF: bass-reflex HF: DOSC waveguide
Nominal impedance	LF: 2 × 8 Ω MF: 8 Ω HF: 8 Ω
Connectors	IN: 1 × 8-point PA-COM LINK: 1 × 8-point PA-COM
Rigging and handling	4 handles integrated into the cabinet inter-enclosure angles: 0°, 0.5°, 1°, 1.5°, 2°, 2.5°, 3°, 4° or 5°
Weight (net)	106 kg / 234 lb
Cabinet	premium grade Baltic birch plywood high density polyethylene
Front	coated steel grill acoustically neutral 3D fabric
Rigging components	high grade steel with anti-corrosion coating
Finish	grey brown RAL 8019
IP	IP43

¹ Peak level measured at 1 m under free field conditions using pink noise with crest factor 4 (preset specified in brackets).

K1 dimensions

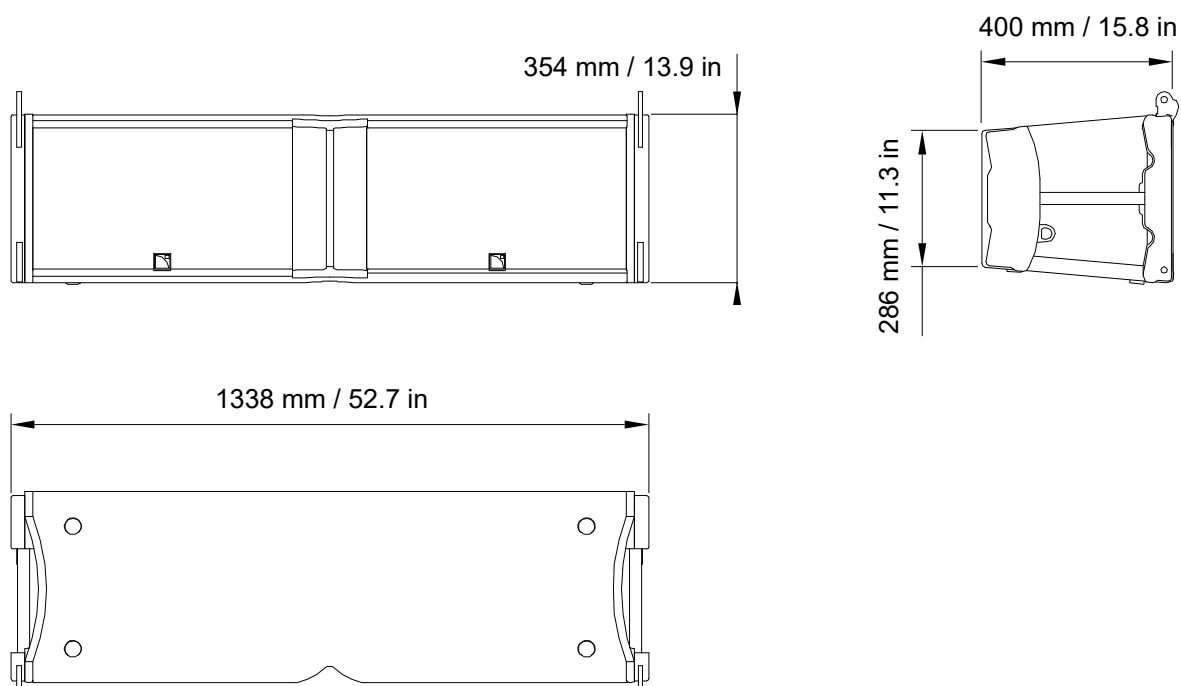


K2 specifications

Description	3-way full-range active WST enclosure, quad-amplified by LA4X / LA12X
Usable bandwidth (-10 dB)	35 Hz - 20 kHz ([K2 70])
Maximum SPL ¹	147 dB ([K2 70])
Nominal directivity (-6 dB)	horizontal: 110°/70° symmetric or 90° asymmetric (35°/55° or 55°/35°) vertical: dependent upon the number of elements and the line source curvature
Transducers	LF: 2 × 12" cone drivers MF: 4 × 6.5" cone drivers HF: 2 × 3" diaphragm compression drivers
Acoustical load	LF: bass-reflex, L-Vents MF: bass-reflex HF: DOSC waveguide
Nominal impedance	LF: 2 × 8 Ω MF: 8 Ω HF: 16 Ω
Connectors	IN: 1 × 8-point PA-COM LINK: 1 × 8-point PA-COM
Rigging and handling	4-point captive rigging system inter-enclosure angles: 0.25°, 1°, 2°, 3°, 4°, 5°, 7.5° or 10°
Weight (net)	56 kg / 123.2 lb
Cabinet	premium grade Baltic birch plywood
Front	coated steel grill acoustically neutral 3D fabric
Rigging components	high grade steel with anti-corrosion coating
Finish	dark grey brown Pantone 426 C
IP	IP55

¹ Peak level measured at 1 m under free field conditions using pink noise with crest factor 4 (preset specified in brackets).

K2 dimensions

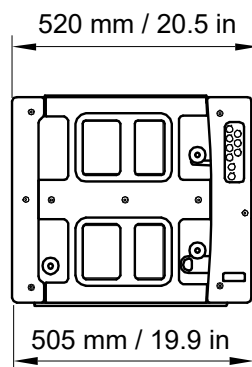
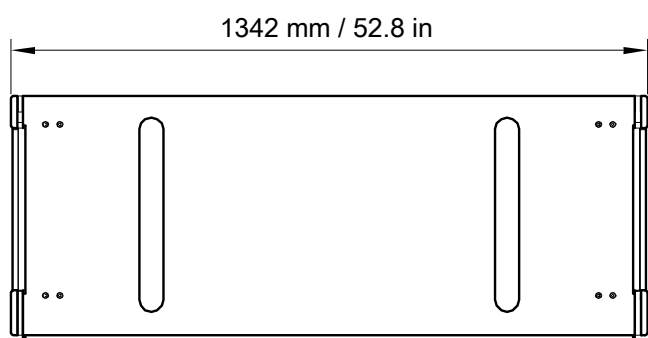
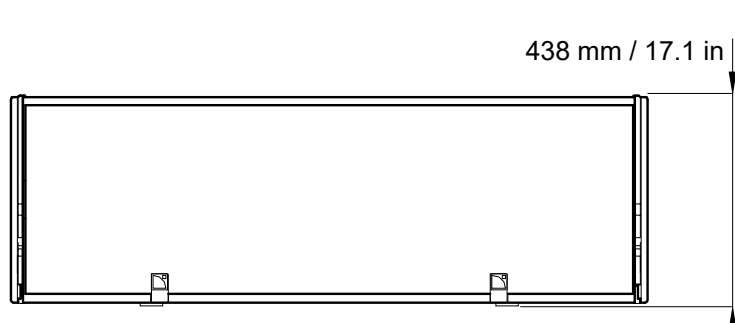
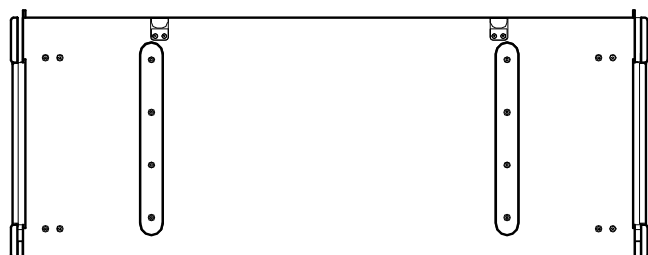


K1-SB specifications

Description	K1 system subwoofer 2×15", amplified by LA12X
Low frequency limit (-10 dB)	30 Hz ([K1SB_60])
Maximum SPL¹	145 dB ([K1SB_X])
Transducers	2 × 15" cone drivers
Acoustical load	bass-reflex, L-Vents
Nominal impedance	4 Ω
Connectors	IN: 1 × 4-point speakON
Rigging and handling	captive rigging system 4-point rigging system inter-enclosure angles: 0°, 0.5°, 1°, 1.5°, 2°, 2.5°, 3°, 4° or 5° 4 handles integrated into the cabinet
Weight (net)	83 kg / 183 lb
Cabinet	premium grade Baltic birch plywood
Front	coated steel grill acoustically neutral 3D fabric
Rigging components	high grade steel with anti-corrosion coating
Finish	dark grey brown Pantone 426 C pure white RAL 9010
IP	IP45

¹ Peak level at 1 m under half space conditions using pink noise with crest factor 4 (preset specified in brackets).

K1-SB dimensions

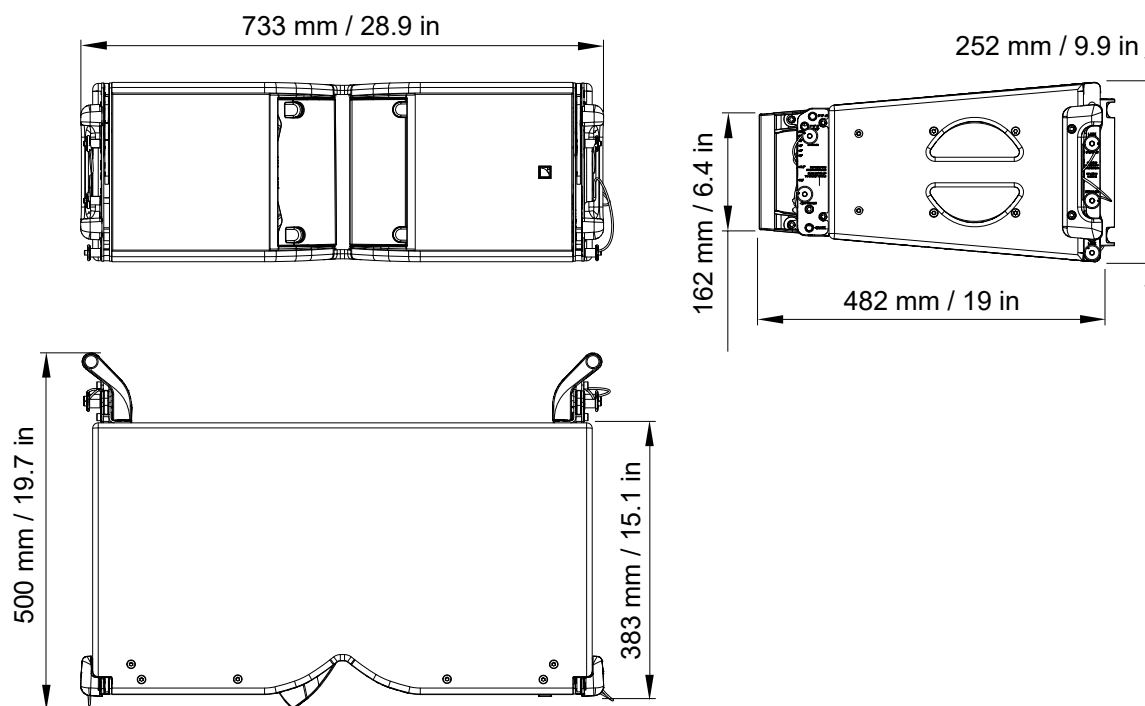


Kara II specifications

Description	2-way active WST® enclosure: 2 × 8" LF + 3" HF diaphragm, amplified by LA4X / LA12X
Usable bandwidth (-10 dB)	55 Hz - 20 kHz ([KARA II 70])
Maximum SPL ¹	142 dB ([KARA II 70])
Nominal directivity (-6 dB)	horizontal: 70° / 110° symmetric or 90° asymmetric (35°/90°) vertical: dependent upon the number of elements and the line source curvature
Transducers	LF: 2 × 8" neodymium cone drivers HF: 1 × 3" neodymium diaphragm compression driver
Acoustical load	LF: bass-reflex HF: DOSC waveguide, L-Fins
Nominal impedance	LF: 8 Ω HF: 8 Ω
Connectors	IN: 1 × 4-point speakON LINK: 1 × 4-point speakON
Rigging and handling	4-point captive rigging system 2 side handles 2 rear handles inter-enclosure angles: 0°, 1°, 2°, 3°, 4°, 5°, 7.5° or 10°
Weight (net)	26 kg / 57 lb
Cabinet	premium grade Baltic birch plywood
Front	coated steel grill acoustically neutral 3D fabric
Rigging components	high grade steel with anti-corrosion coating
Finish	dark grey brown Pantone 426 C
IP	IP55

¹ Peak level measured at 1 m under free field conditions using pink noise with crest factor 4 (preset specified in brackets).

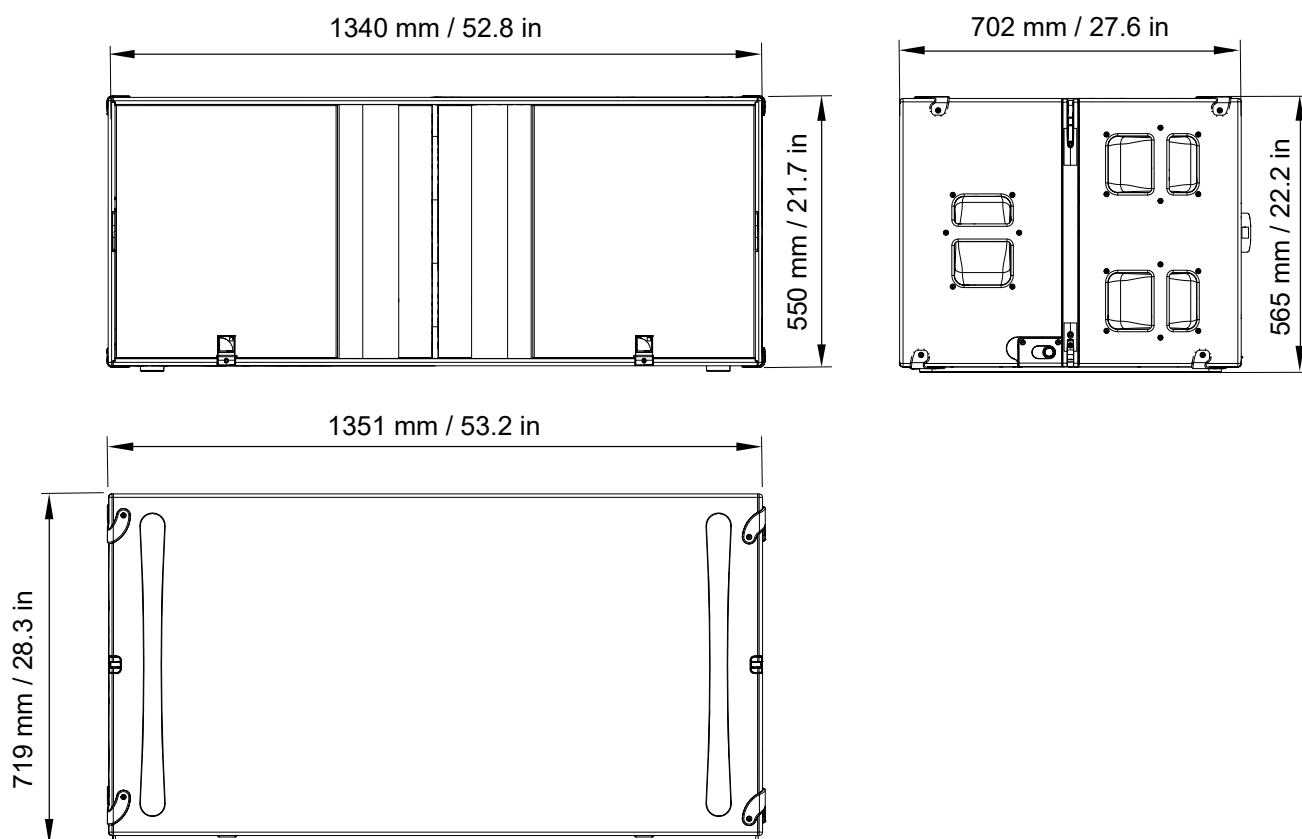
Kara II dimensions



KS28 specifications

Description	Flyable subwoofer 2 × 18", amplified by LA2Xi / LA12X
Low frequency limit (-10 dB)	25 Hz ([KS28_100])
Maximum SPL¹	143 dB ([KS28_100]) with LA2Xi (bridge mode) / LA12X 136 dB ([KS28_100]) with LA2Xi
Directivity	standard or cardioid
Transducers	2 × 18" neodymium cone drivers
Acoustical load	bass-reflex, L-Vents
Nominal impedance	4 Ω
Connectors	IN: 1 × 4-point speakON
Rigging and handling	flush-fitting 2-point rigging system 6 ergonomic handles 2 ground runners 8 side runners
Weight (net)	79 kg / 174 lb
Cabinet	premium grade Baltic beech and birch plywood
Front	coated steel grill acoustically neutral 3D fabric
Rigging components	high grade steel
Finish	dark grey brown Pantone 426 C
IP	IP55

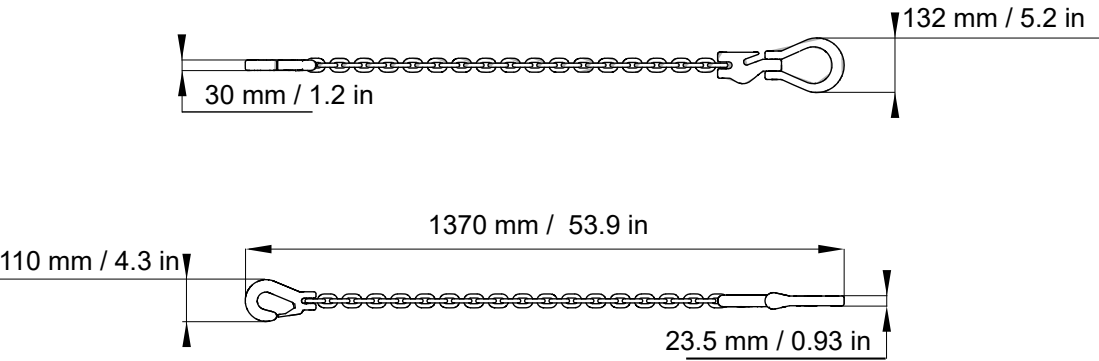
¹ Peak level at 1 m under half space conditions using pink noise with crest factor 4 (preset specified in brackets).

KS28 dimensions

K1-BPCHAIN specifications

Description	Adjustable sling for K1-BUMP or K2-BUMP
Weight (net)	6.3 kg / 13.9 lb
Material	high grade steel with anti-corrosion coating

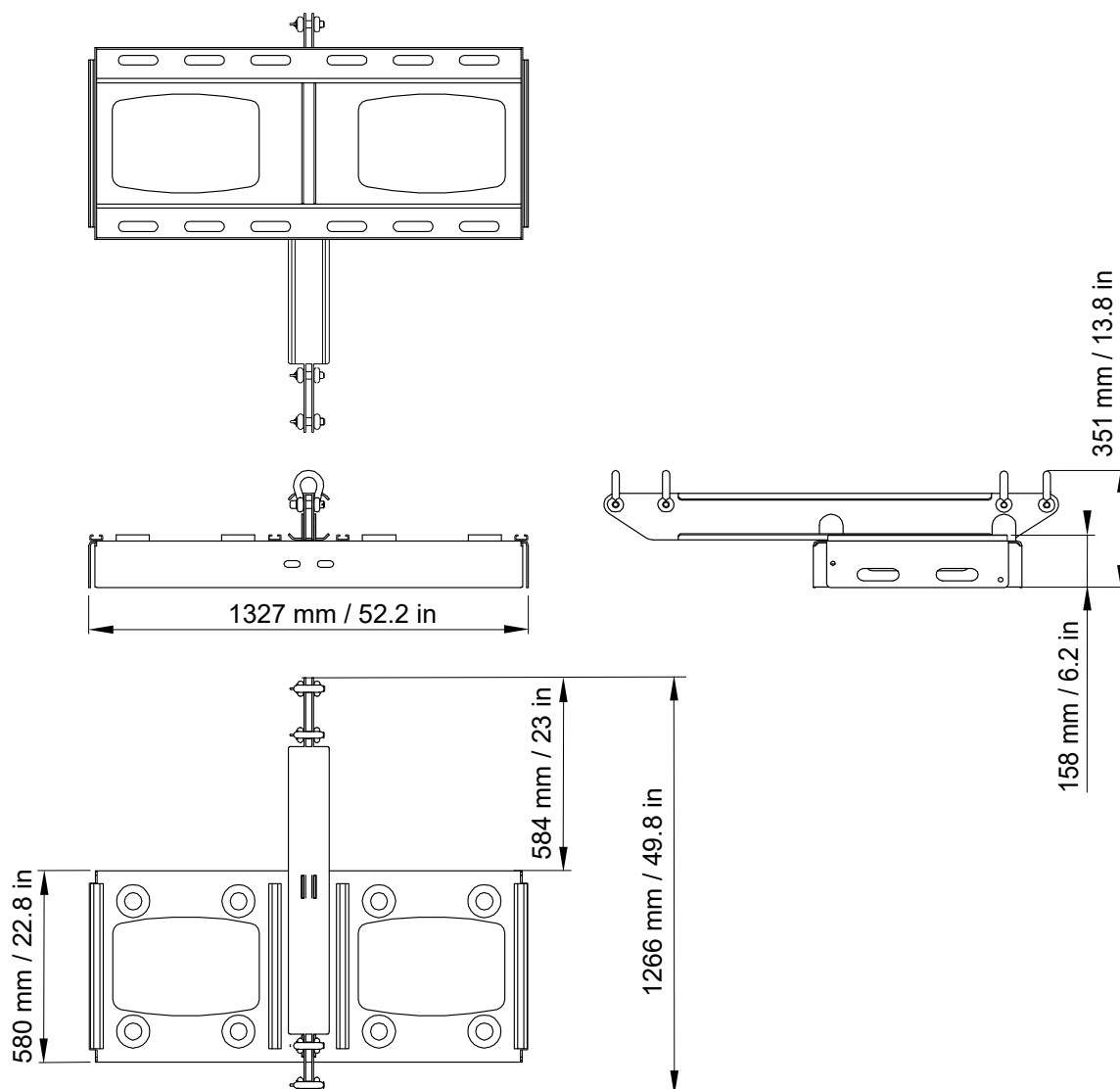
K1-BPCHAIN dimensions



K1-BUMP specifications

Description	Structure for flying and stacking K1 and K1-SB arrays
Weight (net)	98 kg / 216 lb
Material	high grade steel with anti-corrosion coating

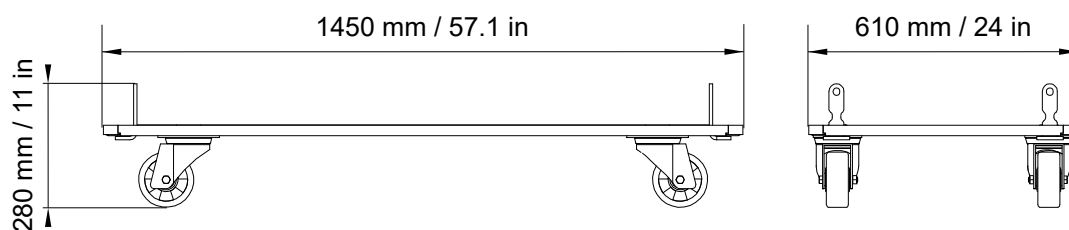
K1-BUMP dimensions



K1-CHARIOT specifications

Description	Chariot for 4 × K1 or 4 × K1-SB
Weight (net)	47 kg / 103 lb
Material	high grade steel with anti-corrosion coating

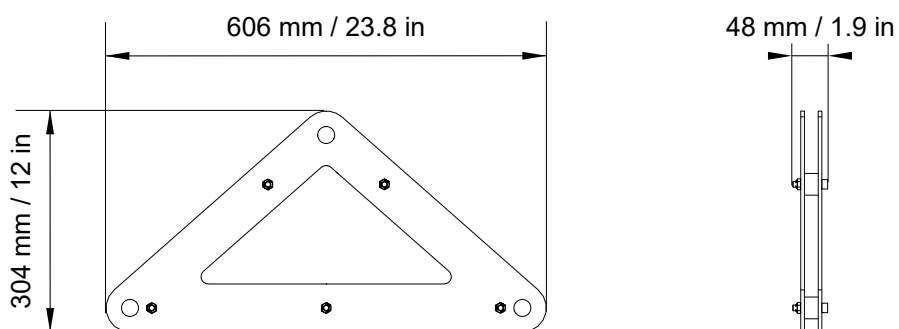
K1-CHARIOT dimensions



K1-DELTA specifications

Description	Rigging accessory for rear attachment of 2 motors to K1-BUMP
Weight (net)	9.5 kg / 21 lb
Material	high grade steel with anti-corrosion coating

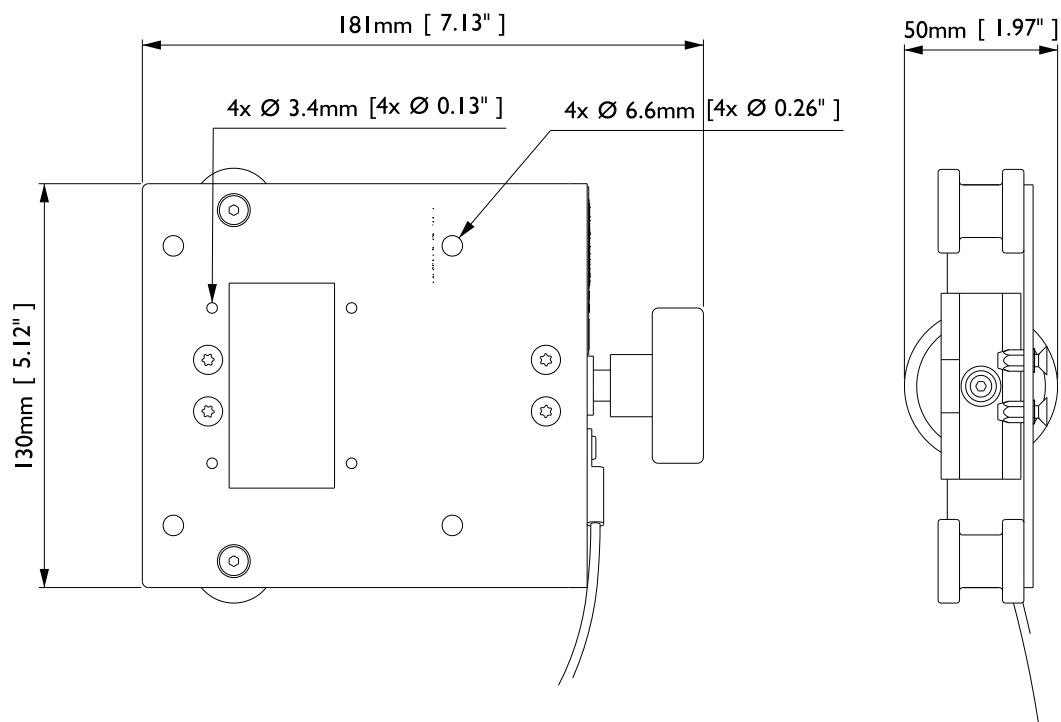
K1-DELTA dimensions



K1-LASERMOUNT specifications

Description	K1 laser support plate (compatible TEQSAS / SSE Prosight / Align Array / KSG)
Weight (net)	0.8 kg / 1.8 lb
Material	high grade steel with anti-corrosion coating

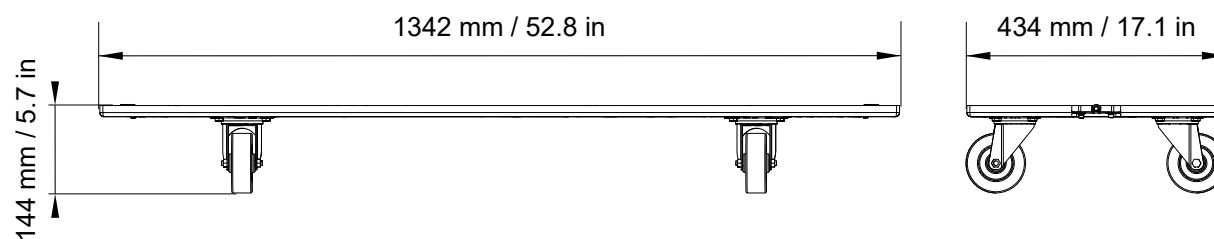
K1-LASERMOUNT dimensions



K1-PLA specifications

Description	Removable front dolly for K1 or K1-SB enclosure
Weight (net)	7.5 kg / 16.5 lb
Material	high grade steel with anti-corrosion coating

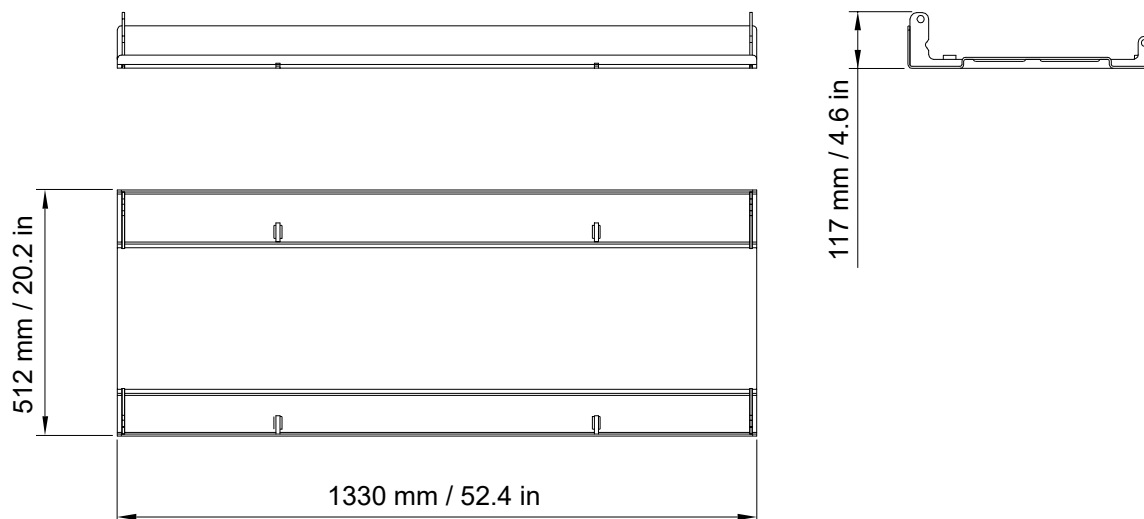
K1-PLA dimensions



KARA-DOWNK1 specifications

Description	Flying bumper for rigging KARA (II) under K1 or K1-SB
Weight (net)	21 kg / 46 lb
Material	high grade steel with anti-corrosion coating

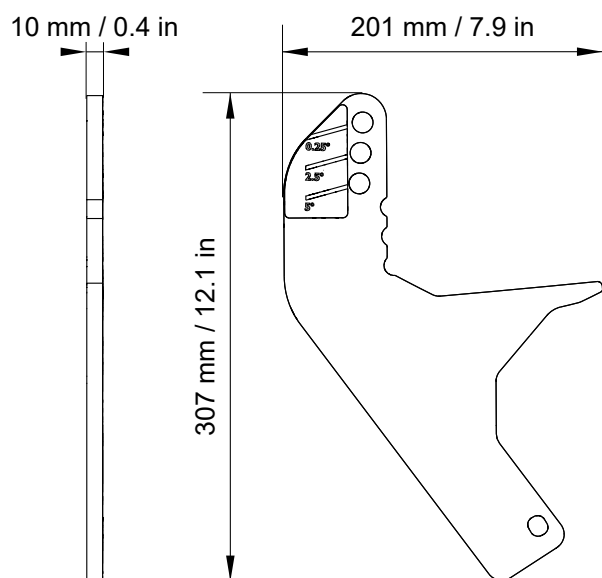
KARA-DOWNK1 dimensions



K2-LINK specifications

Description	Rigging accessory for rear attachment of K2 below K1
Weight (net)	1.8 kg / 4 lb
Material	high grade steel with anti-corrosion coating

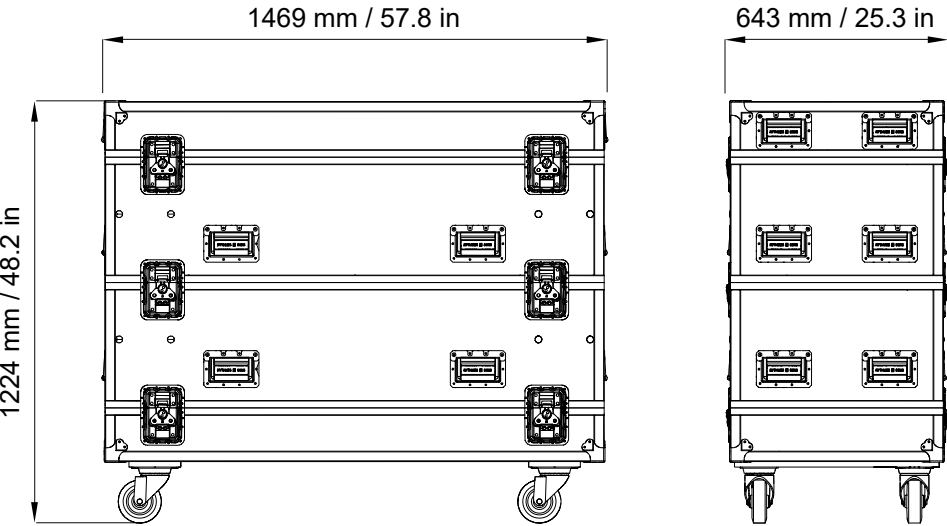
K2-LINK dimensions



K-BUMPFLIGHT specifications

Description	Modular flight case for 2 K1-BUMP or 2 K2-BUMP
	12 pieces of adhesive foam
Weight (net)	105 kg / 231.5 lb

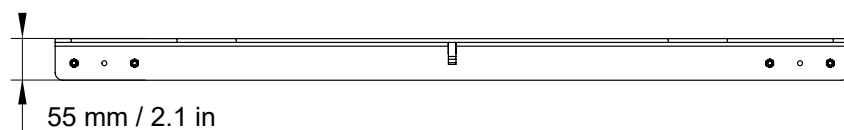
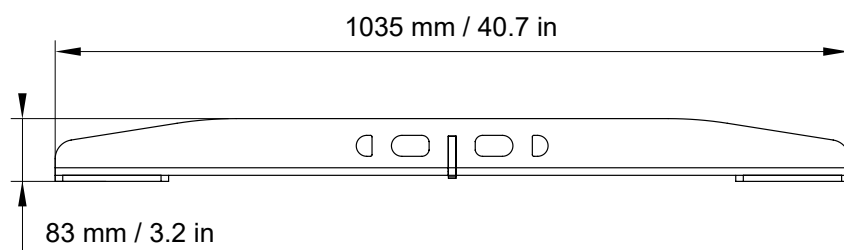
K-BUMPFLIGHT dimensions



KS28-OUTRIG specifications

Description	Stability bars for KS28
Weight (net)	9.4 kg / 20 lb
Material	high grade steel with anti-corrosion coating

KS28-OUTRIG dimensions



Recommendation for speaker cables



Cable quality and resistance

Only use high-quality fully insulated loudspeaker cables made of stranded copper wire.

Use cables with a gauge offering low resistance per unit length and keep the cables as short as possible.

It is good practice to keep loudspeaker cables short to ensure optimal system performance. L-Acoustics strongly recommends using cables of similar type, length, and gauge to address symmetrical deployment of loudspeakers, such as stereo systems, L-ISA frontal systems, or outfill systems.



For more information about cable effect on loudspeaker frequency response, refer to the publication **Demystifying the effects of loudspeaker cables** on the L-Acoustics website, in **Education > Scientific resources > Scientific publications**.

Refer to the following table for recommended cable length for uncompromised performance.

cable gauge			recommended maximum length					
			8 Ω load		4 Ω load		2.7 Ω load	
mm ²	SWG	AWG	m	ft	m	ft	m	ft
1.5	18	16	18	60	9	30	–	–
2.5	15	14	30	100	15	50	10	33
4	13	11	50	160	25	80	17	53
6	11	9	74	240	37	120	25	80

Use the more detailed L-Acoustics calculation tool to evaluate cable length and gauge based on the type and number of loudspeakers connected. The calculation tool is available on our website:

<https://www.l-acoustics.com/installation-tools/>

Installing a laser inclinometer

K1-LASERMOUNT

K1-LASERMOUNT can be used to secure a laser inclinometer vertically or horizontally, using the L-Acoustics Tech Toolcase II accessories.

Five sensor models can be used:

- TEQSAS® LAP-TEQ PLUS (L-Acoustics Tech Toolcase II accessory)
- TEQSAS® LAP-TEQ (legacy model)
- KSG® RECLINE Compact
- UVM sensor
- SSE® ProSight



LAP-TEQ / LAP-TEQ PLUS sensor and display compatibility

The legacy LAP-TEQ sensors are compatible with the new TEQSAS® LAP-TEQ PLUS displays. Conversely, the new sensors are not compatible with the legacy displays.

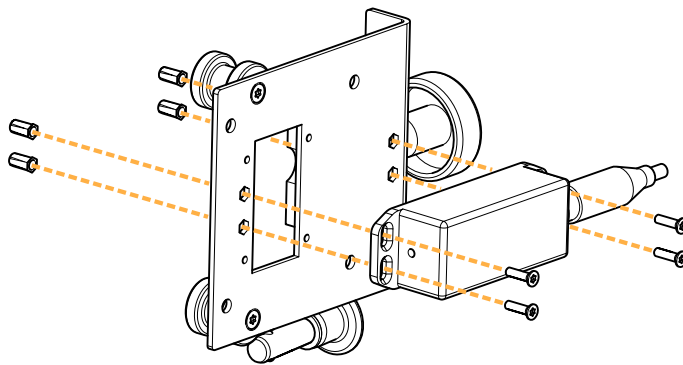


TEQSAS® LAP-TEQ calibration

Make sure the sensor is calibrated for vertical use.

With screws

Mount the inclinometer on K1-LASERMOUNT:



Vertical mount

K1-BUMP

K1-BUMP features a support plate to accommodate a laser inclinometer.

Four sensor models can be used:

- TEQSAS® LAP-TEQ PLUS (L-Acoustics Tech Toolcase II accessory)
- TEQSAS® LAP-TEQ (legacy model)
- KSG® RECLINE Compact
- SSE® ProSight



LAP-TEQ / LAP-TEQ PLUS sensor and display compatibility

The legacy LAP-TEQ sensors are compatible with the new TEQSAS® LAP-TEQ PLUS displays. Conversely, the new sensors are not compatible with the legacy displays.



TEQSAS® LAP-TEQ calibration

Make sure the sensor is calibrated for vertical use

With TEQSAS® LAP-TEQ PLUS / LAP-TEQ / KSG® RECLINE Compact

About this task

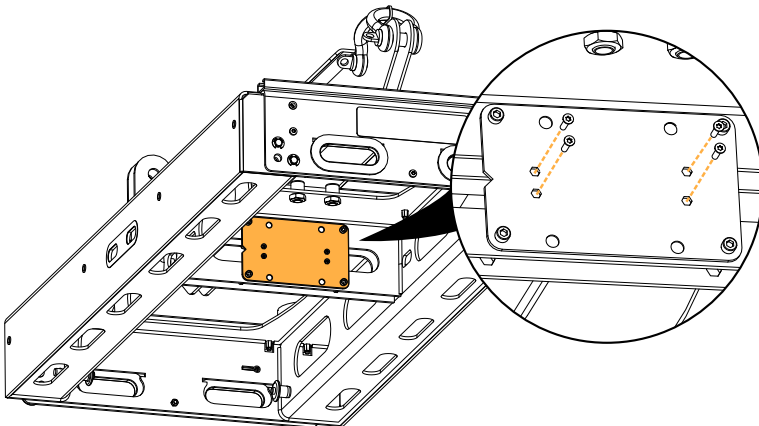


TEQSAS® LAP-TEQ calibration

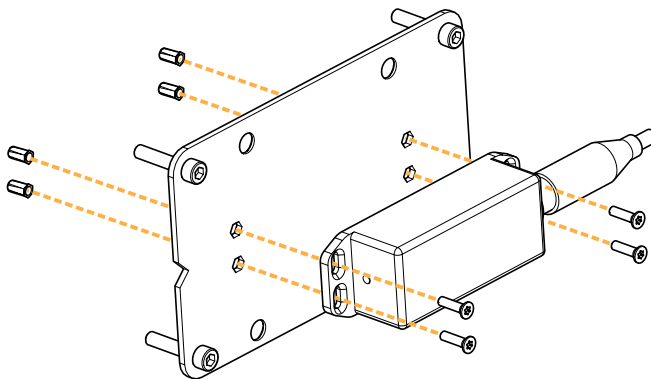
Make sure the sensor is calibrated for vertical use.

Procedure

1. Remove the four M4 Torx screws (T20) from the plate.



2. Put thread-locker in the four threaded inserts.
3. Position the sensor so it points toward the front of the K1-BUMP.
4. Secure the sensor with the four screws.



5. Connect the XLR cable to the sensor.
6. Refer to the **manufacturer instructions** to calibrate the inclinometer.

With SSE® Prosight

SSE Prosight2 is a SSE product. Refer to the [SSE Fitting guides](#) for more information on how to mount SSE Prosight on a K Series bumper.



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